

POLYCOPIÉ

What changed, and what it means for our paper

The breakdown-radius measurements,
August to 16 September 2026

For Chris, from Zé. Everything I ran on our breakdown-radius line since the August pilots, most of which never left my machine: what was measured and on what, what came out, what I had to retract on the way, what it changes in your draft of 16 September, and where each file lives in the repository branch `ze/experiments`.

Written for someone who knows the object. Nothing here re-teaches CPDAGs or Meek closure. Chapter 1 aligns our two vocabularies; Parts II to V are the measurements in the order they happened; Part VI is your draft and the merge; Part VII is the repository.

Every number carries a file. Each green box names the repository file or script that produced it. Where I corrected myself, the correction is in the text.

Contents

Before you start	4
I The shared object	6
1 One object, two vocabularies	7
1.1 Your geometry, on your example	7
1.2 The translation table	9
1.3 The invariant, and the only place the truth enters	10
II August: what the pilots established	12
2 August I: the pilot and the $\mathbf{se}$ criterion	13
2.1 The July pilot: a radius priced against the answer key	13
2.2 $E1'$: the analyst's own standard error replaces τ	14
2.3 ARM 2: what one wrong claim costs a confidence interval	16
2.4 Corrections made before 31 August, and four since	18
3 August II: locality, the firewall, and a wrong skeleton	20
3.1 Prehistory: the three re-rank experiments of 14 August	20
3.2 Locality, measured and then cut down to size	21
3.3 The firewall: what the consistency check buys	23
3.4 X3: a wrong skeleton, and one problem end to end	24
III Early September: what the radius should measure	28
4 Early September: what should the radius measure?	29
4.1 After 3 September: the defect I set out to repair	29
4.2 Thirty-seven candidates, two lemmas, one object	30
4.3 The null radius: Rosenbaum's question, taken literally	32
4.4 9 September: the "never" contrast was an admission rule	33
5 Where to evaluate, and three defects in the old corpus	36
5.1 What a deployed instance is	36
5.2 Defect 1: the analyst's knowledge is read from the true DAG	37
5.3 Defects 2 and 3: what the committed radii count	38
5.4 <i>Beat the BFS First</i> : the protocol of 11 September	40

IV	The protocol, executed (11 to 13 September)	42
6	The observable frame and the frozen questionnaire	43
6.1	The frame: rows chosen from the CPDAG alone	43
6.2	The questionnaire, and the repair that freezes K	45
6.3	The decision table: the claim an estimate rests on	46
6.4	A salted seed, and the calibration rerun	47
7	Knowledge suppliers as error processes	49
7.1	Four sources of wrong claims	49
7.2	Scale buys claims, not validity	50
7.3	Names, a truth-free bridge, and two negative results	52
7.4	The cascade test: a justification withdrawn	53
8	Table 4: the claim unit never over-certifies, the hop unit does	55
8.1	The ledger, five instruments and four buckets	55
8.2	What Table 4 says	57
8.3	Why the zero is structural, and what the hop rows are	58
8.4	The inspection trigger fired	60
9	When the class itself is estimated	63
9.1	The panel	63
9.2	What the panel measured	64
9.3	Why no retraction reaches the error	65
9.4	Predictions, and the sentence the paper can print	66
9.5	The substrate decisions behind Table 2	67
V	Pricing knowledge: the Rosenbaum analogue	69
10	Rosenbaum's Γ, and Table 1 on real structure (M1)	70
10.1	12 September: the subject was wrong	70
10.2	The translation, and where it breaks	71
10.3	M1: Table 1 on real structure	72
10.4	What M1 changed in the report	74
11	M1b: the stop rule and the free screen tie	77
11.1	What the 7 September design asked for	77
11.2	Four policies, one argument, five predictions	78
11.3	The price of the guarantee: 0.910	79
11.4	The starred comparison separates, then ties	80
11.5	What survives, and the column your ranking table lacks	81

12 16 September: what the cells mean, T3, and a bias in our own table	83
12.1 Opening one cell	83
12.2 T3 in full: a cliff, not a curve	85
12.3 A selection bias in our own table	86
12.4 The Table 1 dossier	87
 VI Your draft and ours	 90
13 Your draft, read against the measurements	91
13.1 What the draft gets right	91
13.2 The theory: what the text says	92
13.3 The results section and the manuscript	95
14 One paper by Friday: the merge	98
14.1 Your open items are the runs of 11 to 13 September	98
14.2 What the measurements add, and where the drafts conflict	99
14.3 Three ways to merge, and the one I recommend	100
14.4 Before Friday, and before the 25th	101
 VII The repository	 105
15 The repository, file by file	106
15.1 How the branch is built	106
15.2 Where each chapter lives	107
15.3 Caveats I know about	108
15.4 What is not here	109
 Answers	 111

Before you start

Chris,

this document exists because I kept the work to myself for too long. Between the August pilots and this morning I ran a long series of measurements on our breakdown-radius line. A few pieces reached you: the evaluation plan on the branch `iclr2027-eval-plan`, and what we said to each other on 3 September. Most of it did not. The August package waited on a branch I never pushed, and everything from 7 to 16 September lives in folders on my machine. Your draft of 16 September lists, among the experiments it still owes, the ranking evaluation on the real instances and the elicitation arm. I ran the elicitation arm between 11 and 13 September, and a corrupted-knowledge calibration on the same real rows, though not your τ_b design itself (Chapter 14 is precise about the difference). That overlap is the clearest sign that the gap is mine to close, and this close to the author lock is late to close it.

So here is all of it, in two forms. The code, the results and the pre-registrations are on one branch of `bkrobust`, `ze/experiments`, built on top of your history so that it merges. This polycopié is the explanation: what I ran, why, what came out, what I had to retract, and what it changes in the paper.

The whole document in four sentences

Knowledge buys identifiability, then robustness, never efficiency, and one wrong claim usually costs the whole of what it bought. On the true CPDAG, counted in claims, the certificate never over-certified on 3,014 certifiable rows of eleven knowledge conditions; counted in covering hops, it did on 39. On the one panel where the class itself was estimated, no retraction reached the truth on any certifiable row, so the guarantee is conditional on a correct equivalence class. The paper that survives is a sensitivity analysis for background knowledge in the sense of Rosenbaum's Γ , whose honest baseline is a leave- k -out screen that it ties.

How it is organised

Part I is short. It states the object once in your geometry and gives the translation table to the words I use everywhere else: claim, retraction, hop, r_{claim} , r_{hop} , r_0 , the knowledge interval. If you read only one chapter before our call, read that one and Chapter 14.

Parts II to V are the measurements, in the order they happened. August (the pilot, the `se` criterion, locality, the firewall census, the wrong-skeleton experiment). Early September (what the radius should measure, and three defects in the old corpus). The protocol executed on 11–13 September (the observable frame, the knowledge suppliers, Table 4, the learned CPDAG). And the reframing as a Rosenbaum analogue, with the two measurements that decide it (M1, M1b) and the tables of 16 September.

Part VI is your draft. Chapter 13 reads it against the measurements: what is right, what the text says that the mathematics does not, and a handful of mechanical defects visible in the PDF. Chapter 14 is the merge: the four conflicts, three options, the one I recommend, and what has to happen by Friday.

Part VII maps the repository, file by file, with how to reproduce each table.

How to read the boxes

Green boxes (*What was measured*) hold numbers computed on my machine, and each names the repository file or script that produced it. Orange boxes point to a paper worth reading. Red boxes are errors actually made, ours or the draft's, with what caught them. Purple boxes are sentences in the form I think the paper can print. Aqua boxes are commands and decisions.

The pink boxes are questions to answer without looking; the answers are in the appendix. They are there because this is a lot of material to absorb before Friday, and answering a question fixes a number better than rereading it. Skip them if you prefer; nothing in the argument depends on them.

About the grey boxes

The grey boxes labelled *Digression* are **thematically independent** of the main argument: stories of what went wrong, historical context, dead ends. You can skip all of them and the thread stays whole. They stay because the concrete error teaches, and marking them this way is what cancels the cost of having them in the text.

What I am confident about, and what I am not

Every number in the green boxes was produced by a script and re-read from its output file while writing; the few whose script is not on the branch say so in the box. Several numbers I had quoted earlier turned out to be wrong, and the chapters say which, when and how they were caught; I would rather you see the corrections than meet them in review. The theory of your draft was checked by a re-implementation of the space, the order, the covering relation and the distance from your definitions, which reuses your Meek-closure and extension code, and no counterexample was found. The numbers printed in your draft were not audited.

One file in the repository is older than its data: `results/stage0/STAGE0.md` quotes the calibration run from before a seeding fix, while the calibration files next to it are the rerun (Chapter 6 explains). Chapter 15 lists this and every other caveat about the files.

Zé

16 September 2026

Part I

The shared object

Chapter 1

One object, two vocabularies

Your draft of 16 September and everything in this document measure the same object: an estimated CPDAG $\hat{C}$, a set K of orientations the analyst asserts, the analyst's graph $G_0 = \text{Meek}(\hat{C}, K)$, and the adjustment set $Z = O^*(G_0)$ read off it and then held fixed. We named the pieces differently, and in two places one word carries two meanings. This chapter states the object once in your geometry, gives the translation to the words the other chapters use, and draws your own running example counted both ways. There is no new mathematics here; it is the dictionary the rest of the document assumes.

One object, two ways of counting the distance to failure

Your radius counts covering steps in the space of knowledge states. Most of this document counts the analyst's own claims, withdrawn and re-closed under Meek's rules. The object is the same, and the two counts part exactly where Meek turns a claim into orientations the analyst never asserted.

Before reading

- In your running example, how many orientations does G_0 add to the CPDAG, and how many of them did the analyst assert?
- Your §4.2 says that retracting a claim un-orientates a knowledge-induced edge, and your §6.3 says that r_{claim} retracts asserted claims. Are these the same move?

1.1 Your geometry, on your example

The clearest way to see where the vocabularies meet is your statin example (§4.1 of the draft), which can be drawn exactly from the text: six compelled edges, five undirected ones forming a single component on {Smoke, BMI, Chol, CRP, Statin}, and four asserted claims.

Example 1.1 (Your running example, counted in hops and in claims). The analyst asserts Smoke→BMI, BMI→Chol, Chol→CRP and Statin→CRP. The fifth undirected edge is oriented by the closure, not by the analyst: BMI→Chol–Statin with BMI and Statin non-adjacent forces Chol→Statin by R1. So G_0 adds five orientations to $\hat{C}$, four asserted and one propagated, and $O^*(G_0) = \{\text{Age, Smoke}\}$. Your draft notes that reversing Statin→CRP is, in this instance, a return to the truth, so that claim is the false one.

Your Definition 3 gives $r_{\text{val}} = 3$: the first failing state un-orientates BMI–Chol, BMI–Smoke and Chol–Statin (§4.4). Now count the same state in claims. Withdraw Smoke→BMI and BMI→Chol and re-close. The two claims left, Chol→CRP and Statin→CRP, force nothing: no arrow points into the chain Smoke–BMI–Chol–Statin, and the two arrows into CRP come from adjacent nodes, so no rule fires. The chain stays undirected, and that is your failing state, reached by withdrawing *two* claims. No single withdrawal breaks Z : re-closure restores BMI→Chol or Chol→CRP if either is withdrawn alone, and withdrawing Smoke→BMI or Statin→CRP alone leaves Z valid. The claim radius is 2, the hop radius 3, and the extra hop is Chol→Statin, which nobody asserted.

The two graphs are side by side in Figure 1.1. The same arithmetic runs through your clinician

sentence: of its three statements, «cholesterol drives prescribing» is the propagated arrow, and the analyst never said it.

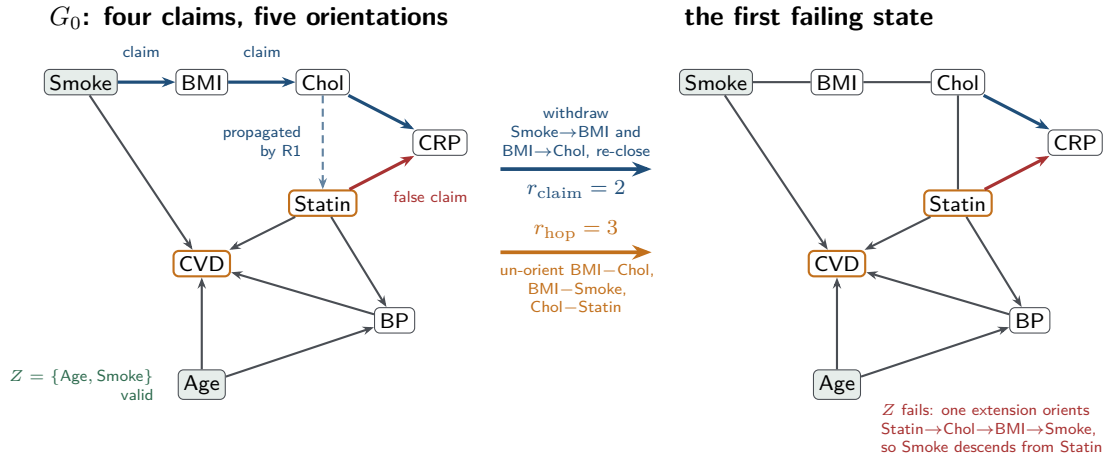


Figure 1.1: The same failing state is two claims and three hops away, because the third hop removes an orientation the closure made and the analyst never asserted. A practitioner told «two safe revisions» can in fact lose only one of the things they said.

The example fixes what the definitions have to say. Your geometry is the first two items below, stated in your notation; the last two are the claim-unit objects the later chapters use.

Definition 1.1 (The object, in your geometry and in claims). Let $\hat{C}$ be a CPDAG, K a set of orientations consistent with it, $G_0 = \text{Meek}(\hat{C}, K)$, $Z = O^*(G_0)$, and for a knowledge state G write $K_G = \text{dir}(G) \setminus \text{dir}(\hat{C})$.

- Knowledge states:** $\mathfrak{G}_{\hat{C}} = \{\text{Meek}(\hat{C}, K') : K' \text{ consistent}\}$, with membership decided by the fixpoint test $\text{Meek}(\hat{C}, K_G) = G$.
- Hops:** order by model inclusion, $G \preceq H$ iff $[G] \subseteq [H]$; d is the shortest path in the graph of covering pairs; the hop radius is $r_{\text{hop}} = \min\{d(G_0, G) : Z \text{ invalid in } G\}$, your r_{val} , and certifies $r_{\text{hop}} - 1$ safe revisions.
- Claims:** the retraction ball is $B_r(K) = \{\text{Meek}(\hat{C}, K \setminus S) : S \subseteq K, |S| \leq r\}$, and the claim radius is $r_{\text{claim}} = \min\{|S| : Z \text{ invalid in } \text{Meek}(\hat{C}, K \setminus S)\}$. A minimising S is the *witness*; the certificate licenses $r_{\text{claim}} - 1$ withdrawn claims.
- Conclusions:** the knowledge interval I_r is the hull of the effects identified over the graphs of $B_r(K)$, estimated with a band from $\hat{\Sigma}$ and n ; the null radius r_0 is the smallest r at which the band contains zero.

One collision in the symbols: in your draft r_{val} is the hop count of item (b). Elsewhere in this document, a bare r_{val} is the validity radius counted in claims, which the ledger computes as r_{claim} . Wherever the unit matters I write r_{hop} or r_{claim} .

Without looking (answer 1, in the appendix)

In your running example, which orientation of G_0 did the analyst not assert, and which rule forces it?

Without looking (answer 2, in the appendix)

State the claim radius and the hop radius of your running example, the claim witness, and the reason the two differ by one.

Your §4.2 also says a flip costs two moves, a retraction and then an assertion. That is true of the distance between two states. It is not the right reading of what a budget covers: by Meek's

theorem, the DAGs represented by the graphs of $B_r(K)$ are exactly the DAGs of the class that contradict at most r of the analyst's claims (lemma L1, Chapter 4). A single withdrawn claim already covers every world in which that claim is reversed.

Before reading

- Which word in your draft maps to our r_{hop} , and which to our r_{claim} ?
- Which adjustment criterion does the repository's fast MPDAG oracle decide?

1.2 The translation table

Table 1.1 is the dictionary. The left column quotes your draft, the middle one gives the word used from Chapter 2 on, and the right one says what to watch for. Instrument labels B2 to B7 are those of `results/ledger/TABLE4.md`.

Table 1.1: Your draft's vocabulary and this document's. Most rows are renamings; rows 4, 6, 12 and 13 are not.

	in your draft	in this document	watch for
1	$\widehat{C}$, K , G_0 , the optimal set	same symbols; $Z = O^*(G_0)$ is the committed set	
2	knowledge state, Def. 1	knowledge state	same object, same fixpoint test
3	K (§4.1); «asserted claims» (§6.3)	claims; one element is a claim	
4	«retracting a claim» (§4.2): unorient an edge of K_G	one hop	K_G holds Meek propagations too, so a hop can remove an orientation nobody asserted
5	elementary revision, covering pair, d (Def. 2)	hop, covering hop, hop distance	
6	r_{val} (Def. 3); $r_{\text{val}} - 1$ safe revisions	r_{hop} , the hop radius (B6)	a bare r_{val} here is counted in claims
7	r_{claim} (§6.3), no definition environment	r_{claim} , the claim certificate (B7)	item (c) of the definition above
8	φ_1	φ_1	same quantity
9	separation of X from the nearest member of the optimal set (§6.1)	the free rule (B2): predict r by the separation, 1 where it is undefined	your finding, turned into an instrument
10	$ K $; SHD to the data-generating DAG	the claim count (B5); SHD is an audit quantity	SHD reads the truth, as your §5.4 says
11	survival AUC and τ_b	not used; the endpoints are the four buckets, coverage and L_{95}	two designs, two panels, never pooled
12	the generalized adjustment criterion (§2)	GAC; back-door is Pearl's	the fast MPDAG oracle decides the back-door version; the two agree on the optimal set of the graph being judged
13	(no counterpart)	knowledge interval I_r , null radius r_0	the conclusion-level objects of Chapters 10 to 12
14	«the data-generating DAG is never on the computation path» (§6)	the observables-only invariant	next section
15	«check whether the certificate held» (§6)	the audit and its four buckets	next section
16	statuses, censoring (§5.4)	statuses; the inspection trigger; the stop rule	the trigger halts a sweep, the stop rule reports or widens

On row 12: `src/bkrobust/mpdag_criterion/criterion.py` decides the back-door version, `src/bkrobust/gac/mpdag_level.py` the GAC, and `THEOREMS.md` §14 proves they coincide on the optimal set of the graph in which validity is judged. The committed set $O^*(G_0)$ judged in another graph, the truth or a retracted state, is an arbitrary set there, and the two can part: the ledger audit found 4 such rows (Chapters 8 and 13).

Without looking (answer 3, in the appendix)

Your draft's r_{val} and this document's bare r_{val} : which unit does each count, and which symbol does this document use to avoid the clash?

Before reading

- Name the two places where the generating DAG is allowed to enter.
- A certificate licenses one safe move and the analyst made one false claim. Is the row covered?

1.3 The invariant, and the only place the truth enters

Your §6 and docs/EVALUATION_PLAN.md state the same invariant in almost the same words.

The observables-only invariant

Every reported quantity (r_{claim} , r_{hop} , φ_1 , the free rule, I_r , r_0) is a function of $(\hat{C}, K, X, Y, \hat{\Sigma}, n)$ alone. The generating DAG enters in two places only: to draw the samples behind $\hat{\Sigma}$, and, after every row is frozen, to audit.

The ledger enforces it row by row. Every line of `results/ledger/rows.csv` carries the three provenance columns `true_dag_on_path`, `chat_source_sha256` and `generating_dag_sha256`. On the oracle panel the CPDAG is the true DAG's, so rows are stamped `via_Chat_only`; on the estimated-class panel of Chapter 9 the two hashes differ.

The audit then asks one question per instrument and per row. Let w be the number of the analyst's claims that are false at the truth. A radius r *covers* the row when $w \leq r - 1$, that is, when the false claims fit inside the moves it licenses; an unreached radius covers everything.

The four buckets, `bucket()` in `experiments/ledger_sweep.py`

	Z invalid at the truth	Z valid at the truth
covered, $w \leq r - 1$	dangerous	held
not covered	safe	slack

Dangerous is the only failure: the certificate licensed as many withdrawn claims as the analyst got wrong, and the set was invalid. Slack is the price of caution.

Chapter 8 shows why the claim certificate cannot land in the dangerous bucket when the class and the validity oracle are correct, and why the hop radius can.

Without looking (answer 4, in the appendix)

Define the four audit buckets, including the covering condition, and say which one is the failure.

Without looking (answer 5, in the appendix)

State the observables-only invariant and name the ledger columns that record it.

Do not move on

On your running example, withdraw BMI→Chol alone and re-close. Name the rule that restores it and the arrow that triggers the rule. If you cannot, redo the closure of Smoke→BMI–Chol before reading Chapter 8.

Exercises

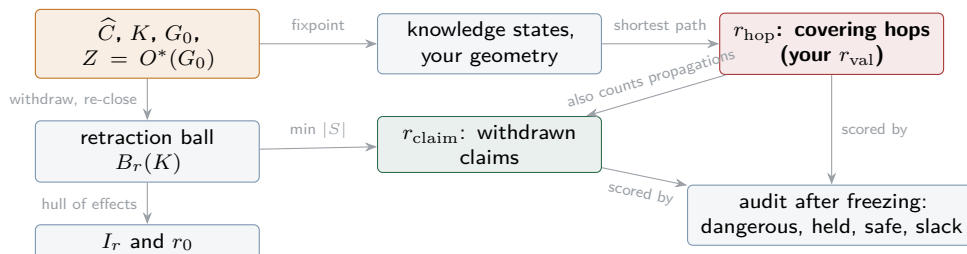
Exercise 1.1. Starting from G_0 of your running example, withdraw only the false claim $\text{Statin} \rightarrow \text{CRP}$ and re-close. Draw the resulting knowledge state, mark every edge that stays oriented with the reason it does, and give its hop distance from G_0 . Is $Z = \{\text{Age}, \text{Smoke}\}$ valid there? Finally, place on your drawing the state your §4.2 calls «two moves out», the one with $\text{Statin} \rightarrow \text{CRP}$ reversed.

Solution. The remaining claims are $\text{Smoke} \rightarrow \text{BMI}$, $\text{BMI} \rightarrow \text{Chol}$ and $\text{Chol} \rightarrow \text{CRP}$. R1 still orients $\text{Chol} \rightarrow \text{Statin}$ ($\text{BMI} \rightarrow \text{Chol} \rightarrow \text{Statin}$, BMI and Statin non-adjacent). $\text{CRP} \rightarrow \text{Statin}$ stays undirected: the only arrow into CRP comes from Chol, which is adjacent to Statin, so R1 does not fire, and $\text{Chol} \rightarrow \text{Statin}$ together with $\text{Chol} \rightarrow \text{CRP}$ is not a directed path between CRP and Statin, so R2 does not fire either. The state has four knowledge-induced orientations against five in G_0 , so it is one hop away. Z stays valid: in the extension with $\text{CRP} \rightarrow \text{Statin}$, every back-door path from Statin to CVD passes through Smoke ($\text{Statin} \leftarrow \text{CRP} \leftarrow \text{Chol} \leftarrow \text{BMI} \leftarrow \text{Smoke} \rightarrow \text{CVD}$, and the same through $\text{Chol} \rightarrow \text{Statin}$), and Smoke is in Z . The state with $\text{CRP} \rightarrow \text{Statin}$ asserted is one of the two extensions of this graph's undirected edge, so the reversal you place two moves out lies inside a single withdrawn claim, which is lemma L1 on your example.

Exercise 1.2. Classify each quantity as (H) hop unit, (C) claim unit, (R) conclusion level, (O) observable and unit-free, or (A) audit only, reading the truth. (a) « $r_{\text{val}} = 3$ » in your §4.4. (b) «the practitioner-facing count of safe revisions is $r_{\text{val}} - 1$ ». (c) «a single claim propagates to orient a 15-node chain, resulting in a structural radius of 14», your §6.3. (d) φ_1 . (e) SHD between G_0 and the data-generating DAG. (f) the smallest r at which the band of I_r contains zero. (g) the separation of Statin from the nearest member of the optimal set. (h) «the committed set is invalid at the truth on 28.8 % of certifiable rows» (a result of Chapter 8).

Solution. (a) H: your Definition 3 counts covering hops. (b) H: it certifies hops, and on your example the analyst can lose only one claim, not two. (c) H for the 14; the same row has $r_{\text{claim}} = 1$, which is why your §6.3 calls the safe budget zero. (d) C: the fraction of single withdrawn claims that break Z . (e) A: it needs the data-generating DAG, which is why your §5.4 calls it an oracle baseline. (f) R: that is r_0 , counted in claims. (g) O: it is read off G_0 with no radius in it, and the free rule (B2) turns it into a predicted radius. (h) A: validity at the truth is only known after the rows are frozen.

Chapter map



Part II

August: what the pilots established

Chapter 2

August I: the pilot and the se criterion

This chapter covers the pilot of 17 July and the E1' run of 19 August. Three things from them still carry weight: a reversed claim is a retraction plus an assertion, so every August budget is worth twice its number in your moves; the radius is censored because most perturbation balls are inert, which no sample size changes; and one wrong claim that passes Meek's check costs a nominal 95 % interval about 23 points of coverage in this ensemble. The chapter ends with the corrections the August package made to itself, and four more found since.

Before reading

- Against what yardstick did the pilot decide that a reversed claim had overturned the conclusion?
- In the atomic moves of your Section 4, how much did the largest pilot perturbation cost?

2.1 The July pilot: a radius priced against the answer key

What the pilot asked

Knowledge that orients an edge the data leave open cannot be tested against the data, so the pilot tried to price it instead: reverse some of the analyst's claims, close again with Meek, and see whether the conclusion survives. The number it found was finite and mostly informative. It was also computed against the true effect.

The design is small and exact: 600 linear SCMs on Erdős–Rényi graphs (p from 5 to 8, expected degree 1.5 to 2.5), in the iSCM parameterisation of Ormaniec et al. with the standardisation frozen, so Σ , τ and every adjusted estimate are closed-form. K holds up to four true orientations of undirected CPDAG edges; the perturbation reverses $\rho \in \{1, 2, 3\}$ of them, checks Meek consistency, closes, and reads O^* off the MPDAG. A density sweep (11 cells, $p \leq 10$, degree up to 6; 1,650 draws, 1,593 kept) and 60 nonlinear SCMs complete the run.

The machinery, `rho_breakdown/pilot/code/test_machinery.py`

Meek consistency and the MPDAG match brute-force enumeration of the class on 7,688 knowledge sets (test T2, re-run while writing: 0 mismatches). Var-sortability is 0.937 under a naive SEM and 0.508 under iSCM on the same graphs (`pilot/results/summary.json`).

Figure 2.1 follows a single reversed claim to the end. Two thirds pass Meek's check, most of those never reach the adjustment set, and every change of O^* that did happen made it invalid. The breakdown radius of the pilot was defined per criterion.

Definition 2.1 (The pilot's ρ^*). For a query (X, Y) , true claims K and a criterion c , ρ_c^* is the smallest $\rho \leq 3$ such that some Meek-consistent set obtained by reversing ρ claims of K overturns the conclusion under c , and 4 (censored) if none does. With $c = \text{sign}$ the member's estimate has the opposite sign to τ ; with $c = \text{rel50}$, $|\hat{\tau}_m - \tau| > |\tau|/2$; with $c = \text{ident}$, the member is not amenable.

Two of the three criteria read τ (`pilot/code/analyse.py`, function `breakdown_radius`). No

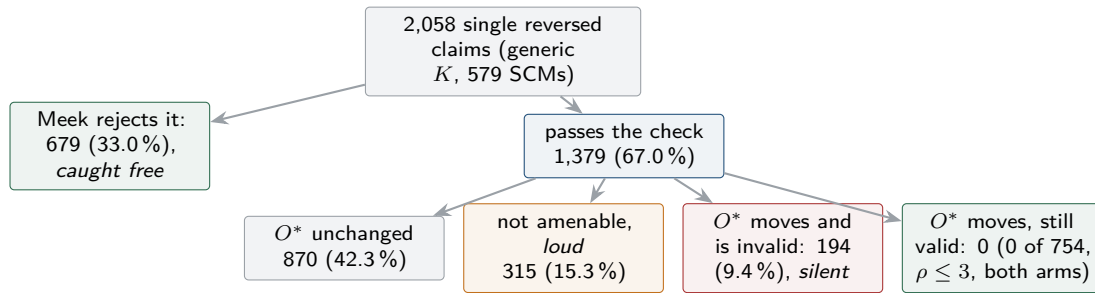


Figure 2.1: Percentages are of all 2,058 reversals. The check catches a third of the errors for free; of the rest, most are inert and nearly a quarter become visible refusals. The empty green box is an empirical zero for claim reversals only: Chapter 3 shows the efficiency-only band is not empty under skeleton errors. Counts from `pilot/results/summary.json`; the 754 is recounted from `linear_raw.json`.

analyst has τ , so the pilot’s radius described the benchmark, not anything a user could compute. On the generic arm it was censored for 90.3 % of SCMs under `sign`, 75.5 % under `rel50` and 30.6 % under `ident`. The pilot also settled what form a theorem could take: when O^* moves, the damage is large, so the only honest statement is a radius, never a continuity bound.

A number I sent you that no file contains

REPORT.md of 31 August says the damage conditional on O^* moving has median 0.59 and goes “up to $128\times$ on scale-free graphs”. The median is right; the pilot ran no scale-free graphs, and 128 is in no result file. Recounted from `pilot/results/linear_raw.json` while writing: over the 274 single reversals that moved O^* (both arms), $|\text{bias}|/|\tau|$ has median 0.591, mean 1.167 and maximum 13.1. The conclusion survives the correction, since the tail is heavy at every scale we have measured (Chapter 3 finds maxima above 1,000 under skeleton deletions), but a maximum quoted in the paper has to be 13.1 with its ensemble.

Your Section 4 counts atomic moves, orient and un-orient, so a flip costs two. The pilot’s operator is a flip (`pilot/code/run_linear.py`, line 133, rewrites a claim $u \rightarrow v$ as $v \rightarrow u$). REPORT.md summarised this as “the pilot ran at radius 8, not 4”; the code says something slightly different. `MAX_K = 4` at line 34 caps the size of K , and the pilot’s ball stopped at three reversals, which is six of your moves. E1’ below enumerated the full ball, four reversals or eight moves, and $\rho^* = 1$ is two moves. Chapter 8 adds a third unit, the covering hop, which differs from both when Meek closure propagates.

Without looking (answer 6, in the appendix)

Why could no analyst compute the pilot’s ρ^* , and which of its three criteria escaped the problem?

Before reading

- If half the problems never break inside the full ball, is that a budget artefact or a property of the problems?
- For what share of problems is the displacement at one reversal exactly zero, and does it depend on n ?

2.2 E1’: the analyst’s own standard error replaces τ

Between the two runs sat E1 (14 August), which tried to repair a radius censored nine times in ten by raising $|K|$ to 8 and switching to `rel50`. Half of its gain came from conditioning on graphs

with at least eight undirected edges (1.5 % of the pilot’s ensemble), it inverted at $p \geq 15$, and both criteria still read τ (`rho_breakdown/rerank_2026-08-14/RESULTS.md`). E1’ went after τ directly, kept $|K| \leq 4$, and enumerated the whole ball so that censoring would be measured rather than imposed by a budget.

The se criterion

Measure how far the estimate can move against the error bar the analyst already publishes. If no set of reversed claims moves it by more than $z \cdot \text{SE}$, the knowledge is not the binding uncertainty.

Written out, with the two companion statistics the run reported beside it:

Definition 2.2 (ρ_{se}^* and its relatives). Let $G_0 = \text{Meek}(C, K)$, $\hat{\tau}_0$ the OLS coefficient of X in $Y \sim X + O^*(G_0)$ and SE_n its standard error at sample size n . Let $d(\rho) = \max_m |\hat{\tau}_m - \hat{\tau}_0|$ over Meek-consistent, amenable members within ρ reversals. Then

$$\rho_{\text{se}}^*(n) = \min\{\rho : d(\rho) > z \text{SE}_n\}, \quad R(\rho, n) = \frac{d(\rho)}{z \text{SE}_n},$$

$\rho_{\text{any}}^* = \min(\rho_{\text{se}}^*, \rho_{\text{ident}}^*)$, where $z = 1.96$, $\rho_{\text{se}}^* = |K| + 1$ when censored, and ρ_{ident}^* the smallest ρ at which a consistent member is not amenable.

Every input is in the analyst’s hands; τ enters only the evaluation (`rho_breakdown/e1prime/code/se.py`). The CPDAG C is still the oracle’s, so the licensed phrase is “computable from $(C, K, \hat{\Sigma}, n)$ ”, never “robust under an estimated CPDAG” (Chapter 9).

Gates on the licensed ensemble, `e1prime/results/arm1_analysis_licensed_K4.json`

2,087 amenable SCMs, p from 5 to 10, degree up to 6. With O^* frozen at $O^*(G_0)$ the instrument censors 1.000 at every n and every z , so it is not a dead switch. Spearman correlation of ρ_{se}^* with the free t -statistic $|\hat{\tau}_0|/\text{SE}$ at $n = 2,000$: 0.199, against a pre-declared collapse bar of 0.90. Against the free ρ_{ident}^* : 0.571, and the `se` half strictly lowers the radius for 30.0 % of problems.

Figure 2.2 is the result that set the direction for the rest of the month. Censoring of ρ_{se}^* falls from 0.614 at $n = 50$ and 0.554 at $n = 200$ to 0.515 at $n = \infty$, and stops there. At $n = \infty$ the standard error is zero, so the floor is exactly the share of problems whose full-ball displacement is zero (gate G6).

The censored bin, decomposed, same file, key `sweep`

At $n = \infty$, of the 1,074 censored problems: none had every perturbation caught by Meek, 5.4 % had every perturbation lose amenability (a loud failure), and 94.6 % had consistent, amenable members whose estimate did not move at all. At one reversal the displacement is exactly zero for 64.6 % of problems, at $n = 200$ and $n = 20,000$ alike. The composite ρ_{any}^* is censored for 0.166 of problems at $n = \infty$, with 69.1 % of the mass at $\rho^* = 1$.

Both statistics failed the pre-registered non-degeneracy profile, in opposite directions: ρ_{se}^* is censored for half the problems and ρ_{any}^* puts two thirds of them at one. The budget was the whole ball, so this is not budget censoring, and an exact zero never exceeds a positive threshold, so it is not a sample-size effect. An inert ball is a property of the problem: the reversed claims do not reach the part of the graph that O^* reads. Chapter 3 measures where those claims sit, and shows that much of the zero is a published class-invariance fact rather than a finding. One label in the

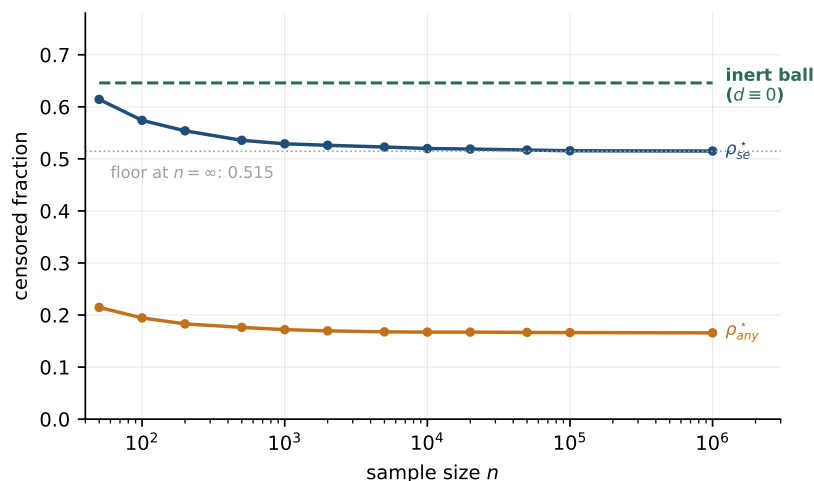


Figure 2.2: Five orders of magnitude of sample size move the censoring of ρ_{se}^* by ten points and then nothing. The dashed line is the share of problems whose one-reversal ball is inert (64.6 %, identical at every n); the dotted floor, 0.515, is the share whose full ball moves no estimate. The composite ρ_{any}^* is rarely censored, but 69 % of its mass sits at one. Source: `elprime/results/arm1_analysis_licensed_K4.json`.

package is wrong here: `REPORT.md` §3 and `NUMBERS.md` call the series 0.554 to 0.515 the censoring of ρ_{any}^* . It is ρ_{se}^* (`SECONDARY_rho_se` in the JSON); ρ_{any}^* goes from 0.183 to 0.166.

Without looking (answer 7, in the appendix)

On the licensed ensemble the censoring of ρ_{se}^* is 0.554 at $n = 200$ and 0.515 at $n = \infty$. What sets the floor, and what does a censored problem almost always mean?

Without looking (answer 8, in the appendix)

Why can no sample size remove the inert-ball mass?

Before reading

- How many points of nominal 95 % coverage does one reversed claim cost, among those Meek does not catch?
- Which width statistic did I pre-register for the robust interval, and why did it flatter the method?

2.3 ARM 2: what one wrong claim costs a confidence interval

ARM 1 kept the knowledge true and asked how far the estimate could move. ARM 2 made it wrong. For 2,500 licensed SCMs, 20 datasets each and $n \in \{200, 1,000, 5,000, 20,000\}$, the analyst receives $(C, K_{\text{asserted}}, \hat{\Sigma}, n)$, where K_{asserted} reverses $r \in \{0, 1, 2\}$ of the true claims. Reversals that Meek rejects are counted as caught free and leave the sample. The analyst reports a naive interval $\hat{\tau}_0 \pm z \text{SE}_n$ and a robust one, the hull of $\hat{\tau}_m \pm z \text{SE}_m$ over consistent, amenable members within ρ reversals. τ decides coverage and does nothing else, and every SCM was rebuilt from its seed with τ matching to 10^{-9} (0 failures in 2,500). This is the calibration experiment that Section 4.3 of your rewrite of 30 August proposed.

Figure 2.3 carries the two numbers August could hand to a reviewer who does not work with MPDAGs.

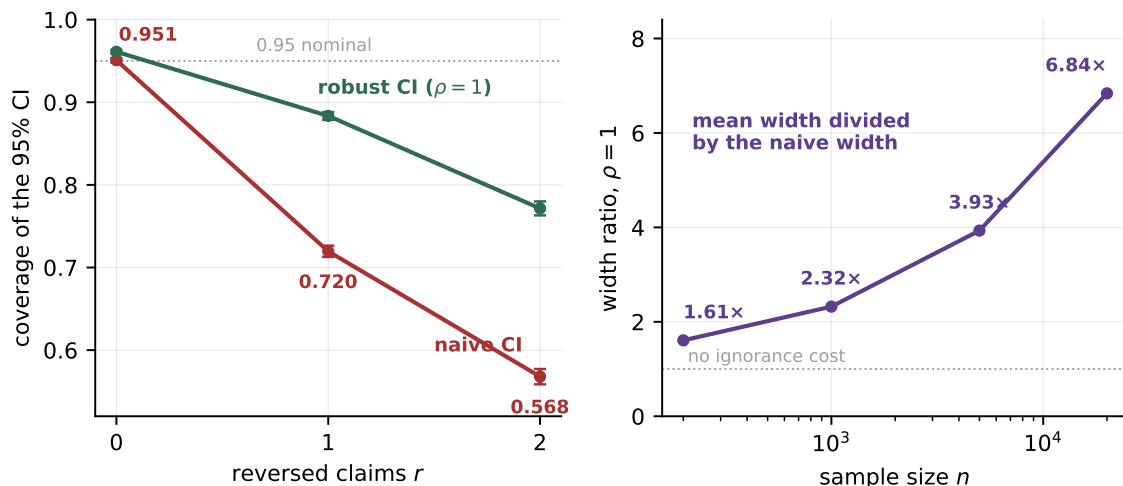


Figure 2.3: Left: with the knowledge right the naive interval is calibrated; one accepted reversal costs it 23 points and two cost 38, and ranging over single reversals recovers most of the first loss. Right: that recovery gets more expensive as data grow, because the standard error shrinks and the knowledge-induced spread does not. Source: `e1prime/results/arm2_analysis.json`.

ARM 2, rho_breakdown/e1prime/results/arm2_analysis.json

At $n = 20,000$ the naive interval covers 0.9509 [0.9481, 0.9533] with no reversal, 0.720 with one and 0.568 with two; Meek rejects 23.6 % of single and 38.6 % of double reversals, and those are excluded. At $r = 1$ the robust interval at $\rho = 1$ covers 0.8835, with width ratio to the naive one of median 1.00, mean 6.84 and 90th percentile 23.3; its ball is inert for 69.3 % of problems. At $\rho = 2$ it covers 0.9245 (mean ratio 11.6), the full ball 0.9306. When two claims are wrong and the ball has radius one, coverage is 0.809 at $n = 200$ and 0.772 from $n = 5,000$ on. The mean width ratio at $r = \rho = 1$ is 1.61, 2.32, 3.93 and 6.84 across the four sample sizes.

The ignorance does not shrink with data: SE_n goes like $n^{-1/2}$ while $d(\rho)$ does not depend on n . Even the full ball stops short of 0.95 (0.931 with one wrong claim, 0.912 with two); the E1' report attributes the residue to cells where the member that restores the truth is itself not amenable and carries no estimate. Chapter 10 prices a wrong claim again on real structure.

The width statistic I pre-registered flattered the method

The rule asked for coverage at least 0.90 at $r = \rho = 1$ and a *median* width ratio below $3\times$. Coverage was 0.8835 and the median 1.00: INCONCLUSIVE. The median is 1.00 by arithmetic, since the ball is inert for 69 % of problems; the mean, $6.84\times$, would have failed the bar. Both belong in the paper.

Without looking (answer 9, in the appendix)

Give the coverage of the naive 95 % interval at $n = 20,000$ with 0, 1 and 2 reversed claims, and the exclusion that must travel with those numbers.

Before reading

- Which August number was retired because the code that would have produced it never existed?
- Was the efficiency-only band, where O^* moves and stays valid, really empty?

2.4 Corrections made before 31 August, and four since

`rho_breakdown/NUMBERS.md` closes with the corrections the package applied to itself. Table 2.1 gives them with the file that settles each, followed by the ones found after it was written. The G1 row compresses two things in `NUMBERS.md` (“0.141 to 0.161”): the first cut divided by all single-reversal members and got 0.1085, and 0.141 is the pilot’s own value on its own grid.

what	first stated	as it stands	settled by
silent bias at distance ≥ 1	0 of 791	retired: no distance function in the pilot code	<code>locality/VERDICT.md</code>
gate G1, one reversal	0.1085	0.161 on consistent members	<code>e1prime/RESULTS.md</code>
robust width ratio	median 1.00	mean 6.84 $\times$ printed beside it	<code>arm2_analysis.json</code>
gate G2 comparator	0.336	0.271 (licensed grid); 0.284 measured, pass	<code>e1prime/RESULTS.md</code>
series 0.554 to 0.515	ρ_{any}^*	ρ_{se}^* ; ρ_{any}^* is 0.183 to 0.166	<code>e1prime/results/arm1_analysis_licensed_K4.json</code>
damage tail	up to 128 $\times$	maximum 13.1 $\times$, median 0.591 ($n = 274$)	<code>pilot/results/linear_raw.json</code>
efficiency-only band	empty, 0 of 754	empty for reversals only; 2.35–5.53 % under skeleton deletions	<code>x3_skeleton/RESULTS.md</code>
pilot budget	radius 8	6 moves in the pilot, 8 in E1'	<code>pilot/code/run_linear.py</code>

Table 2.1: Above the rule, the corrections `NUMBERS.md` lists; below it, those found later. Paths are relative to `rho_breakdown/`. Of the three things `NUMBERS.md` lists as never run, two have run since: an estimated CPDAG (Chapters 3 and 9) and skeleton perturbation; nonlinear mechanisms beyond the pilot’s 60 SCMs remain untested.

What August I licenses

“In linear Gaussian SEMs on Erdős–Rényi graphs with the true CPDAG, one reversed background-knowledge claim that passes Meek’s consistency check lowers the coverage of a nominal 95 % interval to 0.72 at $n = 20,000$. An interval that ranges over all single reversals restores 0.88, at a mean width 6.8 times the naive one that grows with n .” Not licensed: “robust under an estimated CPDAG”, or ρ^* as a well-behaved four-valued diagnostic.

The two exercises rework the gate arithmetic and the censored bin from the files.

Exercise 2.1. (a) The first cut of gate G1 divided silent-bias events by all single-reversal members and got 0.1085; Meek rejects 32.4 % of those members. Recover the rate on consistent members. (b) At $r = \rho = 1$ the mean width ratio is 3.93 at $n = 5,000$ and 6.84 at $n = 20,000$, and the inert share stays near 69 %. Using only the dependence on n , predict the ratio at $n = 2,000,000$. (c) A problem has $\rho_{\text{se}}^* = 2$ in E1'. How many atomic moves of your Section 4 is that?

Solution. (a) The consistent members are $1 - 0.324 = 0.676$ of all, so $0.1085/0.676 = 0.1605$, the 0.161 recorded in the JSON (`G1_silent_bias_rho1`); it sits inside the gate band 0.12–0.18. (b) The inert cells keep ratio 1 at every n , so it is the excess over one that scales. For the other cells the knowledge spread is fixed and the naive width falls like $n^{-1/2}$, so the excess grows like $\sqrt{n}$. Check: from 5,000 to 20,000 the excess goes from 2.93 to 5.84, a factor 1.99 against $\sqrt{4} = 2$. A further factor 100 in n multiplies the excess by 10: about 58.4, so a mean ratio near 59. This is an extrapolation of the functional form; the largest n measured is 20,000. (c) Two reversals, each a retraction and an assertion: four moves.

Exercise 2.2. The censored bin of ρ_{se}^* on the licensed ensemble (2,087 problems, $z = 1.96$), read from `arm1_analysis_licensed_K4.json`, in the table below.

n	censored problems	share loud	share robust	share with $d \equiv 0$
50	1,282	0.0452	0.9548	0.7925
200	1,156	0.0502	0.9498	0.8789
20,000	1,083	0.0536	0.9464	0.9381
∞	1,074	0.0540	0.9460	0.9460

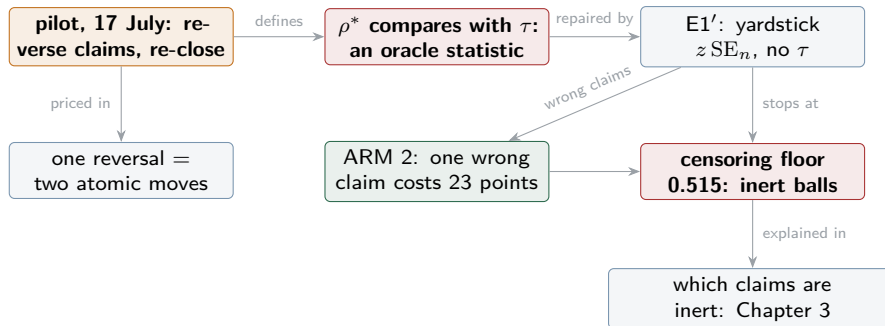
Convert the shares into counts of three kinds of problem: loud, exactly inert, and robust with a displacement that is not zero. Which of the three does the sample size act on, and on how many of the 2,087 problems at most?

Solution. Multiplying out: loud problems are 58 at every n ($1,282 \times 0.0452$, $1,156 \times 0.0502$, and so on). Exactly inert problems are 1,016 at every n . The remainder, robust with a nonzero displacement smaller than $z\text{SE}_n$, is 208 at $n = 50$, 82 at $n = 200$, 9 at 20,000 and 0 at ∞ . Sample size acts only on that last group: at most 208 of 2,087 problems, and 82 from $n = 200$ on. At $n = \infty$ the censored problems are the 1,016 inert ones and the 58 loud ones, 1,074 of 2,087 or 0.515: that is the floor. The shares in the last column rise only because the denominator shrinks.

Do not move on

Say why, at $n = \infty$, a problem is censored exactly when no member of its full ball moves the estimate, and why at $n = 200$ a censored problem can still have a moving estimate. If you cannot, the step to redo is the definition of ρ_{se}^* with $\text{SE}_\infty = 0$.

Chapter map



Chapter 3

August II: locality, the firewall, and a wrong skeleton

One intuition ran through August: a wrong claim far from the query rarely matters. This chapter follows it through five runs and says what is left. Some of it is a result, some is a lemma that Henckel, Perković and Maathuis had already published, and some is dead, including two sentences I sent you on 31 August. Table 3.1 is the ledger; the sections give the evidence. Throughout, *distance* means skeleton distance from a claim’s edge to $\{X, Y\}$, the minimum over both endpoints. The files call it *hop*; it is not the covering hop of Chapter 1.

August claim	status on 16 September
censoring of ρ_{any}^* separates by distance	result, per ensemble: 26.7×, 13.4×, 59.7×, 3.0×
“ O^* is a local functional; 0 of 791 at distance ≥ 1 ”	dead: 142 of 3,639 and 70 of 10,615; 791 has no provenance
one reversal off X and Y never moves O^*	true by enumeration at $p \leq 5$; 95.8 % of it is class invariance
knowledge beyond one hop need not be elicited	dead: identification changes at distance 2
C-FIREWALL	conjecture, complete census at $p \leq 5$
firewall with an estimated $\hat{C}$	result: degrades monotonically with discovery quality
false identification gains	result: not bounded by distance
locality survives a wrong skeleton (X3)	INCONCLUSIVE; the two-hop cut is post hoc
the efficiency-only band is empty	corrected: 2.35–5.53 % under skeleton deletions
only retractions can break a fixed valid Z	lemma, by refinement

Table 3.1: Where each August claim about locality stands. “Dead” means refuted by our own measurements, not merely unproven. Each row’s files are named in the section that discusses it.

Before reading

- Before any locality run, what had the 14 August experiments already said about O^* -relevance?
- Do Guo and Perković ever perturb the background knowledge they enumerate within?

3.1 Prehistory: the three re-rank experiments of 14 August

Three small runs on 14 August shaped what came after. None of them is in the paper, and one of them is a literature read rather than a measurement.

E2, R1-closed finiteness (`rerank_2026-08-14/e2-r1-finiteness/RESULTS.md`). Your draft’s tiered corruption model is close to its T-move arm. When the error is a variable moved between tiers, R1-only closure equals full Meek closure in every member (39,267 of 39,267), which Bang and Didelez entail. When the error is a reversed claim, it holds in 0.8416 of members, so flip errors break the premise of the lemma. Containment is real but only in the tail: the 90th percentile of orientations changed per changed claim is 1 for tier moves against 2 for flips. The same run found that the form of the knowledge buys nothing (tiered against generic under flips, $p = 0.49$); the error model does.

E3, O^* -relevance (`e3-ostar-relevance/RESULTS.md`). A metric weighting wrong claims by their role in $\text{pa}(\text{cn}) \setminus \text{forb}$ beat edit distance at predicting silent bias, AUC 0.842 against 0.637.

Against a plain count of wrong claims touching X or Y the gain fell to $+0.029$, under the pre-declared bar of 0.05 , and the specifically O^* tier was anti-predictive (AUC 0.409). The content was locality, not adjustment-set membership.

E4, a full read of Guo and Perković (arXiv:2010.08611) and Taeb, Guo and Henckel (arXiv:2511.10625v3) (e4-guo-perkovic-read/RESULTS.md). Guo and Perković take K as true throughout (“perturb”, “wrong” and “misspecif” occur zero times); Taeb et al.’s Section 6 perturbs the graph with the knowledge fixed. Two narrowings bind the paper: the separation of identification from estimation is Guo and Perković’s sentence, and their Corollary 8 already gives one adjustment set per member.

Before reading

- The $E1'$ locality table was said to separate by 13 to 27 times. What do its four rows actually give?
- Which two measurements killed the sentence “ O^* is a local functional of the graph”?

3.2 Locality, measured and then cut down to size

The intuition

$O^* = \text{pa}(\text{cn}(X, Y)) \setminus \text{forb}(X, Y)$ reads the causal paths from X to Y and one layer of parents around them. A claim that never reaches that region cannot move it, so an inert ball should mean claims far from the query.

Figure 3.1 is the $E1'$ measurement. Each licensed problem is placed by the smallest distance of any of its claims, and the censoring of ρ_{any}^* at $n = 20,000$ is low when some claim touches the query and near one when none does.

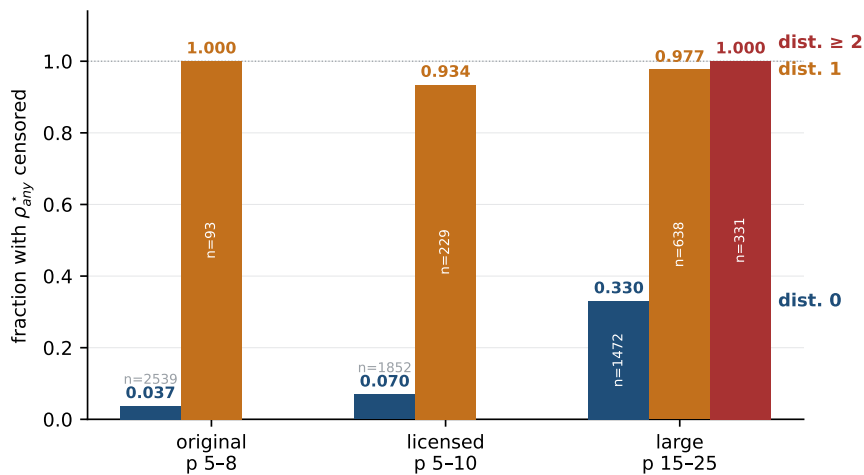


Figure 3.1: The distance of the nearest claim governs whether the radius has anything to say, but the size of the gap depends on the ensemble: $26.7\times$, $13.4\times$ and $3.0\times$ for the three shown. Licensed at distance 1 is 0.9345 in the JSON (NUMBERS.md rounds to 0.935). Of the 331 large SCMs in the red bar, 57 have every claim in a component disconnected from the query. Source: e1prime/results/arm1_analysis_*_K4.json.

Four defects in the table I sent you

The separation ratios are 26.7 , 13.4 , 59.7 (k8, $n = 22$ at distance 1) and 3.0 (large), so “13 to 27 times” holds for two rows of four. The weakest cell, k8/K4 at 0.778 ($n = 27$), was left

out. “Invariant in n to four decimals” is true at distance 1 (0.9345 at $n = 2,000, 20,000$ and ∞); at distance 0 the licensed value moves from 0.0724 to 0.0697 to 0.0680. And ρ_{any}^* folds in loud failures: of the 38 uncensored SCMs whose claims all sit at distance ≥ 1 , 37 lose amenability, 1 has a genuine change at two reversals, and none moves the estimate at one (locality/x2/RESULTS.md §3).

Two runs of 19 August then attacked the sentence “ O^* is a local functional, and no misstatement at distance at least 1 moved it (0 of 791)”. The number went first: the pilot code contains no distance function at all, so 791 could not have been computed (locality/VERDICT.md §4). X1 (locality/x1/) built the operator for spurious required edges on non-adjacent pairs, replacing one true claim with a spurious one under the semantics of b-LOAD, our CIKM paper’s noise operator. On the **original** ensemble such a replacement at distance ≥ 1 moved O^* into invalidity 142 times in 3,639; pooled over three ensembles, 13 in 2,919 at distance ≥ 2 and 0 in 931 at distance ≥ 3 . X2 enumerated instead of sampling. At $p \leq 5$, over every CPDAG, every reachable MPDAG, every query and both orientations of every undirected edge, a single reversal off X and Y never changed O^* : 0 of 136,200 far trials, against 184,332 of 307,584 near ones. Joint reversals did: 70 of 10,615 knowledge sets from the E1’ archive whose every misstatement lies at distance ≥ 1 changed O^* , all 70 into invalidity, each with a misstatement at distance exactly 1, the smallest at $p = 5$. The same enumeration records 360 identification changes at distance 2, which ended “knowledge beyond one hop need not be elicited”.

What the far zero is made of, locality/x2/code/run_lemma.py

At $p = 5$ the far zero rests on 137,040 trials in which both orientations of the claim leave the query amenable (lemma_p5.json, disconnected components included). In 131,280 of them, 95.8 %, the base MPDAG is already amenable. The informative trials, where the claim decides something, number 5,760, all at distance 1; at distance 2 there are none. The split by base is from my notes of 23 August, which recount the strata with the same code.

When the base is amenable, O^* is constant over the whole class, so no refinement of it can move O^* at any distance, near ones included. I traced this class invariance to Henckel, Perković and Maathuis (JRSS-B 84(2), 2022, Lemma E.7 in their appendix); I have not re-read that lemma in the published text, so check the number before citing it. The consequence is plain: most of the far zero is a published fact, the pilot’s “wrong knowledge was harmless in 75 of 75 CPDAG-amenable SCMs” is the same fact, and what distance adds is an incidence condition (the claim does not touch X), not a decay law.

In your draft

The paragraph “Over-claiming does not pay” in your Section 6 (“once a valid set is identified, adding more truthful claims does not improve statistical efficiency”) is this class invariance. Cite it rather than argue it.

Without looking (answer 10, in the appendix)

Which two measurements killed “ O^* is a local functional of the graph”, and what survives in its place?

Without looking (answer 11, in the appendix)

Why is “distance ≥ 1 ” in the X2 zero better read as an incidence condition, and what is the informative denominator at $p = 5$?

Before reading

- For a query already identified by adjustment, can a spurious edge that passes the coherence check make the reported O^* invalid?
- How does the protection change when the CPDAG is estimated by PC?

3.3 The firewall: what the consistency check buys**Checked is not true, but it is not nothing**

Meek's check never decides truth. For a query that adjustment already identifies, though, a false adjacency claim that passes the check appears never to invalidate the set the analyst reports: the check acts as a firewall for validity, and only there. The surprise is that the graph after the claim has a skeleton no member of the true class has, yet the set read from it is valid in the truth.

Stated for a single spurious claim, it reads as follows.

C-FIREWALL, a conjecture

Let G_0 be an MPDAG amenable for (X, Y) , $\{a, b\}$ non-adjacent in its skeleton, and H the Meek closure of G_0 plus $a \rightarrow b$. If H has no directed cycle, then either H is not amenable (a visible refusal) or $O^*(X, Y, H)$ is a valid adjustment set in every DAG of $[G_0]$.

The evidence for it is a census, not a proof.

The census, `locality/x2/cfirewall/results/SUMMARY.json`

0 violations in 70,375,325 trials and 103,105,398 (trial, DAG) checks. The census is complete at $p = 3, 4, 5$ for one claim (and for two claims at $p = 4, 5$, three at $p = 4$); $p = 6, 7, 8$ are samples, with a rule-of-three bound below 4.3×10^{-8} . The detector has power: when the check fails and the edge is stamped anyway, 3,005,488 of 13,791,476 reported sets (21.8 %) are invalid. Pointwise over 791,484 evaluations at $p \leq 5$, the conflict test never fires and “has a directed cycle” coincides with “not extendable”, so the whole hypothesis is acyclicity (`checks_agree.txt`).

It is a conjecture, not a theorem. The first proof handle offered (adding the edge only grows cn) is false; what holds in every enumerated trial is $O^*(H) \cap \text{forb}(G_0) = \emptyset$, which settles the first clause of the adjustment criterion for the whole class, given a further hypothesis that held in every census trial; the second clause is closed by a sufficient condition in 98.0 % of checks and open in the rest. X1 looked like a contradiction (1.1 % silent bias, 64 of 5,815, among coherent spurious claims). My reading is that the operators differ: X1 replaces a true claim, so each of its perturbations is a retraction plus an assertion, closed claim by claim, while the census adds one spurious edge on top of the full closed G_0 .

With PC instead of the oracle, `locality/x2/chat/RESULTS.md`

PC written from scratch; with oracle tests it returns the true CPDAG in 150 of 150 gate draws. 400 SCMs per cell, p from 6 to 10, $\alpha = 0.01$. Among reports valid before the claim whose O^* moved, a checked spurious claim leaves them invalid in 10.9 % at $n = 200$, falling to 2.4 % at $n = 50,000$ and to 0 of 22,282 with oracle tests; stamped past a failed check, 17.4 to 34.8 %. Discovery error alone leaves 35.9 % of base reports invalid at $n = 200$, 17.1 % at

$n = 50,000$.

My first version of that table compared rates on different conditioning sets and reversed the sign of the conclusion; the numbers above condition on a valid base report. Two further arms were written and run on 23 August itself, in the same working session, so they corroborate nothing independently. The false-gain search (`falsegain/results/search_6_7_8_23.json`) adjudicated every report transition against the truth on random DAGs with p from 6 to 8: no report that was already identified became unsound under a checked claim, but when a checked claim turned a refusal into an estimate, the estimate was wrong in 13,988 of 17,936 cases for a false orientation (78.0%) and in 24,895 of 49,467 for a spurious edge (50.3%). Exercise 3.1 builds the family that takes this to any distance. The refutation arm (`refutation/conjectureA/exh6_result.json`) enumerated all 1,067,825 CPDAGs on six nodes, 149,123,280 orientation additions on amenable bases, and found O^* unchanged in every one; that is class invariance again, and it killed nothing.

Without looking (answer 12, in the appendix)

State C-FIREWALL and the two numbers that make its zero meaningful.

Before reading

- If the discovery method misses one edge, does the damage fall off with the distance of that edge from the query?
- In the worked example, how many claims have to be retracted before the reported set stops being valid, and when does the estimate first move?

3.4 X3: a wrong skeleton, and one problem end to end

For the 3 September meeting you asked what happens when the discovery output itself is wrong. X3 answered for one kind of error: $\hat{C}$ is the CPDAG of the true DAG with one edge deleted, which models a missed weak edge rather than what a CI test actually returns. The run covers 9,721 SCMs and all 130,120 single-edge deletions, with population Σ . The claims of K are kept where their edge is still undirected. A smoke test found the true CPDAG not amenable for the query in 18 of 19 draws, so before any outcome was computed the arm with knowledge became the primary one.

Figure 3.2 carries the verdict. The rule asked for a ratio of at least 5 between the silent-bias rate at distance 0 and at distance ≥ 1 , and a rate below 0.02 at distance ≥ 1 . The ratio is 7.10; the rate is 0.0231. Pooled over the three ensembles, distance 0 gives 5,066 of 36,829 (0.1376) and distance ≥ 2 gives 15 of 35,501 (0.000423), a 326-fold drop, but that cut is post hoc and has to be re-registered and re-run before it goes near the paper. The sentence the paper can print today is that the pre-registered test was inconclusive and the data suggest a threshold at two edges rather than one; and “a distant skeleton error rarely does damage”, never “does little damage”.

Figure 3.3 holds what does not depend on the verdict. Without any knowledge the CPDAG answers the query by adjustment in 8.2, 16.9 and 55.8% of problems, so on most of them there is nothing to be robust about until knowledge arrives. The efficiency-only band exists: O^* moves and stays valid in 2.35, 2.65 and 5.53% of deletions. The median absolute weight of the deleted edge is the same for damaging and harmless deletions (0.5726 against 0.5583 on licensed, inverted on **large**), so strength does not predict damage and distance does. Severity does not fall with distance: among silent-bias events on licensed at distance ≥ 1 the relative bias has median 0.39, 90th percentile 1.75 and maximum 1,204.

The worked example (`x3_skeleton/results/worked_example.txt`, seed 4906, $p = 7$) was

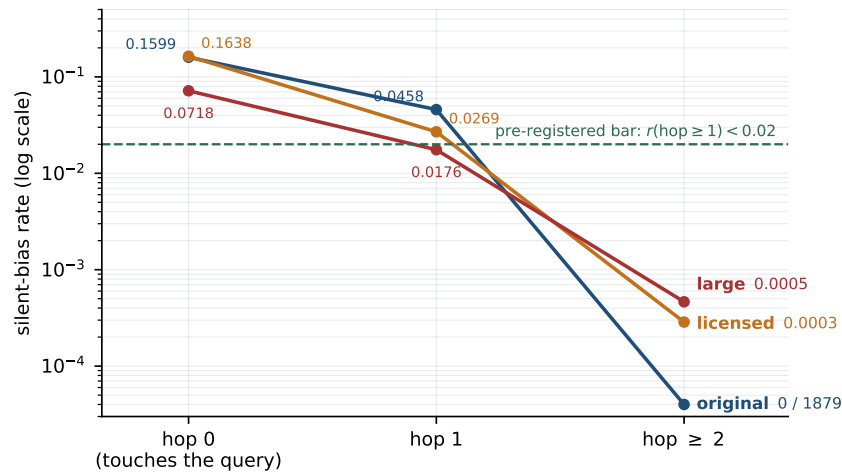


Figure 3.2: A missed edge at distance 1 still breaks the set in 2.7 % of amenable cases on the licensed ensemble; pooled with distance ≥ 2 that is 2.3 %, above the pre-registered 0.02, so the rule returns INCONCLUSIVE. The cut at two hops looks clean but was chosen after seeing these lines. The **original** zero (0 of 1,879) is drawn at the axis floor. Source: `x3_skeleton/results/x3_analysis.json`.

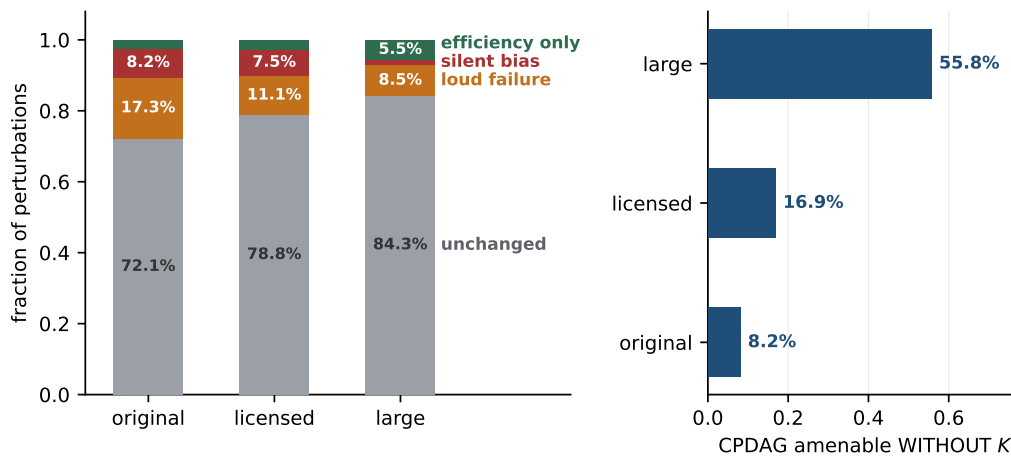


Figure 3.3: Left: the efficiency-only band (green) is not empty under skeleton deletions, which corrects the 0 of 754 we measured with claim reversals. Right: without knowledge the question of robustness rarely arises, because the CPDAG seldom identifies the effect. Source: `x3_skeleton/results/x3_analysis.json`.

built for the 3 September meeting. The names are attached by topological position and claim nothing empirical. The query is the effect of **Statin** on **CardiacEvent**; the analyst asserts $\text{Exercise} \rightarrow \text{BMI}$ and $\text{BMI} \rightarrow \text{BloodPressure}$, Meek propagates $\text{BloodPressure} \rightarrow \text{Statin}$, and $O^* = \{\text{BMI}, \text{Exercise}\}$ gives -0.4191 , which is also the true effect. A move is adding or retracting one claim and closing again (`worked_example.py`, function `neighbours`).

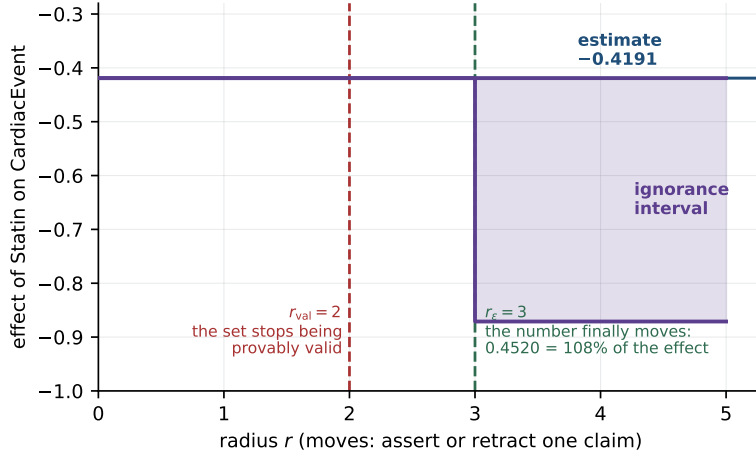


Figure 3.4: The set stops being provably valid after two retracted claims, but the estimate does not move until radius three, where the interval opens by 0.4520, 108 % of the effect. One radius would have raised the alarm one step early. The axis counts claim additions and retractions. Source: `worked_example.txt`.

Figure 3.4 is the case that justifies reporting validity and estimate separately: fragile identification, stable estimate. The witness reads “the estimate holds unless $\text{BMI} \rightarrow \text{BloodPressure}$ is retracted and $\text{Exercise} \rightarrow \text{BMI}$ is retracted”. Building it also produced a lemma. If G' refines G then $[G'] \subseteq [G]$, so a Z valid in every DAG of $[G]$ stays valid in every DAG of $[G']$: asserting a consistent claim can never break a fixed valid set, only retracting can. It is the same monotonicity that Chapter 8 audits on the ledger.

Without looking (answer 13, in the appendix)

What did X3 return under its pre-registered rule, and why is the two-hop cut not a result?

Without looking (answer 14, in the appendix)

In the worked example, what does “fragile identification, stable estimate” mean in numbers?

Exercise 3.1. *Refute with a construction: “a false claim that passes Meek’s check and lies two or more edges from the query can at worst make the analyst refuse to report.” Use a path of five nodes.*

Solution. Take the skeleton $c_0 - c_1 - c_2 - X - Y$ and the true DAG $Y \rightarrow X \rightarrow c_2 \rightarrow c_1 \rightarrow c_0$, so the true effect of X on Y is $\tau = 0$. A directed path has no v-structure, so the CPDAG is fully undirected; the path $X - Y$ starts undirected and the query is not amenable: the analyst refuses. Now assert the false claim $c_0 \rightarrow c_1$, at distance 2 from $\{X, Y\}$. It is consistent, and R1 propagates it along the path:

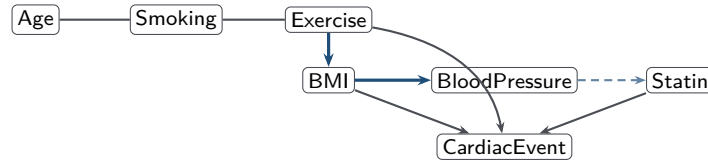


Now the query is amenable, $O^* = \text{pa}(Y) \setminus \text{forb} = \emptyset$, and the analyst reports the regression of Y on X , which is not zero because Y causes X . The same construction works at distance k on $k + 3$

nodes; `locality/x2/falsegain/results/chain_k0_k8.log` checks $k = 0, \dots, 8$ (all unsound, absolute bias 0.85 to 0.95 against $\tau = 0$). The firewall does not apply: the base was not identified.

Exercise 3.2. Draw G_0 of the worked example, marking asserted, propagated, compelled and undirected edges. Then retract each claim alone and say what the closure returns and whether $Z = \{BMI, Exercise\}$ stays valid. Why is $r_{val} = 2$?

Solution.

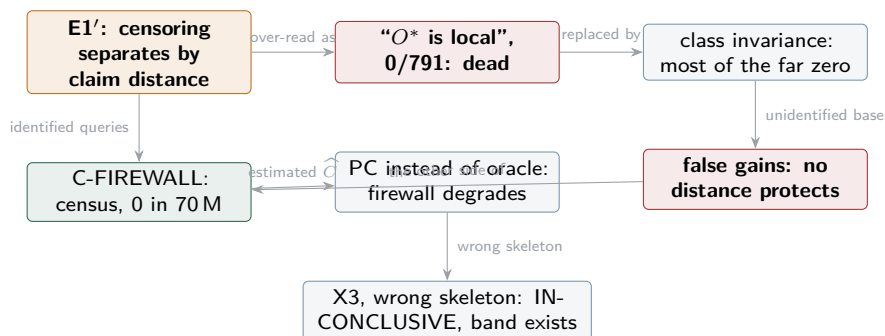


Thick blue: the claims; dashed: propagated; grey arrows: compelled; plain lines: undirected. Retracting $BMI \rightarrow BloodPressure$ alone returns the same graph: R1 re-derives it from $Exercise \rightarrow BMI$, since $Exercise$ and $BloodPressure$ are not adjacent. Retracting $Exercise \rightarrow BMI$ alone leaves that edge undirected, but $Statin$ keeps no undirected edge, so the query stays amenable and Z valid. Retracting both returns the CPDAG: $Statin - BloodPressure$ is undirected, the query is not amenable, and some DAG of the class has $Statin \rightarrow BloodPressure \rightarrow BMI$ with BMI a mediator. So $r_{claim} = 2$, and the witness's first step moves nothing, a step a covering-hop count would not charge. The hop walk is not cheaper for that: it pays three steps here, one of them for the propagated orientation between $BloodPressure$ and $Statin$ ($r_{hop} = 3$, Chapter 1; the unit comparison is Chapter 8). I checked the three closures with `x3_skeleton/code/`.

Do not move on

Name the difference in operator that lets X1 find silent bias from coherent spurious claims while the firewall census finds none. If you cannot, the step to redo is what X1 does to the claim its spurious edge replaces.

Chapter map



Part III

**Early September: what the radius
should measure**

Chapter 4

Early September: what should the radius measure?

This chapter covers 3 to 9 September: the week in which I tried to repair the degeneracy of r_{val} , ended up changing the question the radius answers, and then found that one contrast I had drawn, between my random draws and your corpus, came from how the two pipelines admitted queries rather than from structure. The mass at one retraction itself is real: it returns on the frozen frame in Chapters 5 to 12. Unit: the validity radius of my harness and of your corpus is a breadth-first distance in covering hops, so I write it r_{hop} here; the null radius counts retracted claims. The files of that week are in `rho_breakdown/radius_redef_2026-09-07/` unless a path says otherwise.

Before reading

- Your deck of 3 September closed the WP0 fork. Which branch did it support, and which did it reject?
- If a statistic puts almost all of its mass on two values, can a quantile or a mean of it repair that?

4.1 After 3 September: the defect I set out to repair

Your deck worked exactly on one eight-variable clinical graph, all 48 MPDAGs enumerated, and closed the WP0 fork in its own words. The diagnostic-tool branch was supported ($r_{\text{val}} = 3, 3, 2$, no saturation at 1). The trade-off paper was not, because all 14 valid adjustment sets share $r_{\text{val}} = 3$ and the efficiency-robustness frontier is flat. A week later your Phase 2 on `experiments/survival-and-pareto` falsified the Pareto premise twice and abandoned it under its pre-registered trigger: truthful knowledge that identifies a set always identifies the same one, so precision never moves, and what knowledge buys is identifiability (385 of 1,431 proposals, 26.9%, justified no valid set). Nothing in this document disputes either conclusion.

In the same days I had the opposite picture on small random problems. On 7 September I re-measured the validity radius, in covering hops (r_{hop}), with an independent harness: random DAGs on six nodes, one to six undirected CPDAG edges, up to three true claims, every query with an identified optimal set kept, the whole space enumerated around G_0 .

r_{hop} on 400 random problems, `harness.py`

$r_{\text{hop}} = 1$ on 190 (47.5%), 2 or 3 on 9 (2.3%), unreached on 201 (50.2%): 97.7% of the mass on $\{1, \infty\}$ (`EXPERIMENTS.md`, §1).

I read this as worse than a bias toward 1. A biased statistic can be recentred; a nearly binary one answers a single question (does some single retraction break the set?), and any quantile or mean of it inherits the two values. That reading drove the week. Section 4.4 shows which part of it described my harness.

Before reading

- Retracting a claim and reversing it look like two ways of doubting it. Are they two objects?
- If the hedge is restricted to the claims that matter, does the interval get narrower?

4.2 Thirty-seven candidates, two lemmas, one object

The target was a definition that is not degenerate, keeps a soundness statement, and gives a narrow interval where no damage is present. I wrote down 37 candidate redefinitions and audited each against measurements before refining any (`plan.json`). Nineteen were killed, most because on almost every problem they returned r_{val} , $r_{\text{val}} - 1$ or a contradiction-count radius that moves with them; all seven quantile or mass radii and all six weighted minima died that way. The arithmetic is short: first shells held at most 7 elements, so any quantile above 0.857 returned the minimum (160 of 160 problems), and for a well-posed centre the broken share one hop out never exceeded one half (maximum 0.5000 over 840 problems). The survivors had replaced the minimum by a range of effects. The object they share is easiest to see on a five-node example (Figure 4.1).

Example 4.1 (Three claims, three behaviours). True DAG $C \rightarrow X$, $C \rightarrow Y$, $C \rightarrow W$, $X \rightarrow M$, $M \rightarrow Y$; query (X, Y) ; true effect $1.2623 \times 0.7518 = 0.949$. The CPDAG keeps $C \rightarrow Y \leftarrow M$ and leaves $C \rightarrow W$, $C \rightarrow X$, $M \rightarrow X$ undirected; its four extensions give the effects $\{-0.203, 0.746, 0.949\}$, whose hull $[-0.203, 0.949]$ (width 1.152) is the *blanket*. The analyst asserts $K = \{C \rightarrow W, C \rightarrow X, X \rightarrow M\}$ and reports the point 0.949. Retracting $C \rightarrow X$ opens the hull to $[0.746, 0.949]$; retracting $C \rightarrow W$ changes the graph but not the effect; retracting $X \rightarrow M$ changes nothing, since R1 forces it back from $C \rightarrow X$. The union over single retractions is $[0.746, 0.949]$, 0.176 of the blanket. Asserting $X \rightarrow C$ instead would give the point 0.746, off the truth.

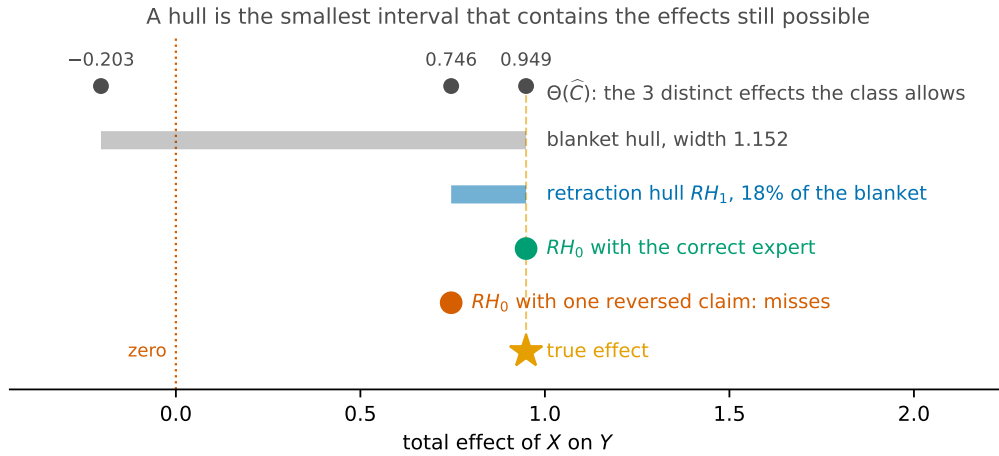


Figure 4.1: With the claims right the report is a point on the truth and one retraction widens it to under a fifth of the blanket; with one claim reversed the report misses, and nothing at budget zero can show it.

Definition 4.1 (Retraction hull). With $\Theta(G)$ the set of effects of X on Y over the DAG extensions of G (adjusting for the parents of X in each), $\text{RH}_j = \text{hull} \bigcup_{|S| \leq j} \Theta(\text{Meek}(\hat{C}, K \setminus S))$. RH_0 is the report and $\text{RH}_{|K|}$ the blanket. If $\hat{C}$ is the true class and at most j claims are false, RH_j contains the true effect, with no continuity assumption. Later chapters call it the knowledge interval I_j .

Two lemmas turned most of the surviving candidates into this one object.

Lemma 4.1 (L1 and L2). **L1, retraction equals reversal.** For every budget b , retracting at most b claims reaches exactly the worlds reached by reversing at most b claims: a world compatible

with $\text{Meek}(\hat{C}, K \setminus S)$ contradicts K on exactly one $\sigma \subseteq S$. **L2, screens are width-neutral.** If $\Theta(G_0)$ is a point and $Z_0 = O^*(G_0)$ is valid, a claim whose retraction leaves Z_0 valid leaves Θ unchanged; the hull over the load-bearing claims equals the hull over all of K .

Both were checked numerically before anything was built on them.

Lemmas and frontier, EXPERIMENTS.md

L1: 0 differences over 573 problems. L2: 250 of 250 and 300 of 300. Frontier (`confirm.py`: seven nodes, up to four claims, 300 problems per arm): the first budget reaching 95 % coverage costs 0.710 of the blanket's width with one reversed claim and 0.903 with two, and at one reversed claim the one-retraction hull is the blanket on 67.2 % of problems (§§2–3).

L1 makes “flip up to b claims” and “retract up to b claims” one object: reversal models the error, retraction hedges it. L2 takes my fourth proposed direction, identifying which claims are false, out of any width table. Locality measured in coverage on 8 September is the same fact in another currency: at a matched budget of one retraction, retracting only claims that touch the query covers 0.998 at 0.621 of the blanket's width against 1.000 at 0.630 for any claim, with 3.13 instead of 4.31 enumerations (`local_hedge.py`). It pays in computation and in what to ask the expert, not in width.

Coverage 1.000 in every cell

The first run of `confirm.py` covered at every budget, budget zero included: the claims were drawn truthfully, so there was nothing to be robust to. Asking what could have made the number differ from 1 caught it; the fix reverses some claims and keeps only draws that survive Meek closure, the errors the consistency check misses.

An audit of the frontier also asked what the width is made of. On informative problems with one reversed claim the class disputes whether X causes Y at all in 86.9 %; among the worlds that agree on direction the width is 0.314 of the blanket, and exactly zero in 60.7 % (EXPERIMENTS.md, §4). About two thirds of the hedged width is a dispute about causal order. Only 20 to 24 % of drawn problems were informative at all.

Without looking (answer 15, in the appendix)

State L1 and give the one-line reason it holds. What does it merge?

Without looking (answer 16, in the appendix)

A colleague proposes to rank which claims are false and hedge only those, to get a narrower interval. What does L2 say, and in which currency can the idea still pay?

Without looking (answer 17, in the appendix)

With one reversed claim the mean width ratio of the one-retraction hull is 0.710, and the hull equals the blanket on 67.2 % of problems. What is the mean ratio on the rest?

Before reading

- Rosenbaum's Γ does not ask whether the identifying assumption holds. What does it ask?
- If four problems in five leave no doubt about the effect, what does a robustness statistic measure on them?

4.3 The null radius: Rosenbaum’s question, taken literally

What Γ asks

Not whether there is unmeasured confounding, but how strong it would have to be before the conclusion falls. For background knowledge: how many of my claims would have to fall before I can no longer claim an effect?

On 8 September I wrote that down with the machinery above. Two things change with respect to the harness radius: the unit (r_{hop} walks covering hops, the new radius counts retracted claims) and, more importantly, the predicate that stops the search (r_{hop} stops where some graph invalidates Z , the new radius where the hull reaches zero). The audit of 7 September had rejected a hop-ball ancestor, unreached on 183 of 200 draws; this one is read off the claim-indexed hull.

Definition 4.2 (Null radius). $r_0 = \min\{j : 0 \in \text{RH}_j\}$, and $r_0 = \infty$ if no budget puts zero in the hull. Every effect compatible with any retraction of at most $r_0 - 1$ claims has the same sign.

Its first measurement looked like a failure: over all drawn problems with a non-zero effect, r_0 was “never” on 84% (`signradius.py`). The population was the problem. Most drawn problems have a blanket of width zero, where nothing can move; conditioning on the *informative* stratum, the problems with positive blanket width, swaps the two statistics (Figure 4.2).

The informative stratum, `stratum.py`

1,800 problems, one reversed claim	all	informative (381, 21.2 %)
$r_{\text{hop}} = 1$	43.9 %	89.5 %
effective values 2^H of r_{hop} (hops) / of r_0 (claims)	2.28 / 2.66	1.51 / 3.46

Conditioning on the informative stratum worsens one and improves the other

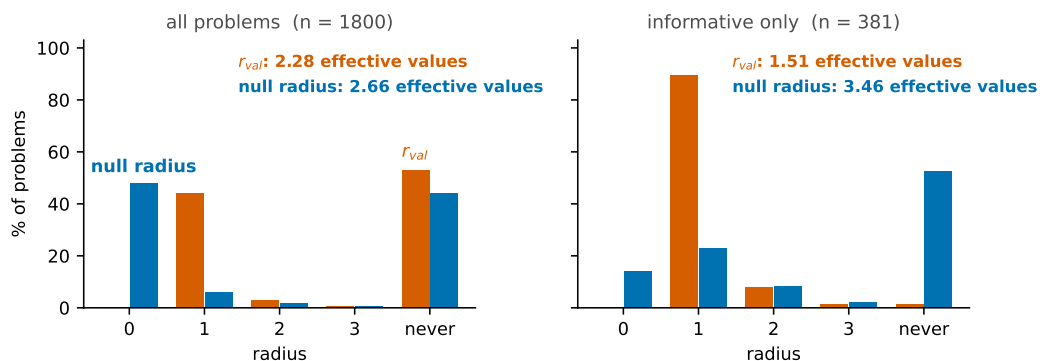


Figure 4.2: Where the class leaves any doubt about the effect, the validity radius (r_{hop}) is almost a constant and the null radius spreads over four levels.

The stratum is defined by the blanket, which reads neither K nor the planted errors, so this is not selection on the outcome. But 1.51 against 3.46 is conditional on a selection and does not show that the redefinition improved the statistic. Your `LOG.md` of session 6 retracts a coverage conclusion for the same reason (543, 182, 106 admissible rows: a composition effect).

The null radius, as it can be printed

Under the stated equivalence class, every effect compatible with retracting any $r_0 - 1$ of the analyst's orientation claims has the same sign; some set of r_0 claims, once retracted, admits a zero or opposite-sign effect.

Without looking (answer 18, in the appendix)

The harness's validity radius and r_0 differ in two ways. Name both, and say which one changes what the number means.

Before reading

- On your real networks at full coverage, is $r_{\text{hop}} = 1$ more or less frequent than on my random problems?
- Computed from $\hat{\Sigma}$, in which direction does a point-hull null radius err?

4.4 9 September: the “never” contrast was an admission rule

On 9 September I recounted your `results/axisa3/instances.jsonl` instead of copying the session 6 report, matching knowledge coverage at 1.0.

Two populations, `stratum.py` against `results/axisa3/instances.jsonl`

	$r_{\text{hop}} = 1$	on $\{1, \infty\}$	never	largest	eff. values
my random draws (1,800)	43.9 %	96.8 %	52.9 %	4	2.28
39 real networks (543 rows)	69.8 %	69.8 %	0.0 %	14	2.61

Neither 52.9 % nor 0 % “never” describes structure; each describes an admission rule. The mass at one retraction is a different matter and does not go away: on the frozen frame of 11–13 September $r_{\text{claim}} = 1$ on all 540 full-coverage rows (Chapter 5), r_0 is nearly binary (Chapter 10), and one retraction returns the whole-class interval on all 26 informative queries of T3's 37 eligible (Chapter 12). Your `THEOREMS.md` §14 already says the saturation at 1 is not a definitional artefact.

Two comparisons of mine, both across populations

I had first noted that $r_{\text{hop}} = 1$ falls from about 80 % to 61.1 % on real structure; the 61.1 % pools three coverage levels, and at coverage 1.0 it is 69.8 %. The table repeats the error at a deeper level. My draws kept every query whose claims survive Meek closure, including queries with no causal path, and `stratum.py` counted a query with no identified set as “never”. Your `fast_gate` rejects such queries and, at full coverage, admits a pair only if one retraction breaks O^* , so “never” is impossible by construction, as your `LOG.md` notes. Your random arm, gated the same way, has 0 unreached radii in 2,334 draws (`results/axisa2/random_analysis.txt`).

Within your corpus, $r_{\text{hop}} = 1$ on 69.8 % of full-coverage rows, with a tail up to 14 that Chapter 5 explains. Every width, coverage and direction number of Section 4.2 remains a measurement on my small random problems.

The same day I computed r_0 from $\hat{\Sigma}$ on your four Gaussian networks, whose coefficients are fitted to real data. The routine reads no DAG, but its inputs were the oracle CPDAG and claims from `select_knowledge`, read from the truth (Chapter 5): the run tests the estimator, not a

deployment. `arth150` hit the 20-minute wall at coverage 0.5 and is left out.

r_0 from $\hat{\Sigma}$, `rho_breakdown/rnull_real_2026-09-09/direcao_fast.py`

Coverage 1.0, 145 queries: oracle $r_0 = 1$ on 49 (34 %), ∞ on 96; the zero is structural (some extension has an effect of exactly zero) on 40 of the 49. Share of replicates whose estimate exceeds the oracle, claiming robustness that is not there:

n	200	1,000	5,000	20,000
point hull	24.7 %	22.3 %	20.6 %	18.1 %
95 % band	1.5 %	0.1 %	1.7 %	1.0 %

Without looking (answer 19, in the appendix)

Why do the 52.9 % “never” of my draws and the 0 % of your corpus not show that random graphs make the radius degenerate?

Do not move on

Write down the single condition in `fast_gate` that makes an unreachable radius impossible at full coverage. If you cannot, reread the red box above: Chapter 5 builds on it.

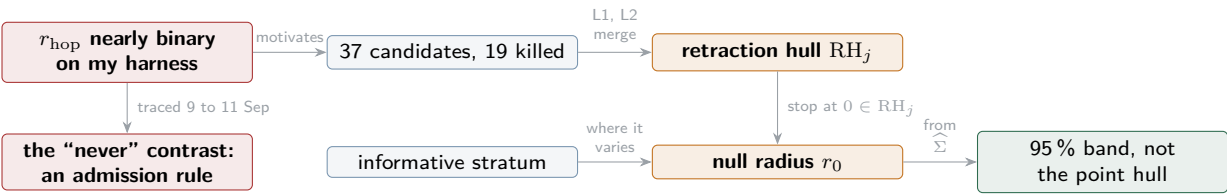
Exercise 4.1. A draft sentence reads: “On *ecoli70*, from $n = 20,000$ samples, the null radius of the query is infinite, so the conclusion that X affects Y survives the retraction of every claim.” The radius was computed with the point hull. Diagnose the sentence and rewrite it.

Solution. The estimator: the point hull overstates the null radius on 18.1 % of replicates even at $n = 20,000$, because a structural zero becomes $\pm\epsilon$ in $\hat{\Sigma}$ and lands on either side; recompute with the 95 % band (at most 1.7 %). The scope: the run used the oracle CPDAG and knowledge read from the truth, so nothing follows about errors in the class. The reading: $r_0 = \infty$ says no budget puts zero in the hull, not that the effect is large. Rewrite: “Under the stated equivalence class, and with the 95 % band from $\hat{\Sigma}$, no retraction of the analyst’s orientation claims admits a zero or opposite-sign effect of X on Y .”

Exercise 4.2. Five-node example. (a) RH_1 and its share of the blanket. (b) r_0 , given RH_2 is the blanket. (c) r_0 with $X \rightarrow C$ asserted (report 0.746, RH_1 the blanket), and did the report cover the truth? (d) With the right claims $r_{\text{hop}} = 1$; compare with (b).

Solution. (a) The retractions give $\{0.949\}$, $[0.746, 0.949]$, $\{0.949\}$, so $\text{RH}_1 = [0.746, 0.949]$, width 0.203, and $0.203/1.152 = 0.176$. (b) RH_0 and RH_1 exclude zero and $\text{RH}_2 = [-0.203, 0.949]$ contains it: $r_0 = 2$. (c) $\{0.746\}$ misses 0.949; RH_1 contains zero, so $r_0 = 1$, and the same single retraction restores coverage. (d) Validity of Z breaks one hop away while the sign of the effect survives any single retraction: $r_{\text{hop}} = 1$ raises an alarm that $r_0 = 2$ does not share.

Chapter map



Chapter 5

Where to evaluate, and three defects in the old corpus

On 10 September I stopped to ask where the radius should be evaluated at all, since every result so far rested on sampled graphs that do not look like a deployment. Answering it meant reading the committed pipeline end to end, and the reading found three defects in the corpus of `results/axisa3/`. Two of my own first diagnoses were wrong, and both corrections are below. The notes and scripts of these two days are in `rho_breakdown/eval_protocol_2026-09-11/`; the protocol they produced is the one executed in Chapters 6 to 9.

Before reading

- An evaluation instance has five coordinates. Which of them does a list of datasets fix?
- In which two places may the true DAG appear without breaking a deployment claim?

5.1 What a deployed instance is

Not a list of datasets

An evaluation methodology is a sampling distribution over instances $I = (G_{\text{true}}, \hat{C}, (X, Y), K, D_n)$. Swapping Erdős–Rényi graphs for real networks fixes one coordinate; if K is still read from the true DAG, the ball stays centred on the truth, over better graphs.

What a deployed analyst holds fixes the rule for every coordinate.

The invariant

$r = r(\hat{C}, K, X, Y, \hat{\Sigma}, n)$: a functional of analyst-observable quantities only. The true DAG enters twice, both times after the deployment rows are frozen: to generate the sample, as nature does, and to audit whether the promise held. A row is in the audit arm when the hash of its CPDAG source equals the hash of its generating DAG, decided by comparing bytes rather than reading a label.

With the invariant written down, the comparison that motivated the question reads differently. Section 6.2 of your draft contrasts separation of at least 2 in 45.4 % of real instances with 14.2 % of synthetic ones. Both numbers are in your files, over different denominators.

Separation ≥ 2 , `results/axisa3/instances.jsonl`

Real: 210 rows, 45.4 % of the 463 with defined separation, 25.3 % of all 831 admissible. Random arm (`results/axisa2/random_control.jsonl`): 14.2 % of 1,587 measurable draws pooled over five generators; Erdős–Rényi alone 12.5 %, and 3.6 %, 14.7 %, 24.1 % in its dense, medium and sparse cells. Undefined separation: 44.3 % of real rows, 17.1 % of random ones.

The sparse cell, 24.1 %, sits next to the real corpus over its full denominator, 25.3 %; den-

sity drives most of the gap, and the real corpus (mean degree 1.60 to 9.86, median 2.92, `results/axisa3/descriptive_structure.csv`) was never matched to a density cell. The reason to evaluate on real structure is the invariant, not a degeneracy of random graphs.

Without looking (answer 20, in the appendix)

Where may the true DAG appear without breaking a deployment claim, and what makes those two places harmless?

Before reading

- At coverage 1.0, what is G_0 in the committed corpus?
- 518 queries are rejected at partial coverage as `o_g0_not_identified`. Did the optimal-set criterion kill them?

5.2 Defect 1: the analyst's knowledge is read from the true DAG

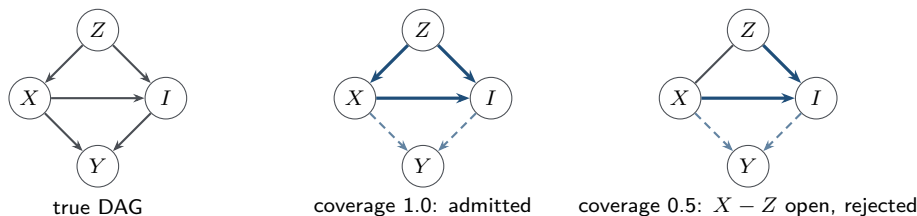
In `src/bkrobust/benchmarks/measure.py`, `select_knowledge(dag, cpdag, coverage)` takes the true DAG and calls `knowledge_to_recover` (`src/bkrobust/demo/example.py`), which repeatedly asserts the true orientation whose Meek closure orients the most edges; it sorts the result and keeps an evenly spaced fraction. Every claim is true, and at full coverage the closure is the true DAG.

Correction 1: a default read as a measurement

I first wrote that G_0 had no undirected edge in all 831 admissible rows, reading `g0_undirected_edges = 0`. The field is assigned only in the intractable-extensions branch (line 240 of `measure.py`) and keeps the dataclass default elsewhere. Rebuilt row by row: the 543 rows at coverage 1.0 are fully oriented, hence equal to the true DAG, and all 288 partial-coverage rows keep an undirected edge. The same silent default sits in `k_g0` on rejected rows.

The rejection `o_g0_not_identified` fires 286 times at coverage 0.25, 232 at 0.5, never at 1.0. The four-node mediator network shows the mechanism.

Example 5.1 (mediator, query $X \rightarrow I$). The DAG $Z \rightarrow X, Z \rightarrow I, X \rightarrow I, X \rightarrow Y, I \rightarrow Y$ has no v-structure, so the CPDAG is fully undirected. The sorted recovering set is $(X \rightarrow I, Z \rightarrow I, Z \rightarrow X)$. At coverage 1.0, G_0 is the DAG, $O^* = \{Z\}$, admitted. At 0.5, $\text{round}(3 \times 0.5) = 2$ and the stride keeps the first two claims; $X - Z$ stays open, one extension makes Z a confounder ($O^* = \{Z\}$) and the other a mediator ($O^* = \emptyset$), and the row is rejected.



Correction 2: Lever 0 recovers nothing

I first read this example as the optimal-set definition deleting the realistic case, since it enumerates extensions and gives up when they disagree, and proposed Lever 0: the closed form $O^* = \text{pa}(\text{cn}(X, Y)) \setminus \text{forb}(X, Y)$ would bring the rows back. It does not. All 518 rejected

rows are non-amenable; here $X - Z \rightarrow I$ is a possibly causal path starting with an undirected edge. The deletion is right and the label is wrong. Caught by running `is_amenable` and the closed form on every rejected row before writing the fix.

The numbers behind both halves of that correction:

The diagnosis, `diagnostico_amenabilidade.py`

518 of 518 rejections non-amenable, 0 recoverable. Over every same-size subset of the recovering set on the eight smallest networks (1,238 subsets), 29 of 40 non-degenerate rejected cases have another subset of that size that keeps the query amenable, 11 have none, and 61 more are degenerate ($\text{round}(1 \times 0.5) = 0$ leaves no knowledge). The 12 September differential agrees: the closed form returns a set where enumeration said “not identified” on 0 rows (`results/stage0/lever0_differential.json`).

So the defect is the `sorted()` in the thinning, not the optimal criterion: where it can be tested, the query usually dies because the analyst holds the wrong claims, chosen by alphabetical order. The stride is not even nested ($K(0.25) \not\subseteq K(0.5)$ on `diabetes`, `ecoli70`, `magic-irri`, `munin1`), so a coverage sweep is not a retraction sequence. The 518 are not missing data either: their radius is already zero, and Stage 0 added an identifiability radius (retractions until adjustment stops identifying the effect) that is defined exactly where the current radius dies.

What landed on 12 September, and what no patch fixes

Additive patches to `measure.py`, committed behaviour as default: a `nested` selection (prefix of one seeded permutation); `not_amenable` split from `o_g0_not_identified`; a three-valued verdict for O^* ; both silent fields assigned on every branch; the dispatch leg as a column. On 213 recomputed committed rows no admissible row moved (`experiments/stage0_differential.py`). A better stride fixes the size of K , never its origin; that takes knowledge that does not come from the truth (Chapter 6).

Without looking (answer 21, in the appendix)

What did Lever 0 recover of the 518 rejected rows, and what is the actual defect?

Before reading

- The gate admits a pair only if something happens under one retraction. What, and what does that fix?
- How often does a hop count from X to the adjustment set, computable from $\hat{C}$ and K , reproduce the committed radius?

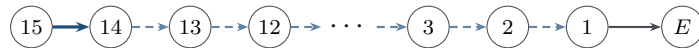
5.3 Defects 2 and 3: what the committed radii count

`fast_gate` in `measure.py` reads the true DAG five times: whether Y descends from X , the optimal set on the DAG, the validity of that set and of the empty set, and a loop that admits a pair only if retracting a single claim of the recovering set, with Meek re-closure, makes the optimal set invalid. At coverage 1.0 the asserted set is the recovering set, so admission is the definition of $r_{\text{claim}} = 1$.

The claim unit at full coverage, stage0/rclaim_cov1.py

540 rows (`pathfinder` excluded): $r_{\text{claim}} = 1$ on all 540. Committed hop radius 1 on 377, 2 on 102, 3 on 43, 4 to 14 on 18. Median asserted claims 5, median closure $|K_{G_0}|$ 12. The share of single retractions that break the set, ϕ_1 , runs from 0.034 to 1, median 0.30.

Section 6.3 of your draft names the sharpest case. In `paths` (17 nodes), hop radius 14, the analyst asserts one claim, $15 \rightarrow 14$, at the top of the fifteen-node chain component, and R1 orients the thirteen other chain edges from it. Retract it and the chain un-oriens at once: the hop count is the length of a Meek cascade and the tolerable number of wrong claims is zero. The paragraph commented out in your `06_simulation.tex`, saying $r_{\text{claim}} = 1$ on all 540 rows by construction, is correct and should go back in.



one asserted claim (thick), thirteen chain edges compelled by R1 (dashed): hop radius 14, claim radius 1

The third defect is how little the hop radius adds over the free rule: predict the separation s , the hop distance from X to the nearest member of O^* in the undirected component of X , or 1 if no member lies there. It is a fifteen-line search on $(\hat{C}, K)$.

The free rule against the committed radius, stage0/verify_ladder.py

Exact on 636 of 831 rows (76.5%), within one on 813 (97.8%), two or more apart on 18 (`paths` 13, `Kampen_2014` 3, `insurance` 2). Over-certifies on 16 rows, all $r = 1$, $s = 2$; under-certifies on 179, 69 of them in the 368-row undefined stratum. $|K_{G_0}|$ alone matches on 15. The law transfers as $r \geq \min(s, |K_{G_0}|)$ on 447 of 463 measurable rows. Also in `results/stage0/STAGE0.md`.

A hop count reproduces the radius and a claim count does not: the radius follows where knowledge sits. Your session 5 law $r = \min(s, |K_{G_0}|)$ held on 1,572 of 1,572 designed instances, so this confirms the theory; what it rules out is a headline the search reproduces (outside the teaching files, five rows on two networks differ by two or more). One more confound sits in the same rows: `local_up_fast` answered 811 rows, all at radii 1 to 3, and `e1_ladder` 20, all at 4 to 14, because `DEFAULT_SEARCH_BUDGET` is 3 in `src/bkrobust/hybrid.py`. A cost curve against radius on these rows is a cost curve against algorithm.

Without looking (answer 22, in the appendix)

Why is $r_{\text{claim}} = 1$ on all 540 full-coverage rows, and what does that make of “safe moves = $r_{\text{hop}} - 1$ ” on `paths`?

Without looking (answer 23, in the appendix)

State the free rule, its agreement with the committed radius, and the direction of its errors.

Before reading

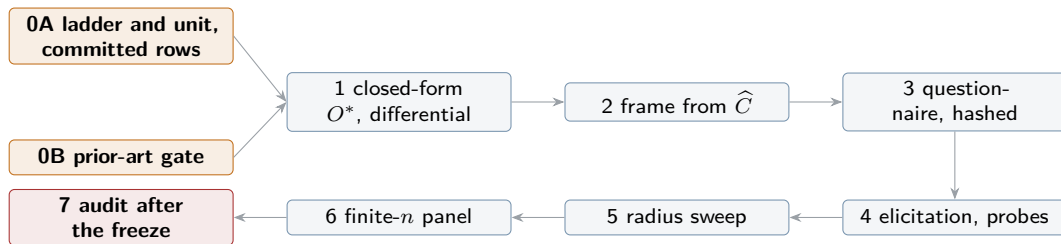
- Which experiment should a protocol run first: the most informative, or the one that can retire the framework most cheaply?
- The smallest subset of asserted constraints whose removal breaks a property: does that object already have a name?

5.4 Beat the BFS First: the protocol of 11 September

The protocol is named after its thesis: measure whether a certificate counted in the analyst's own claims, with no true-DAG quantity on its path, beats the free statistics an analyst already has, and whether its promise survives the truth; until then every other axis re-derives a hop count at great expense. One decision per design axis (02_PROTOCOL.md):

-
- A1 **Substrate.** Four labelled tiers never pooled into a headline; random graphs only as instrument calibration, matched on expected degree; the network as unit of evidence (intraclass correlation 0.398 on the $r = 1$ indicator over the 831 committed rows on 25 networks, about 60 effective rows).
 - A2 **Queries.** One frame drawn from the CPDAG alone, frozen before any knowledge; conditions on K become printed statuses, never filters; the five true-DAG reads leave the deployment path.
 - A3 **Knowledge.** `select_knowledge` off the deployment path; a frozen, hashed questionnaire asking causal order per chain component, answered by independent suppliers.
 - A4 **Geometry.** Centre at Meek($\hat{C}, K$); r_{claim} by exhaustive subsets to depth 3, printed beside r_{hop} .
 - A5 **Statistics.** Sound one-sided validity tests with censoring as an explicit outcome; deployment claims only on an estimated-CPDAG panel; the 95 % band, never the point hull.
 - A6 **Audit.** Truth-free probes first, then the truth-revealed buckets (dangerous, held, safe, slack), opened after the freeze.
-

Deployment arms (a language-model supplier and its replicates, harvested and human knowledge, a matched random control, empty knowledge as a status line) sit above a rule, and audit arms (truthful but incomplete knowledge, synthetic corruption, the old recovering set) below it. Nine tables run from the applicability funnel (Table 0) through the certificate ledger (Table 4) to the decision table naming the first-failing retraction set (Table 8), and the build order puts everything free first.



The protocol as written still carries my two wrong claims

02_PROTOCOL.md was assembled before the code reading of the same day. It still says G_0 has no undirected edge in all 831 rows, makes Lever 0 a blocking recovery step, and proposes safe moves $r - 1$ as a headline, which Defect 2 makes zero by selection. Read the protocol for its design and 06_CODE_FIXES.md and docs/EVALUATION_PLAN.md for the facts.

The prior-art gate ran on 12 September and retired a claim. The object we report, the size of the smallest subset of asserted constraints whose removal makes a property fail when adding constraints can never break it, is a minimal correction set, named and studied in satisfiability diagnosis. Eight areas were searched, from stability radii to AGM contraction, and none applied the object to causal graphs, Meek closure or adjustment validity (docs/PRIOR_ART.md). The contribution has to live in that instantiation and in the measurement of what it buys over the free statistics.

Minimal correction sets

Liffiton and Sakallah, *Algorithms for computing minimal unsatisfiable subsets of constraints*, J. Automated Reasoning 40(1), 2008; Bacchus and Katsirelos, *Using minimal correction sets to more efficiently compute minimal unsatisfiable sets*, SAT 2015. For the hitting-set duality our depth-3 enumeration naively re-implements; both still to be checked against the papers.

Without looking (answer 24, in the appendix)

What is r_{claim} called in the satisfiability literature, and what is left for our paper to contribute?

Do not move on

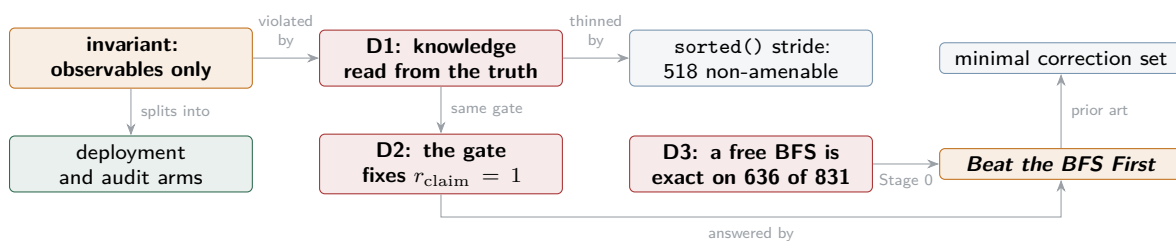
Write the two lines of `select_knowledge` that decide which claims a partial-coverage analyst keeps, and what the `nested` mode puts in their place. If you cannot, redo the `mediator` example.

Exercise 5.1. Classify each item as deployment-legal, audit-only, or a defect on the deployment path: (a) `select_knowledge`; (b) the check `y not in dag.descendants(x)` in `fast_gate`; (c) simulating D_n from the true DAG; (d) the free rule; (e) a questionnaire restricted to the pairs that `fast_gate` admitted; (f) the share of a supplier's claims that are false.

Solution. (a) Defect: reads orientations from the truth; kept only as an audit yardstick. (b) Defect: a filter choosing who is measured by looking at the truth; it becomes a post-hoc audit label. (c) Deployment-legal: nature's role. (d) Deployment-legal: computed from $\hat{C}$ and K alone, hence the baseline to beat. (e) Defect one level up: the questions would depend on the truth through the gate; the protocol scopes them from the CPDAG's chain components. (f) Audit-only, after every deployment row is frozen.

Exercise 5.2. Diagnose and rewrite this passage of Section 6.2 of your draft: “The topological separation [...] is 2 or greater in 45.4 % of real instances, compared to just 14.2 % synthetically. [...] while a radius of 1 still occurs in about 60 % of real instances, the scores stretch as high as 14.”

Solution. Right: about 60 % at radius 1 matches the corpus (61.1 %). To fix: 45.4 % is over the 463 rows with defined separation (25.3 % over all 831); 14.2 % pools five generators, Erdős–Rényi's sparse cell alone gives 24.1 %, and the undefined rate differs 2.6-fold, so the contrast is mostly denominator and density. “As high as 14” is `paths`, one asserted claim, claim radius 1, and every radius above 3 came from the ladder, not the search. Rewrite: “Over all 831 admissible real instances the hop radius is 1 on 61.1 %, separation is at least 2 on 25.3 %, and a tail reaches 14. The tail measures Meek cascades rather than tolerable errors: on every full-coverage row, one retracted claim breaks the adjustment set.”

Chapter map

Part IV

The protocol, executed (11 to 13 September)

Chapter 6

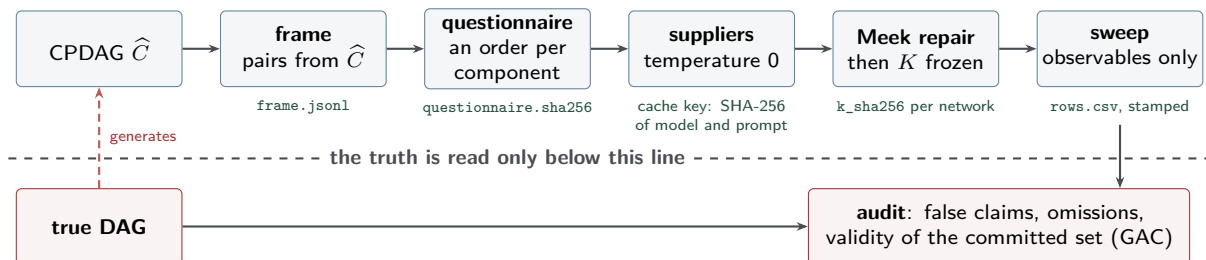
The observable frame and the frozen questionnaire

Chapter 5 ended with a protocol on paper. This chapter is how its population and its knowledge were built on 11 and 12 September, before any number of Chapters 7 to 9 existed. The second `\todo` of your Section 6 asks for this machine almost word for word: knowledge “collected by asking for a causal order over each chain component”, measured “on the same frozen instance frame, with a matched random control and an audit opened only after the rows are frozen”. It exists; its entry point is `docs/PROTOCOL_RUN.md`.

The idea in one sentence

Choose the rows from the CPDAG alone and freeze them. From then on a knowledge source may change the *status* of a row, never whether the row is in the table, so every source is scored on the same 528 rows and every contrast between sources is paired by construction.

What matters most is the order of the freezes. Everything above the dashed line is a function of $\hat{C}$, the claims and the query; the true DAG enters below it, once to generate $\hat{C}$ and once, after `rows.csv` is written, to score.



Before reading

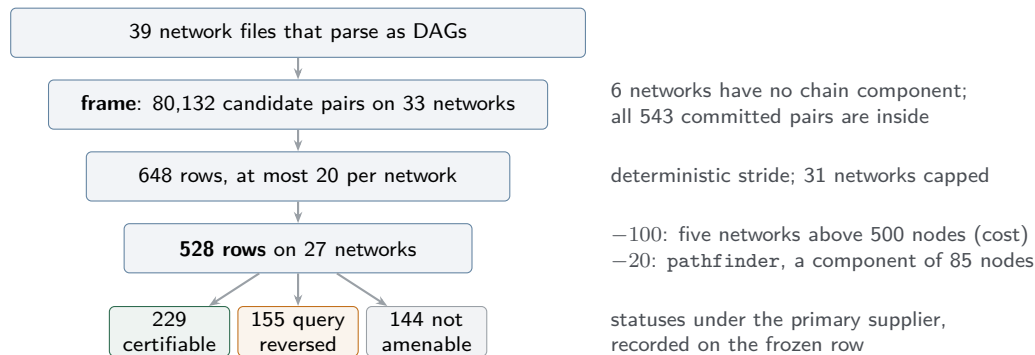
- How does one get from 80,132 candidate pairs to 528 rows, and does any cut read the truth?
- Why are “query reversed” and “not amenable” statuses of a row rather than reasons to remove it?

6.1 The frame: rows chosen from the CPDAG alone

The rule in `experiments/frame_build.py` is one sentence: the treatment lies in a chain component of $\hat{C}$ with at least two nodes, or next to one, and the outcome is a possible descendant of the treatment in $\hat{C}$. Nothing else filters. The file’s docstring lists the five reads of the true DAG the committed gate made before admitting a pair; each is now a status or an audit column. Per network, a deterministic stride over the sorted pairs keeps at most twenty and records the fraction kept.

Example 6.1 (The fraction kept, on three networks). `child`: 20 of 228 pairs, fraction 0.0877. `munin2`: 20 of 22,016, fraction 0.000908. `mediator`: all 12. A rate pooled over rows weights each network by its cap, not by its size (`results/frame/per_network.csv`).

That, and the structure, supplier and repair that rows of one network share, is why the network is the unit of every interval in the ledger. Table 0 prints the price: on the certifiable indicator, the 260 benchmark rows carry an intraclass correlation of 0.216 and a design effect of 5.1, so they are worth about 51 independent rows. The funnel, with each cut named:



The frame, results/frame/frame_summary.json

80,132 pairs on 33 of 39 networks; all 543 committed pairs inside, none outside, an expansion of **147.6×**; eight networks yield a frame and yielded no committed instance. Both cuts from 648 to 528 use sizes of networks and components, facts of $\hat{C}$ (TABLE0_FUNNEL.md).

The frame keeps the rows an arm cannot certify. What varies between arms is the composition of statuses on the same frozen rows (per tier in results/ledger/TABLE3_STATUS.md).

Statuses on the same rows, results/ledger/sweep_summary.json

arm	rows	certifiable	query reversed	not amenable
no knowledge (D_DEGEN)	528	68	0	460
primary supplier, 7B (D_LLM)	528	229	155	144
scale replicate, 72B (D_LLM_72B)	528	285	191	52
chance on the 7B's edges (D_RAND, 3 draws)	1,444	649	401	394
greedy recovering set (A_ORACLE)	528	294	234	0

The oracle arm's graph is fully oriented and true, so its statuses are the truth's: on 234 of the 528 rows the outcome is not a descendant of the treatment. The committed gate removed such pairs by reading the truth; here they stay, and a supplier that orients the query away is right to. "Query reversed" under a supplier is information about its claims, not a missing row, and a filter that depended on K would have given each arm a population of its own. (Rows of frame.jsonl say true_dag_on_path: false, true of the selection step; rows.csv restamps them via_Chat_only, the stamp to quote.)

Without looking (answer 25, in the appendix)

From 648 sampled rows to 528: name the two cuts, the rows each removes, and why neither reads the truth.

Before reading

- Why ask for a causal order per chain component rather than for a direction per edge?
- When an elicited set has no consistent Meek closure, which claims go first, and where does a claim with no stated confidence go?

6.2 The questionnaire, and the repair that freezes K

`experiments/questionnaire_build.py` asks one question per chain component that contains or touches a frame treatment, capped at twenty nodes, and the question is a causal *order*. Pairwise directions from an imperfect source form cycles at scale; an order cannot, and the only inconsistency left is a v-structure that $\hat{C}$ does not have. On the fixed skeleton an order also carries required and forbidden edges; ancestral claims, the one granularity it cannot express, have their own block in Chapter 7. The parser in `experiments/elicitor_run.py` reads each undirected pair as `a->b`, `b->a` or `DECLINE`. A `DECLINE` stands; a stated direction that contradicts the model's own order is counted and the order wins; a pair neither stated nor ordered is `not_reached`; a pair only ordered becomes a claim with no confidence.

Every component is rendered under real names, with a domain sentence that never names the benchmark and, where the file has them, the variables' states, and under scrambled tokens with a neutral sentence; each in two seeded presentation orders, asserted distinct at build time. Each network also gets a control block of up to eight edges that $\hat{C}$ already orients, asked as if open: a wrong answer there is commission visible without the truth.

The questionnaire, `results/elicitor/questionnaire.json`

120 components on 32 networks, **369 open pairs**, **540 items**: 120×2 namings $\times 2$ orders = 480, plus 60 control items (30 networks, both namings, first order), 206 control edges. Dropped for size: `pathfinder`'s 85-node component; two of 35 and 25 nodes on each of `mumin2` and `mumin3`. SHA-256 `a62896791a7b...`, recorded by every condition in `knowledge.json`. It was hashed before the first supplier call.

Per network, the claims are closed with Meek's rules against $\hat{C}$. A set with no consistent closure is recorded, which is the rejection rate the protocol asked for, and then repaired as a practitioner would: drop claims in increasing stated confidence until the closure succeeds. `repair()` puts unscored claims at the back of the queue. The result is frozen with its hash before any radius.

Example 6.2 (The repair on `child`, primary supplier). Twelve pairs asked. The 7B declines `Disease-Sick`, states nine directions at confidence 1, and places `Age → Disease` and `C02 → LungParench` only through its order, with no confidence (`results/elicitor/answers_D_LLM.json`). Eleven claims, eight true at the truth. With `Disease → LungParench`, the claim `C02 → LungParench` makes a v-structure between non-adjacent parents that $\hat{C}$ does not have. The repair drops the nine confident claims (eight true), then `Age → Disease`, and the closure succeeds. The frozen K is $\{C02 \rightarrow LungParench\}$, which is false: `LungParench` is the parent of `C02` in the network.

The rule did what it says; it rests on a confidence that is 1 almost everywhere and on an unargued tie-break for unscored claims.

The repair, `results/ledger/PREREGISTRATION.md`, A.1

Primary supplier: 214 claims, **43 dropped**, 171 kept; 6 of 32 networks repaired. Order replicate 57 of 196 (7 networks); second family 41 of 264 (8); scrambled names 12 of 163 (4). A minority of networks every time (L2 supported), about one claim in five where it fires.

Without looking (answer 26, in the appendix)

Derive the 540 items from the 120 components, and say why the question is an order.

Without looking (answer 27, in the appendix)

On child, why did the repair keep C02 → LungParench, and was it true?

Before reading

- On how many exact rows is the break named by a single claim?
- When the named claim is true at the truth, is the committed set always valid?

6.3 The decision table: the claim an estimate rests on

The certificate’s search stops at the first retraction set that breaks the committed set, and that set, a minimal correction set in the sense of `docs/PRIOR_ART.md`, is written in the analyst’s variable names. Table 8 (`experiments/decision_table.py`) turns it into a sentence: the estimate holds unless this claim is retracted.

network	query	r_{claim}	holds unless retracted	named false?	set at truth
Acid_1996	x1 → x10	1	x4 → x1	yes	invalid
Didelez_2010	Occ → HRT	1	Age → Occ	no	valid
Kampen_2014	AIS → DET	3	ALN → SAN, APA → SAN, SAN → AIS	yes	invalid
Kampen_2014	AFF → AIS	1	AFF → ALN	no	invalid

The columns left of “named false?” are observables; the last two are the audit. The fourth row is not a bug. The search enumerates retraction sets in a fixed order and names the *first* that breaks, so when several single claims break the set, only one is named. On **AFF → AIS** the named claim is true; the set is invalid at the truth because of the four false claims, out of eight, that the supplier made on that network.

Table 8, results/ledger/TABLE8_DECISION.md, decision_table.csv

Primary supplier: 229 certifiable rows, **157 with an exact claim radius** (of the other 72, 44 survive every retraction to depth three, 27 survive every retraction, 1 is censored by the budget). A **single claim names the break on 141 of 157** (8 need two, 8 three). Audit: named claim false and set invalid 58, false and valid 10, true and invalid 4 (all **Kampen_2014**, treatment **AFF**), true and valid 85.

On 58 of the 62 invalid rows the sentence points at a claim that is in fact false. That is the form of report Chapter 10 asks for: the conclusion rests on none, or on one named claim that the analyst can go and check.

The decision sentence

“For the effect of x1 on x10, the adjustment set rests on a single background claim, x4 → x1; without it, the set no longer satisfies the adjustment criterion.” Always with its premise: the equivalence class is correct.

Without looking (answer 28, in the appendix)

Why can the claim named by Table 8 be true while the committed set is invalid at the truth, and where does it happen?

Before reading

- Why does a seed built from Python's `hash()` of a network name differ in every process?
- In the stage-0 calibration rerun, which instrument lost its zero, and which kept it?

6.4 A salted seed, and the calibration rerun

Python salts the hash of a string per interpreter process, so a seed built from `hash("child")` changes in every run and in every worker of a split sweep. I made that mistake more than once. It was found first in the chance arm of the sweep, fixed before the ledger was rerun (`results/ledger/PREREGISTRATION.md`, A.0), then in the source-overlap probe of Chapter 7. On 13 September, re-reading every script against its output, I found it in four more scripts under `experiments/`: `lever0_differential.py`, `stage0_calibration.py`, `frame_status.py` and `questionnaire_build.py`. All now seed from the SHA-256 of a stable key.

The Lever 0 differential was rerun: 834 agreements of 845, where the salted run printed 818 of 822, and no invalid set on either (`results/stage0/lever0_differential.json`). The first status run of the frame is superseded by the ledger. The questionnaire cannot be rerun: every per-network permutation in it went through the salted seed. It stays authoritative, because it is what the suppliers answered and each item carries the SHA-256 of its prompt; the builder now refuses to overwrite it and writes a rebuild beside it. The corrupted-knowledge calibration (claims of the recovering set flipped at rates 0 to 0.5, coverages 1.0 and 0.5, on the committed rows) was rerun on 13 September. `results/stage0/STAGE0.md` still prints the salted run, and `docs/PROTOCOL_RUN.md` says the rerun is in progress; both predate it.

Stage-0 calibration, `results/stage0/calibration_summary.json`

16,496 draws on 24 networks; **8,911 scored** (salted run: 8,553). Set invalid at the truth on 1,009 rows (salted: 963). Corrupted rows only, salted run in parentheses:

instrument	dangerous rows	detection	false alarm	certified moves
claim radius (B7)	0 (0)	1.000 (1.000)	0.473 (0.501)	0.062 (0.083)
hop radius (B6)	3 (0)	0.997 (1.000)	0.386 (0.404)	0.764 (0.792)
free rule (B2)	9 (8)	0.991 (0.992)	0.458 (0.498)	0.294 (0.305)

The three dangerous hop rows are all at corruption rate 0.5.

Most cells moved by a few hundredths, as a new set of draws should. One moved in kind.

"The hop radius never over-certified" was a property of one draw

I read the salted run as a hop radius that never over-certifies and detects every invalid set. Under stable seeds it still beats the free rule on all four columns (3 dangerous rows against 9), but its zero is gone. The claim radius kept its zero, for the structural reason of Chapter 8: a zero that moves with the draw is a statistic; one that cannot move is a property of the instrument.

Without looking (answer 29, in the appendix)

In the calibration rerun with stable seeds, which instrument went from zero to non-zero dangerous rows, by how many, and why is the claim radius's zero on both runs a different kind of number?

Do not move on

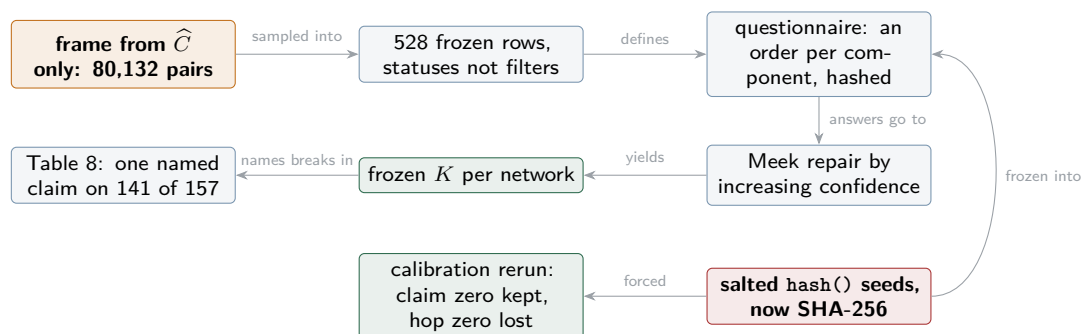
Start from the 7B's answer on **child** and write the two steps that turn its eleven orientations into the frozen K : why the set is inconsistent, and the order in which claims are dropped. If you cannot say why the survivor is false, reread the example in Section 6.2.

Exercise 6.1. (a) Check the funnel: rows lost from the sample to the sweep, and the expansion from the 543 committed pairs to the frame. (b) From the audit of Table 8, compute the share of exact rows named by a single claim, the share of invalid sets when the named claim is false, the share of valid sets when it is true, and the base rate of invalid sets among exact rows. (c) Say in one sentence what (b) means for an analyst without the truth.

Solution. (a) $648 - 100 - 20 = 528$, so 120 rows; $80,132/543 = 147.6$. (b) $141/157 = 0.898$; $58/68 = 0.853$; $85/89 = 0.955$ (so $4/89 = 0.045$ invalid); $62/157 = 0.395$. (c) Checking the one named claim moves the chance that the set is invalid from 0.40 to 0.85 if the claim is false and to 0.045 if it is true: the table says which single claim to verify, given a correct equivalence class.

Exercise 6.2. Refute each statement with a specific case from this chapter. (a) “If the claim Table 8 names is true, the committed set is valid at the truth.” (b) “Eliciting an order rather than directions guarantees a Meek-consistent set.” (c) “Repairing by increasing confidence keeps the claims most likely to be true.” (d) “A seed derived from the network’s name is reproducible.”

Solution. (a) Kampen_2014, AFF $\rightarrow$ AIS: the named AFF $\rightarrow$ ALN is true and the set is invalid, through the four false claims the supplier made there. (b) **child** under the 7B: the implied C02 $\rightarrow$ LungParench with Disease $\rightarrow$ LungParench is a v-structure $\hat{C}$ does not have. An order excludes cycles, not new v-structures. (c) **child** again: eight true claims at confidence 1 dropped, a false unscored one kept. (d) Python salts string hashes per process; the source-overlap probe re-asked its suppliers on every run because the prompt changed.

Chapter map

Chapter 7

Knowledge suppliers as error processes

Chapter 6 built the rows and the questions; this chapter is about who answered them. The question is not whether a language model understands causality. It is whether the wrong claims the certificate was tested against came from processes varied and realistic enough that the verdict of Chapter 8 does not depend on one way of being wrong. The models are suppliers of claims, one of four knowledge processes, and nothing in the paper ranks them.

The idea in one sentence

A certificate against wrong claims can only be tested with wrong claims, and whoever chooses *which* claims are wrong decides what the test can see. So the evaluation uses several error processes, including one whose errors nobody chose.

Before reading

- Which four processes produce the wrong claims of the evaluation, and what does each answer that the others cannot?
- Why are commission and omission two columns and never a sum?

7.1 Four sources of wrong claims

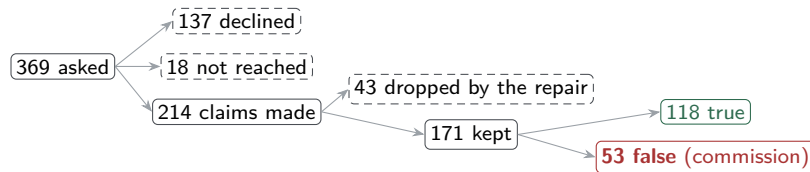
A synthetic reversal chooses the error: the experimenter knows how many claims are wrong and where, which is what the knowledge interval needs, so the controlled arms $b = 0, 1, 2$ feed Table 1 (`results/interval/PREREGISTRATION.md`; Chapter 10). But a reversal never decides what to talk about, never correlates neighbouring errors and never declines. A real source does all three. The evaluation therefore has four processes, each answering one question:

- controlled reversals: what exactly b wrong claims cost;
- chance (`D_RANDOM`): orientations drawn among the Meek-consistent choices on the primary supplier's own edges, that is, what a coin does where the source spoke;
- language-model elicitation: an uncontrolled error process that chooses what to assert, what to decline and where to err;
- truthful controls (`A_TRUE`, and `A_ORACLE`), read off the truth and printed below the rule: where one asserts, separated from asserting wrongly.

An arm with no knowledge (`D_DEGEN`) is the floor.

The local suppliers ran at temperature zero and seed zero, each answer cached under the SHA-256 of model and prompt (`experiments/elicitation_run.py`). The primary supplier, `qwen2.5:7b-instruct`, answered on real names (`D_LLM`), on the second presentation order (`D_LLM_INSTR`) and on scrambled names (`D_SCRAMBLED`); a second family, `llama3.1:8b`, on real names (`D_LLM_SRC`). A twenty-billion reasoning model on the same host spent its whole budget reasoning, returned nothing, and was dropped.

Example 7.1 (What the primary supplier did with 369 open pairs). Every open pair of the questionnaire ends in exactly one box.



Dashed boxes end with no kept claim: $137 + 18 + 43 = 198$ pairs, the omission. Counts from TABLE4.md (elicitation block) and knowledge_audit.json.

The tree separates two quantities, and the protocol names them so that nobody adds them.

Definition 7.1 (Commission and omission, at the knowledge level). For one supplier and one network, *commission* is the fraction of claims kept after the repair that are false at the truth, and *omission* is the fraction of asked pairs that end with no kept claim: declined, not reached or dropped. Both are averaged over networks with a network bootstrap and printed in separate columns (experiments/knowledge_audit.py).

A decline leaves the edge open and the analyst’s graph wider; a false claim closes the edge the wrong way and can invalidate the set. The denominators also differ, so the sum is not the fraction of anything.

Primary supplier, results/elicit/knowledge_audit.json

Commission **0.325** [0.210, 0.449] by network mean (53 of 171 pooled); omission 0.450 [0.361, 0.545]. By tier: 0.27 on the benchmarks, 0.28 on the applied-paper diagrams, 0.44 on the four Gaussian networks, **0.75** on the textbook diagrams, whose variables are abstract symbols. L1 registered a band of [0.10, 0.45]: supported.

The protocol’s two other suppliers, claims harvested from the source publications and human analysts, are absent: both cost human hours, and docs/PROTOCOL_RUN.md records the deferral with that reason.

Without looking (answer 30, in the appendix)

The primary supplier has commission 0.325 and omission 0.450. Why is there no “total error” column?

Before reading

- Of the predictions S1 to S10, which failure changes what the paper says?
- How can a supplier that is wrong less often per claim have its committed set invalid as often?

7.2 Scale buys claims, not validity

On 12 September four checkpoints on one node of four H100 answered the same 540 frozen items through vLLM, with each model’s chat template, temperature zero, seed zero and 700 tokens: Qwen2.5-72B-Instruct (real names at both orders, and scrambled), Qwen2.5-32B-Instruct, Llama-3.3-70B-Instruct and gemma-3-27b-it. The 2,160 answers entered the same cache and went through the same parser, repair and sweep: 8,308 ledger rows, 3,014 certifiable rows on the true CPDAG. Predictions S1 to S10 were written after the jobs were submitted and before any answer was read, with a decision rule: if S1 and S6 both held, the 72B row would lead.

	prediction	verdict	number
S1	scale lowers commission	not as registered	72B – 7B: $-0.100 [-0.224, +0.017]$
S2	scale lowers omission	supported	72B – 7B: $-0.280 [-0.390, -0.164]$
S3	at 72B, names carry information	not supported	scrambled – real $+0.088 [-0.064, +0.238]$
S4	claim certificate never dangerous	supported, structural	0 of 3,014 certifiable rows
S5	the unit gap does not close	supported, thinly	hop: 1 dangerous row at 72B, 0 on gemma
S6	scale lowers invalid sets	not supported	0.298 at 72B, 0.288 at 7B
S7	moves do not order by truth	unresolved, as predicted	0.846 at 72B, 1.086 on the truthful arm
S8	the third family is no outlier	not supported	gemma – 72B: $+0.112 [+0.029, +0.213]$
S9	less source variance at scale	direction only	0.182 against 0.220
S10	bridge improves, stays biased	not supported	72B: control – open $+0.004 [-0.153, +0.159]$

S6 is the row that decides (`results/elicit/PREREGISTRATION_SCALE.md`, Appendix A). The 72B keeps claims on 281 of the 369 pairs and declines 27, where the 7B kept 171 and declined 137. Commission falls from 0.325 to 0.233 by network mean without a separated paired interval, while the committed set is invalid at the truth on 85 of 285 certifiable rows at 72B and on 66 of 229 at 7B. S1 did not separate and S6 failed, so the 7B stays the primary supplier and the 72B is the scale replicate. Figure 7.1 puts all ten conditions side by side.

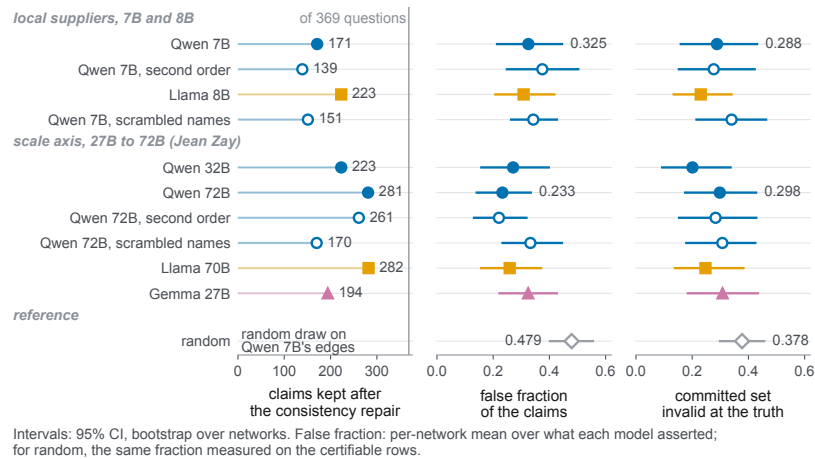


Figure 7.1: Scale buys claims and accuracy per claim, not validity: across the Qwen family the number of kept claims rises from 171 to 223 to 281 and the false fraction falls, while the share of certifiable rows whose committed set is invalid at the truth goes 0.288, 0.201, 0.298, in no order and with overlapping intervals.

The mechanism is counting. A set is invalid when one wrong claim sits where the query needs it, and a supplier that asserts more puts more claims near each query. On commission, gemma is the high outlier against the 72B. At 72B the model is also far more stable under presentation order: instrument disagreement **0.047** [0.019, 0.079] on shared edges, against 0.160 at 7B (`TABLE4.md`). The batch script is `experiments/remote/elicit_jz.slurm` (one node, four H100, every cache on scratch because vLLM's compile cache overflowed the home quota, eager mode); `elicit_import.py` puts the answers under the local cache key, and `elicit_run.py` refuses to call a cluster model.

Without looking (answer 31, in the appendix)

Without numbers: why can a supplier that is wrong less often per claim leave the committed set invalid just as often?

Before reading

- Asked cold for the publication behind each network, how many do the local suppliers name?
- Is the error on compelled edges, which needs no truth, an unbiased estimate of commission on open edges?

7.3 Names, a truth-free bridge, and two negative results

Scrambled names. At 7B, scrambling gives commission 0.343 [0.260, 0.431] against 0.325; what changes is declining, omission 0.568 against 0.450 (L3 not supported). At 72B the gap is +0.088 [−0.064, +0.238] and the model declines 96 pairs against 27 (S3). A source reciting the benchmarks would collapse without the names, and neither does; by invariance, that licenses “no recitation detected” and nothing stronger.

The compelled-edge bridge. The control block asks about edges the data already orient, so its error needs no truth. If it estimated open-edge commission without bias, a tier with no known truth could be scored. At 7B it does not: 0.191 [0.094, 0.293] against 0.325, a correction of about 0.13 that varies by network (L9 not supported).

Control minus open-edge commission, results/elicit/knowledge_audit.json

Paired by network: 7B −0.153 [−0.324, +0.009] on 23 networks; 72B +0.004 [−0.153, +0.159] on 26. The 72B’s own control error is 0.232 [0.121, 0.358], higher than the 7B’s 0.191, so size did not improve the control; it removed the bias. At 7B the paired interval reaches +0.009: the bias rests on the registered comparison and a point estimate of −0.15.

The bridge is therefore reported per supplier: usable at 72B, biased at 7B.

The source-overlap probe. The scrambled arm answers the recitation question only by invariance, so a second probe asks it directly: shown the variable list alone, name the publication behind the network and its declared exposure, outcome and adjustment set. “Recited” requires the source and the query.

Source overlap, results/elicit/SOURCE_OVERLAP.md

32 networks, 29 with a scorable source, 10 with a declared query. Source named: **0 of 29** for qwen2.5:7b, which declines on 28 and misattributes the one it answers; **0 of 29** for llama3.1:8b, which answers all 29 and matches none. Declared query reproduced on 2 of 10 by each, never with the source. **Recited: none**, for both.

Matching a query without the source is reading the names. The pre-registration’s appendix scores an earlier pass on prompts rendered with the salted seed of Chapter 6 (27 declines, same verdict); the box is the rerun on stable prompts. The cluster models have not been probed.

Ancestral claims. Asked about 156 non-adjacent pairs at skeleton distance two, the 7B asserted ancestry on **6**, said “neither” on 45 and declined 105; the six reduce soundly to ten orientations, four false, which change one true claim on one swept network and no certificate (results/elicit/PREREGISTRATION_ANCESTRAL.md). All three predictions fell: a negative result.

Without looking (answer 32, in the appendix)

At which supplier size is the compelled-edge bridge unbiased, what is the number, and why does that not mean the larger model is better on compelled edges?

Before reading

- Does the greedy recovering set, which the old pipeline used as knowledge, orient more edges per claim than the 7B supplier?
- Which justification for the redesign was withdrawn, and what still carries it?

7.4 The cascade test: a justification withdrawn

The committed pipeline’s knowledge was the greedy recovering set, now **A_ORACLE**: few claims, chosen from the truth so that Meek’s closure recovers as much as possible. The redesign had two reasons to drop it: the invariant, and “the old generator cascades unrealistically”, registered to be withdrawn unless the oracle set compels more edges per claim than an elicited set, paired by network.

Example 7.2 (Three sources on **child**). The 7B keeps one claim, the false **C02** → **LungParench** of Chapter 6, and the closure orients 10 edges the CPDAG left open. The oracle set asserts 2 and orients 12; the 72B asserts 8 and orients 9. Per claim: 10, 6 and 1.125. The same edge oriented truthfully, **LungParench** → **C02**, orients 2 (**results/ledger/rows.csv**, columns **n_k** and **k_g0**). On this network the 7B cascades most.

The quantity is *closure per claim*, $k_{G_0}/|K|$: edges the analyst’s graph orients that $\hat{C}$ left open, asserted or propagated, over kept claims. It is constant within a network and an arm, so the test pairs arms by network (**experiments/cascade_test.py**).

Cascade test, **results/ledger/cascade_test.json**

Oracle – 7B: +1.13 [−0.26, +2.86] on 27 networks, larger on 14: **not separated**. Oracle – 72B: +1.28 [+0.43, +2.50]; oracle – Llama 70B: +1.62 [+0.32, +3.27]; both separated. Truthful orientations on the 7B’s edges – 7B: −0.63 [−1.52, −0.02]. Network means: oracle 2.97, 7B 1.84, 70B 1.35, 72B 1.15.

The justification is withdrawn as written (**results/ledger/CHECKS.md**). Figure 7.2 shows why “unrealistic” had no fixed referent: the 7B declines a lot, keeps few claims, and each drags others along, as the oracle’s minimal set does; the large models assert almost every edge of a component, and then the closure has little left to add.

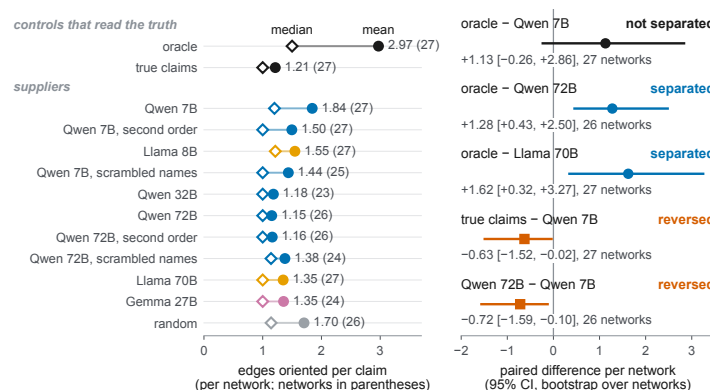


Figure 7.2: Whether the old generator “cascades too much” depends on the supplier it is compared with: the oracle set does not separate from the 7B supplier in edges oriented per claim, and does separate from the 72B and 70B suppliers, which assert many claims that propagate little. The redesign keeps only the invariant as its reason.

Without looking (answer 33, in the appendix)

Explain without contradiction why the same oracle set does not cascade more than the 7B supplier but does cascade more than the 72B.

Do not move on

Write the chain from “the 72B keeps 281 claims” to “its set is invalid on 0.298 of certifiable rows”. If the step from more claims to more invalid sets does not come, reread Section 7.2.

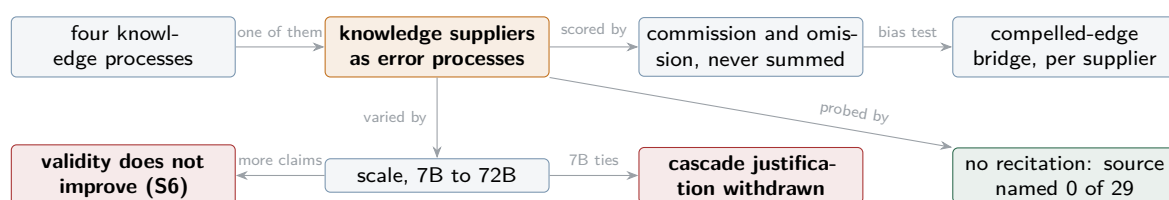
Exercise 7.1. From *TABLE4.md* and *knowledge_audit.json*, the six real-name suppliers at the first order. (a) Within the Qwen family, does the invalid-set rate fall with size? (b) Which column explains gemma? (c) Check that the first four columns account for all 369 pairs for the 72B and for Llama 70B. (d) Which supplier is “best”, and why does that question not belong in the paper?

supplier	kept	dropped	declined	not reached	commission	certifiable	invalid
Qwen 7B	171	43	137	18	0.325	229	0.288
Qwen 32B	223	26	83	37	0.271	204	0.201
Qwen 72B	281	24	27	37	0.233	285	0.298
Llama 8B	223	41	92	13	0.308	243	0.230
Llama 70B	282	78	9	0	0.259	267	0.247
gemma 27B	194	68	14	93	0.324	263	0.308

Solution. (a) No: 0.288, 0.201, 0.298, in no order, and Table 4’s intervals overlap. (b) “Not reached”: gemma leaves 93 pairs unanswered, drops 68 in the repair and keeps a commission as high as the 7B’s. (c) 72B: $281 + 24 + 27 + 37 = 369$. Llama 70B: $282 + 78 + 9 + 0 = 369$. (d) None: the suppliers are error sources for testing instruments, and what the paper uses is that validity does not improve with scale.

Exercise 7.2. Model a certifiable row as depending on m claims, each false independently with probability equal to the supplier’s commission c , so the set is invalid with probability $1 - (1 - c)^m$. (a) Estimate the effective m that reproduces the invalid-set rate for the 7B ($c = 0.325$, rate 0.288) and for the 72B ($c = 0.233$, rate 0.298). Use $\ln 0.675 \approx -0.393$, $\ln 0.712 \approx -0.340$, $\ln 0.767 \approx -0.265$, $\ln 0.702 \approx -0.354$. (b) Compare the ratio of the two m with the ratio of pairs the two suppliers kept. (c) Name two reasons the model is too crude to report.

Solution. (a) 7B: $m = \ln 0.712 / \ln 0.675 \approx 0.340 / 0.393 \approx 0.86$. 72B: $m = \ln 0.702 / \ln 0.767 \approx 0.354 / 0.265 \approx 1.33$. (b) $1.33 / 0.86 \approx 1.5$, against $281 / 171 \approx 1.6$ kept claims: the rise in effective claims per query is about the rise in claims, which is the counting mechanism in numbers. (c) Errors are not independent (one answer, one repair, clustering by network); validity depends on the claims on the query’s paths, not on a random m ; and the rates come from different certifiable rows, 229 and 285. An order-of-magnitude check, not a result.

Chapter map

Chapter 8

Table 4: the claim unit never over-certifies, the hop unit does

Table 4 is the audit that closes the protocol run. On every certifiable row it asks each instrument one question: when you promised that $r - 1$ revisions were safe, had the analyst already made more wrong claims than that, on a set that was invalid? This chapter reads `results/ledger/TABLE4.md`, shows why the claim radius's zero is a consequence and not an estimate, and tells what the pre-registered inspection trigger found on 12 September. Three findings touch code in your repository; I state each one as exactly as I can.

Before reading

- A radius promises that $r - 1$ revisions are safe. Against which two facts, known only to the truth, does the audit check that promise?
- On the `child` row of this section, how many claims and how many covering hops separate G_0 from the first state where the committed set fails?

8.1 The ledger, five instruments and four buckets

The idea in one sentence

A radius is a promise: you may be wrong in up to $r - 1$ of your moves and the adjustment set still holds. The audit checks it against the two things only the truth knows, the number w of asserted claims that are false and whether Z is valid; two yes-or-no answers give four buckets.

The ledger crosses the frozen frame of Chapter 6 with every knowledge condition of Chapter 7.

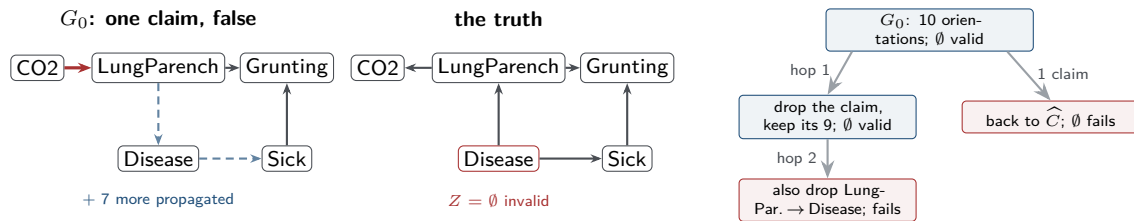
The ledger, `results/ledger/sweep_summary.json` and `TABLE4.md`

528 frame rows on 27 networks, 14 knowledge conditions, 8,308 evaluations by `experiments/ledger_sweep.py`. The eleven deployment conditions that assert knowledge (the 7B primary supplier with its order and scrambled-name replicates, the 8B second family, six conditions on four larger checkpoints, chance with three draws) certify **3,014** rows; no knowledge certifies 68, none invalid at the truth; the two audit conditions, which read the truth, 503. The primary supplier D_LLM (qwen2.5 7B, real names) certifies 229 of 528; on 155 its own graph reverses the query, and 144 are not amenable.

Every CPDAG in the ledger is the true DAG's (stamp `via_Chat_only`); Chapter 9 removes that idealisation. Five instruments answer from observables alone.

free rule (B2)	hop distance inside X 's chain component to the nearest member of Z ; 1 when none is there
closure count k_{G_0} (B4)	orientations of G_0 absent from $\hat{C}$, asserted and propagated
claim count $ K $ (B5)	number of claims
hop radius r_{hop} (B6)	your r_{val} as <code>hybrid.breakdown_radius</code> computes it: fewest covering hops up from G_0 to a failing state; a floor when the walk is budgeted
claim radius r_{claim} (B7)	fewest claims whose retraction and re-closure make Z fail, exhaustive to depth 3 under 300 subsets; a floor when censored

Example 8.1 (`child`, `LungParench` $\rightarrow$ `Grunting`, primary supplier). One claim, `CO2` $\rightarrow$ `LungParench`, and it is false ($w = 1$). Meek's rules turn it into ten orientations: `LungParench` $\rightarrow$ `Disease`, seven edges out of `Disease` (`Sick` among them), `LVH` $\rightarrow$ `LVHreport`. In G_0 `Disease` is a mediator and the committed set $Z = \emptyset$ is valid; in the truth `Disease` causes both `LungParench` and `Sick`, and $\emptyset$ leaves that path open.



Retracting the claim returns to $\hat{C}$, where $\emptyset$ fails: $r_{\text{claim}} = 1$. The hop walk moves one orientation at a time; its only first step keeps the nine propagated orientations, a closed state nobody asserted, and only the second step breaks Z : $r_{\text{hop}} = 2$. Free rule, closure count, claim count, hop and claim radius certify 1, 10, 1, 2, 1 (`results/ledger/rows.csv`); covering $w = 1$ needs $r \geq 2$, so the closure count and the hop radius promise safety on a broken set.

Every instrument is scored against w counted in claims, since claims are what the analyst asserted. In its own unit the hop radius is right; the danger is hearing «one revision is safe» as «one wrong claim is safe», when undoing this one wrong claim takes ten orientations.

The buckets, `experiments/ledger_sweep.py`, function `bucket`

An instrument certifies r on a row with w false claims. It *covers* the row iff $r = \infty$ or $w \leq r - 1$. **dangerous**: covers, Z invalid at the truth; **held**: covers, valid; **safe**: does not cover, invalid; **slack**: does not cover, valid. Dangerous rate over certifiable rows; detection = safe over invalid rows; false alarm = slack over valid rows. A budgeted search enters with the floor it certifies (`claim_lower_bound`), never as ∞ .

Dangerous is the verdict an analyst cannot recover from; slack is a refused move that was fine. So validity is read as absolute and sharpness only among safe instruments, as one reads a conformal method: coverage first, size second.

Without looking (answer 34, in the appendix)

Explain to yourself, then check: why can a row on which no retraction breaks Z ($r = \infty$) only be held or dangerous, and what would a dangerous row of that kind mean for the claim radius?

Before reading

- Is the primary supplier better than chance at what it asserts, and is its committed set invalid less often?

- On how many deployment rows is each of the free rule, the hop radius and the claim radius dangerous, and over which population?

8.2 What Table 4 says

Table 4, blocks 1 and 2, results/ledger/TABLE4.md

Primary supplier, 229 certifiable rows: false claims over asserted **0.329** [0.184, 0.494]; set invalid at the truth **0.288** [0.155, 0.435], 66 rows. Chance on the same edges, 649 rows: 0.479 [0.398, 0.559] and 0.378 [0.295, 0.460]. Paired on 24 networks, chance minus supplier, post hoc: +0.143 [+0.006, +0.287] and +0.082 [−0.020, +0.176]. Network cluster bootstrap, $B = 4000$. The false-claim rate here is over the 229 certifiable rows on 27 networks; Chapter 7 quotes 0.325 [0.210, 0.449], the network mean over all kept claims on 32 networks (results/elicited/knowledge_audit.json). Both are right for their population.

The supplier beats a coin on what it says, in a contrast declared afterwards; on whether the set breaks the difference is not shown. L4 asked for separated intervals and did not get them, so the abstract leads with the identifiability step (68 certifiable rows without knowledge, 229 with it) and with the dangerous counts, each of which needs its population.

Dangerous rows, TABLE4.md ladder block, recounted from rows.csv

	primary supplier	five conditions, 12 Sep	eleven conditions	audit conditions
certifiable rows	229	1,524	3,014	503
free rule	0	15	15	0
hop radius	4	29	39	0
claim radius	0	0	0	0

The five conditions of 12 September: primary supplier, order replicate, second family, scrambled names, chance. There the closure count is dangerous on 0.22 to 0.34 of rows, the claim count on 0.17 to 0.34. The audit zeros hold by construction ($w = 0$, true claims keep Z valid) and never enter a sentence about instruments.

On the primary supplier the free rule and the claim radius are both safe, so what separates them is sharpness. Figure 8.1 shows it bucket by bucket, and the paired ladder measures it.

The paired ladder, TABLE4.md, last block

False alarm, claim radius minus the other instrument, difference of network means on the same rows. Against the free rule: primary supplier **−0.206** [−0.380, −0.053] on 25 networks, radius wins; order replicate −0.188, scrambled −0.246, chance −0.308, wins; second family −0.129 [−0.294, +0.038], underpowered; the six scale-axis conditions −0.19 to −0.27, six wins. Against the hop radius: primary supplier **+0.144** [+0.025, +0.297], radius loses; +0.12 to +0.19 on the other four conditions of 12 September.

Appendix A.2 of results/ledger/PREREGISTRATION.md writes the abstract from one sentence: the free rule is safe and blunt, the hop radius sharp and unsafe, the claim radius safe and sharper than the safe alternative. Its errata of 13 September qualify «safe» for the free rule, strict only on the primary supplier and the scale axis and dangerous on 3 to 6 rows in each of the other four conditions.

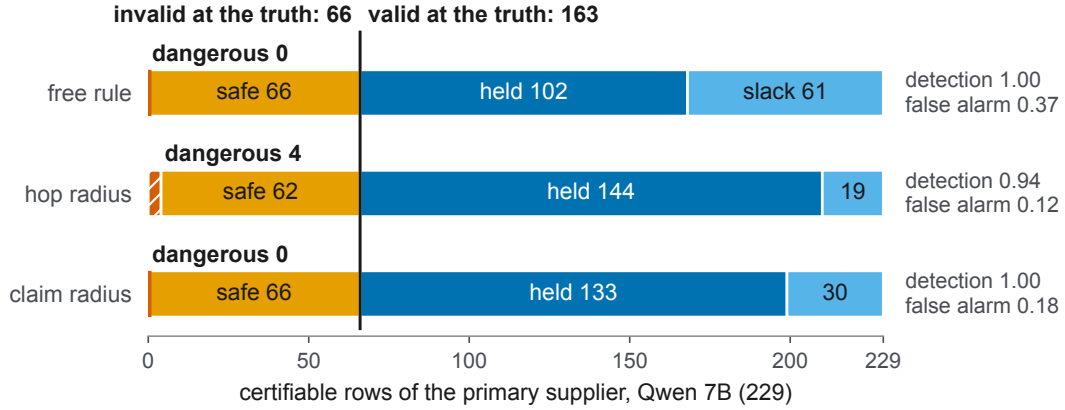


Figure 8.1: Safety is read left of the rule and sharpness right of it. Of the 66 sets invalid at the truth, only the hop radius leaves any in the dangerous bucket. Of the 163 valid sets, the claim radius refuses a move on 30 and the free rule, equally safe on this supplier, on 61.

Without looking (answer 35, in the appendix)

Is the primary supplier better than chance? Answer on both measures with the numbers, and say what the pre-registration asked.

Before reading

- Why does the claim radius's zero get no confidence interval?
- Do the claim radius and the hop radius in the ledger decide validity with the same criterion?

8.3 Why the zero is structural, and what the hop rows are

More rows would not tighten the zero. It follows from one move: retracting exactly the false claims.

Proposition 8.1 (The claim radius cannot over-certify, given a correct class). *Let Z be committed in $G_0 = \text{Meek}(\hat{C}, K)$, let certificate and audit decide validity with the same generalised adjustment criterion, and suppose the true DAG belongs to the class of $\hat{C}$. If Z is invalid at the truth and the false claims form $W \subseteq K$, $|W| = w$, then retracting W breaks Z . Hence $r_{\text{claim}} \leq w$ whenever the search reaches depth w , every floor certified below depth w is at most w , and the claim radius never covers w .*

Proof. The claims in $K \setminus W$ are true and the truth is in the class of $\hat{C}$, so the truth extends $\text{Meek}(\hat{C}, K \setminus W)$. Validity in an MPDAG is validity in every extension, and Z fails in this one. $\square$

The zero is what a sound criterion and an audit asking the same question must return, so it gets no interval; the rule-of-three floor Table 4 prints beside it (0.0131 for the primary supplier, $3/229$) would turn a consequence into a rate. Reading the radius as a minimal correction set needs failure to be upward-closed under retraction, your Theorem 1, and that was audited as implemented.

Monotonicity, experiments/monotonicity_audit.py

5,000 pairs (a failing retraction set, a random superset) on 469 rows of the elicited, truthful and oracle conditions: **0 violations**; one-sided 95 % bound 0.0006, treating pairs as inde-

pendent although they cluster within rows (`results/ledger/monotonicity_audit.json`). Proposition 1 of your draft rests on this theorem.

The hop radius's dangerous rows are where the two units part.

The unit gap, PREREGISTRATION.md A.0b and A.2, recounted from rows.csv

Of the hop radius's 29 dangerous rows on the five conditions of 12 September, **22** have $r_{\text{claim}} = 1$, $r_{\text{hop}} = 2$, one false claim and an invalid set, the shape of Example 8.1; of its 39 on all eleven conditions, 28. Leverage $r_{\text{hop}}/r_{\text{claim}}$ on the 1,390 rows of the 12 September ledger with both radii exact and finite: median 1, above 5 on 8 rows, maximum 12 (oracle) and 9 (deployment).

One confound had to be ruled out before quoting the gap, and I found it only while writing this chapter.

Two criteria inside one ladder

In the ledger the claim radius decides failure with the generalised criterion, `is_gac_valid_mpdag` in `src/bkrobust/gac/mpdag_level.py`. The hop radius, through `breakdown_radius`, uses `is_valid_mpdag` in `src/bkrobust/mpdag_criterion/criterion.py`, which carries the back-door strengthening. Your THEOREMS.md §13 gives the direction, $r_{\text{val}}(\text{GAC}) \geq r_{\text{val}}(\text{back-door})$, so a GAC hop radius would be dangerous on at least the same rows and raise no more false alarms. Both hop conclusions are conservative; the columns should still share one criterion before the paper sets them side by side.

For your draft, this is the measurement behind the abstract's distance that «counts elementary revisions of the analyst's own claims». An elementary revision is a covering step, which on every case checked changes exactly one orientation, propagated or asserted alike (THEOREMS.md, Lemma R). Read as a count of wrong claims, the hop radius over-certified on 39 of 3,014 rows where the claim radius never did. Chapter 14 takes up the unit the paper reports in.

The licensed form, for the audit table

On the 3,014 certifiable queries of eleven elicited and chance knowledge conditions, on the true CPDAG, the claim radius never certified more wrong claims than the analyst had made on an invalid adjustment set, which a sound criterion guarantees when the equivalence class is correct. Counted in covering hops, the radius did so on 39 queries, and the free hop-count rule on 15. Where the free rule was also safe, the claim radius raised fewer false alarms. Never: an interval on the zero, «robust», «in deployment», «beats a leave-k-out screen» (Chapter 11).

Without looking (answer 36, in the appendix)

State the move that makes the claim radius's zero structural, its premise, and why the rule-of-three column must not be quoted as its bound.

Without looking (answer 37, in the appendix)

In the ledger the two radii decide failure with different predicates. Which ones, and why can that not have produced any of the 39 dangerous hop rows?

Before reading

- What does the clause attached to L6 order when one dangerous row of the claim radius appears, and what does it forbid until then?
- A claim search stopped by its 300-subset budget: what did it certify?
- Which function in your repository is Pearl’s back-door criterion, and which test now pins where it differs from the generalised criterion?

8.4 The inspection trigger fired

L6 predicted no dangerous row for the claim radius, and the clause beside it said: one such row, halt and inspect it before reading anything else, because it is a bug or the result. The first complete table showed seven, six under the order replicate and one under the second family, and 29 for the hop radius (PREREGISTRATION.md, A.0b). The inspection found two bugs and one result.

Six rows: a floor read as infinity. All six are on `paths`, the 17-node textbook network (its chain component holds 15 nodes) chain, where the order replicate asserted 13 claims, all false. The search stops before exceeding 300 subsets: $13 + 78 = 91$, and size three would add 286, so it stops at `censored_at_depth_3`, certifying $r_{\text{claim}} \geq 3$. That floor covers two wrong claims against $w = 13$: safe. The first bucket function scored the censored radius as unbounded, hence dangerous. Every censored radius now enters at its floor, and a search that tried every subset without a failure is stamped `unreached` instead of `gt_d`. The six rows are still in `rows.csv`, now safe; the ledger holds 66 rows censored at depth 3 and 3 at depth 2.

One row: the audit asked a different question. On `ecoli70`, second family, query `yedE → b1963`: four claims, three false (`nmpC → pspA`, `pspA → cspG`, `pspA → pspB`), committed set `{atpD, pspA}`. The certificate found no break up to depth three; the audit called the set invalid. By the proposition both cannot hold, since the three false claims were a subset the search tried. An independent implementation of the generalised criterion on `networkx` sided with the certificate. The audit had used `is_valid_adjustment_set_dag` (`src/bkrobust/demo/evaluate.py`), which is Pearl’s back-door criterion and says so in its docstring: it forbids every descendant of the treatment, and `pspA` descends from `yedE` on no causal path to `b1963`. Your `THEOREMS.md` §13 and §14 already state the relation (the generalised criterion accepts a superset and coincides on O^*). The error was ours, in choosing that function for the audit.

The oracle, PREREGISTRATION.md A.0b, recounted from rows.csv

On the 1,169 committed sets of the four elicited conditions of 12 September and the oracle condition, the back-door function rejected 4 that the generalised criterion accepts, each holding a descendant of the treatment off every causal path, and agreed on the other 1,165. On all 3,585 certifiable rows of the current ledger: 6, always in that direction.

Your function was not touched. The audit calls `is_valid_adjustment_set_gac_dag` in `src/bkrobust/benchmarks/audit.py`, which shares no code with the instrument, and keeps the back-door verdict as the column `z_backdoor_at_truth`. The guard test `tests/benchmarks/test_audit_oracle.py` pins the smallest case: $U \rightarrow X \rightarrow C \rightarrow Y$, $U \rightarrow Y$, $X \rightarrow A$, where $\{U, A\}$ is valid under the generalised criterion and rejected by the back-door function, while all three oracles agree on $\{U\}$, $\emptyset$, $\{C\}$, $\{U, C\}$. The 29 hop rows showed no defect: they are the unit gap, and they stayed. The sweep was rerun with the two fixes and nothing else.

A zero reached by fixing code is suspect by default. This one rests on checkable facts: the halt

clause predates the first row, each fix is recorded in A.0b with the row that triggered it and a reason independent of the outcome, the unwelcome hop rows were left alone, and the next fix (the severity label) was rerun and diffed, changing only that column on 113 of 5,140 rows.

The ladder that never returned

The first sweep hung on `andes` (223 nodes) inside the CP-SAT ladder: `add_failure` (`src/bkrobust/sat/failure.py`) loops, at each of the path's node positions, over every ordered node triple, about 2.4×10^9 iterations at 223 nodes against 1.2×10^7 at 60. The 5-second limit bounds the solver, not the build, which had not returned after twelve minutes; those rows were discarded unread (A.0). The hop walk is now exhaustive up to 12 retractable edges and depth 3 beyond, under 4,000 closures, the ladder runs only up to 60 nodes, and a budgeted walk enters the bucket as $r_{\text{hop}} \geq \text{depth} + 1$. The gate is a workaround; the build itself is unchanged.

Do not move on

Write the three facts that made the `ecoli70` row a contradiction: how many claims were false, how deep the search went, and what that implied for r_{claim} if Z were invalid. If one is missing, reread that paragraph before the exercises.

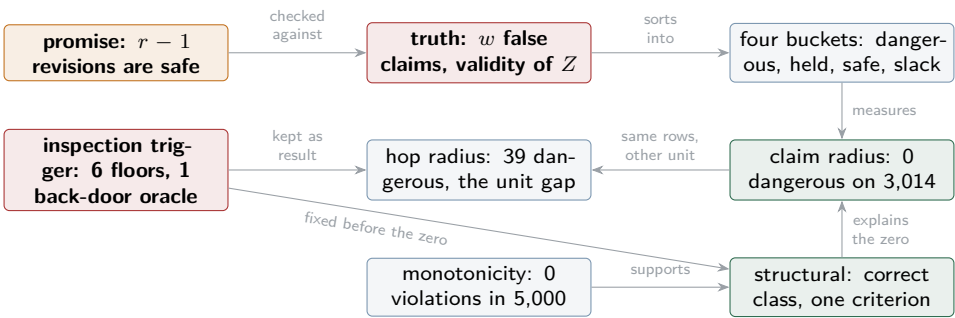
Exercise 8.1. *A row built for this exercise. Oracle CPDAG, claims a, b, c with a and b false; $r_{\text{claim_status}}$ is `unreached`; $z_{\text{valid_at_truth}} = 0$ and $z_{\text{backdoor_at_truth}} = 0$; the claim bucket reads `dangerous`. (a) Can either defect of 12 September explain it? (b) What should the search have found, and at which depth? (c) Name, in order, the three checks you would run and the function each calls. (d) Could this row be a result on an estimated CPDAG?*

Solution. (a) No: `unreached` means all $3 + 3 + 1 = 7$ subsets were tried, and both oracles call Z invalid. (b) Retracting $\{a, b\}$ leaves the true claim c , the truth extends $\text{Meek}(\hat{C}, \{c\})$, so Z must fail at depth at most 2. (c) `apply_orientations`($\hat{C}, [c]$), checking the truth is a consistent extension (the closure); `is_gac_valid_mpdag` on that graph against `is_valid_adjustment_set_gac_dag` on the truth (the two implementations of the criterion); the count of w in `audit_row`. On the oracle ledger it is a defect in one of these, never a result. (d) Yes: once the class may miss the truth, a dangerous row is a measurement (Chapter 9).

Exercise 8.2. *Table 4, scrambled names (`D_SCRAMBLE`D, 197 certifiable rows): free rule held 43, dangerous 3, safe 64, slack 87; hop radius held 99, dangerous 7, safe 60, slack 31; claim radius held 76, safe 67, slack 54. (a) How many rows are invalid at the truth? (b) Detection and false alarm of each. (c) The ladder gives claim minus free rule $-0.246 [-0.405, -0.103]$ and claim minus hop $+0.171 [+0.074, +0.279]$; compare with the pooled differences and state each verdict. (d) Why is the printed rule-of-three floor 0.0152 not the claim radius's bound?*

Solution. (a) Safe plus dangerous: $64 + 3 = 60 + 7 = 67$; valid 130. (b) Free rule 0.955 and $87/130 = 0.669$; hop radius $60/67 = 0.896$ and $31/130 = 0.238$; claim radius 1 and $54/130 = 0.415$. (c) Pooled -0.254 and $+0.177$, close to the paired values (network means first). The claim radius dominates the free rule (safe where it is dangerous on 3, and radius wins on false alarm); against the hop radius it is safe where the hop is dangerous on 7 but less sharp (radius loses). (d) $3/197$, the bound a zero would carry if it were a rate; this zero is a consequence of the proposition.

Chapter map



Chapter 9

When the class itself is estimated

Every row of Chapter 8 was computed on the CPDAG of the true DAG, which is what the stamp `via_Chat_only` says. The claim radius's zero rests on Proposition 8.1, and that proposition has a premise: the true DAG belongs to the class of $\hat{C}$. This chapter reads `results/pest/TABLE7.md`, the panel that drops the premise by learning the class from data. What it measured limits the scope of the paper's main claim. The substrate decisions behind Table 2 close the chapter.

Before reading

- Which algorithms, sample sizes and seeds did the panel use, and how was the reference class chosen?
- At $n = 200$, does a learned CPDAG leave more edges open than the oracle's?

9.1 The panel

The idea in one sentence

The claim radius answers how many of your claims can be wrong before the set breaks. It does not ask whether the graph the claims were made on is right, and it has no way to ask.

The invariant of Chapter 1 says the analyst sees $\hat{C}$, K , the query and the data. In the ledger $\hat{C}$ was the oracle's; in an analysis it comes out of a discovery algorithm. Only four networks in the corpus carry fitted Gaussian parameters from which data can be simulated: `ecoli70` and `arth150` (gene expression), `magic-niab` and `magic-irri` (plant genetics). P-EST uses those four.

The design was registered in `results/pest/PREREGISTRATION.md` on 12 September, after the ledger was read and before any discovery run: samples from each linear-Gaussian SEM at $n \in \{200, 1000, 5000\}$, ten seeds; PC with a Fisher- z test at $\alpha = 0.01$ and GES with BIC (`causal-learn`), a 900-second wall per run, GES on `arth150` limited to three seeds. The reference class $\hat{C}_E$ of each network is PC at $n = 5000$, seed 0, or the next seed if its Meek closure fails. On $\hat{C}_E$ everything was rebuilt: frame, questionnaire, the primary 7B supplier, sweep and audit. The truth draws the samples and audits the frozen rows, nothing else (`true_dag_on_path = false`). Fed the oracle CPDAG instead of $\hat{C}_E$, the pipeline reproduced all 420 comparable ledger rows with no difference in any field (A.0).

What discovery returned, TABLE7.md blocks 1 and 2, experiments/pest_table.py

219 runs, 7 at the wall. The Meek closure failed on 36 of 120 PC runs, 24 of which returned a graph that already held a directed cycle; the references of `arth150` and `magic-niab` fell back to seeds 1 and 5. Reference skeletons against the truth, edges missing and extra: 12 and 2 (`ecoli70`), 29 and 6 (`arth150`), 0 and 1 (`magic-niab`), 5 and 4 (`magic-irri`). Compelled edges pointing the wrong way: 12 of 52, 40 of 86, 14 of 62, 25 of 90. Open edges, estimate against oracle: 8/25, 41/45, 5/10, 11/12.

At $n = 200$ the two algorithms fail in opposite directions: PC leaves more edges open than the oracle on all four networks, GES returns a sparse graph that is almost fully oriented.

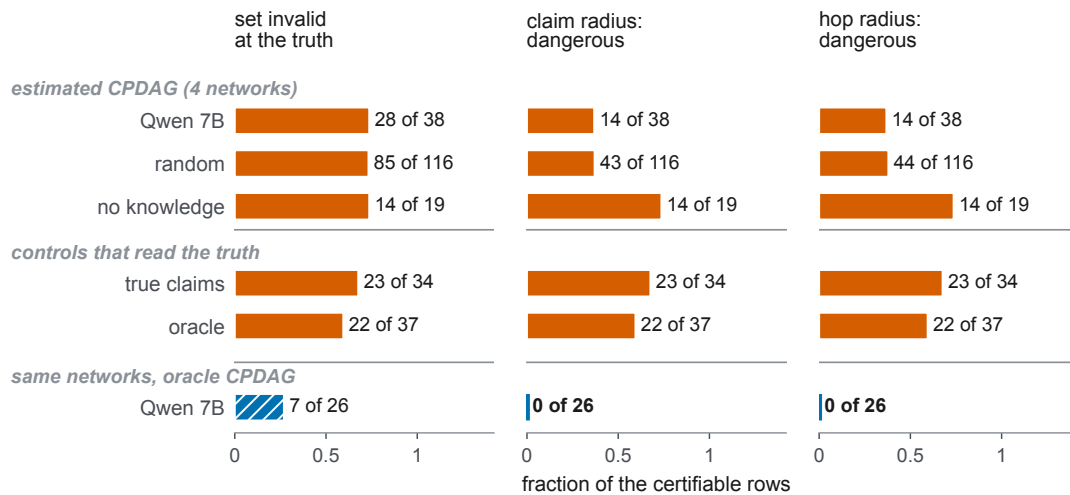


Figure 9.1: On the learned CPDAG the claim radius and the hop radius are dangerous on the same rows, and truthful claims do not rescue them. On the oracle CPDAG of the same four networks the supplier’s two radii are dangerous on none of its 26 rows. The error that breaks the set is in the learned graph, where no retraction of claims reaches it.

Without looking (answer 38, in the appendix)

At $n = 200$ GES leaves only 3 to 4 % of its edges open, far fewer than the oracle CPDAG. Why is that bad news for the analyst rather than good news?

Before reading

- On how many of the 244 certifiable rows does the truth extend the analyst’s graph?
- For the primary supplier, on how many rows is the claim radius dangerous, and the hop radius?
- Which network has no dangerous row on any deployment condition, and why?

9.2 What the panel measured

The number that decides is the premise itself, checked row by row: is the generating DAG a consistent extension of the analyst’s graph? On **0 of the 244** certifiable rows, summed over all conditions. Every reference skeleton differs from the truth, so Proposition 8.1 has no row to apply to.

Table 7, results/pest/TABLE7.md blocks 3 to 7, from rows.csv

Primary supplier: committed set invalid at the truth on **28 of 38** certifiable rows (0.737, against 0.269 on the oracle panel of the same networks). Claim radius dangerous on **14 of 38**; hop radius on the same 14 rows. Chance: 43 and 44 of 116. No knowledge: 14 of 19. Truthful claims on the supplier’s edges: 23 of 34. On the oracle panel of the same four networks, neither radius is dangerous on any row of any condition. Primary supplier by network: 6 of 8 (ecoli70), 5 of 7 (arth150), 0 of 14 (magic-niab), 3 of 9 (magic-irri).

Figure 9.1 sets the two panels side by side. Without knowledge there is no claim to retract, the certificate reports no break, and every invalid set becomes dangerous. The truthful condition asserts nothing false and is still dangerous on 23 of 34 rows, because the error is in the graph. In the ledger the two radii differed by the unit; here both reason inside the wrong class and certify

the same broken sets, and over the whole panel they differ on one row.

The certifiable share rose instead of falling, which falsified P2: the primary supplier certifies 38 of 80 frame rows (0.475) against 26 of 80 (0.325) on the oracle panel. The learned graph orients more than the truth does, and identifiability is bought with wrong orientations. **magic-niab**, whose reference misses no skeleton edge and adds one, is the only network where neither the supplier nor chance is ever dangerous, and all their invalid sets (6 and 14) are detected. The audit conditions are still dangerous there on 3 and 4 rows, because 14 of its 62 compelled edges point the wrong way.

Without looking (answer 39, in the appendix)

magic-niab is the one network with no dangerous row on the deployment conditions. Why does that confirm the scope limit rather than weaken it, and why are the audit conditions still dangerous there?

Before reading

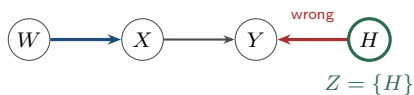
- Name the three mechanisms the inspection found on the dangerous rows of the supplier and of no knowledge.
- Why can no retraction of claims repair an edge the data got wrong?

9.3 Why no retraction reaches the error

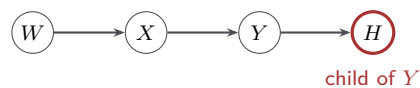
P3 carried the ledger’s decision rule, and it fired: the six dangerous rows of the supplier and of no knowledge that passed its test (three queries, two networks) were inspected before anything else was read. No bug: each committed set is valid on the analyst’s graph and contains a node the truth forbids.

Example 9.1 (A true claim, a dangerous row). The data suggest a collider that does not exist: the learned CPDAG compels $X \rightarrow Y \leftarrow H$ and leaves $W - X$ open. The analyst asserts $W \rightarrow X$, which is true.

analyst’s graph G_0 : learned class + one true claim



the truth



On G_0 the optimal set for $X \rightarrow Y$ is $\text{pa}(Y) \setminus \{X\} = \{H\}$, and it is valid. Retracting $W \rightarrow X$ reopens $W - X$ and leaves $H \rightarrow Y$ alone, since no claim created it, so $\{H\}$ stays valid and the certificate reports no break. At the truth H is a child of Y , hence forbidden, and $\{H\}$ is invalid. The row is dangerous with no false claim at all.

The measured rows follow the same pattern with real variables. On **magic-irri**, for the effect on FT, $\hat{C}_E$ compels $HT \rightarrow FT$ and $YLD \rightarrow FT$ against the truth, so both variables, descendants of the outcome in the truth, enter the set; that is the example above. The other two mechanisms are missing edges: on **magic-irri**, for the effect of **GTEMP**, $\hat{C}_E$ lacks **GTEMP** – **GW**, so **GW**, which the truth forbids, enters the set; on **arth150**, $\hat{C}_E$ lacks three of the treatment’s edges and 11 members of the set are forbidden at the truth (**PREREGISTRATION.md**, A.1, P3).

The inspection also showed that P3 had been registered on the wrong test: it looked only at the claims and the query, and an error in the skeleton or in a compelled edge passes it. The exact test, **truth_extends_g0**, was added after the first table and is marked post hoc; the frozen instrument rows were byte-identical across both runs. Written with its premise in view, Proposition 8.1 says:

if the true DAG belongs to the class of $\hat{C}$, retracting the false claims lands on a graph the truth extends. Without that clause nothing follows, and on this panel the clause held on no row.

Without looking (answer 40, in the appendix)

State the structural property of the claim radius with the premise P-EST removed, and say on how many certifiable rows of the panel the premise held.

Before reading

- Which of P1 to P5 turned out vacuous, and why?
- What is the licensed sentence for this panel, and which sentence does it forbid?

9.4 Predictions, and the sentence the paper can print

	prediction	verdict	what decided it
P1	estimation leaves more open than the oracle, less as n grows	not supported as registered	holds for PC (non-increasing on three networks); GES at $n = 200$ is sparse and almost fully oriented
P2	the certifiable share falls	falsified	0.475 against 0.325
P3	no dangerous row where the truth extends the graph	falsified as operationalised; premise held on no row	truth extends G_0 on 0 of 244; the inspection found no bug
P4	wrong-skeleton rows are counted, not dropped	partly supported	nothing dropped; one of the two statuses never fires
P5	realised bias larger at certified floor 1	supported as registered, not robust	0.110 against 0.063; medians reverse

P5 is supported only as pooled means (absolute bias of the OLS estimate 0.110 on 71 rows at floor 1, 0.063 on 102 at floor 2 or more, deployment conditions): the medians reverse, and so do the means without `magic-irri`. On an invalid set the bias is structural, so the certified floor is not a bias proxy on this panel.

The licensed form, docs/PROTOCOL_RUN.md, stage 6

The claim radius certifies against error in the analyst's claims given a correct equivalence class. It does not certify against discovery error, which on these four networks at $n = 5000$ invalidates more committed sets than the claims do. It goes in the main text (one results paragraph, one sentence in the limitations, Table 7 in the appendix), as `results/pest/PREREGISTRATION.md` A.2 says. Never «robust under an estimated CPDAG», for the claim radius or for the hop radius.

The wrong use of this panel is the usual limitations sentence, and your draft has one.

A scope limit, not a softened idealisation

Section 7 of your draft says: «The skeleton and the v-structures are treated as given . . . Real discovery on finite data returns a different and usually sparser skeleton, and a missing weak edge is precisely the error that breaks back-door blocking; nothing here speaks to it.» The mechanism is right, and it is one of the two the inspection found; the other is a compelled edge pointing the wrong way. «Nothing here speaks to it» is no longer true: P-EST measured that the guarantee fails, not an idealisation to relax later. A deployment claim for either radius needs a certificate that also retracts the discovery output; the panel sizes that gap

and does not close it.

Without looking (answer 41, in the appendix)

P5 came out supported as registered: mean bias 0.110 at certified floor 1 against 0.063 at floor 2 or more. Give the three readings that stop anyone calling the floor a bias proxy.

Before reading

- How close to one are var-sortability and R^2 -sortability on the four Gaussian networks?
- What exactly is the **sachs** network in the repository, and what word must the paper never use for it?

9.5 The substrate decisions behind Table 2

Table 2 describes the corpus network by network (`results/substrate/TABLE2_SUBSTRATE.md`). Four decisions were closed on 12 September in `results/substrate/DECISIONS.md`, and the first concerns the four networks of this chapter.

Sortability, `results/substrate/sortability.json`

`experiments/sortability_probe.py`, $n = 5000$, ten seeds, definitions of Reisach et al. (2021, 2023) implemented from the papers. Var-sortability 0.62 (**arth150**), 0.71 (**ecoli70**), 0.59 (**magic-irri**), 0.33 (**magic-niab**); R^2 -sortability 0.25, 0.38, 0.61, 0.76. Seed dispersion at most 0.015; every mean within about one standard deviation of its population value.

None of the four sits near one on either criterion, and the two do not track each other (**arth150** has the second-highest var-sortability and the lowest R^2 -sortability). A learned CPDAG on these networks cannot be read off the variances, so the discovery errors of P-EST are not an artefact of an easy ordering; every other network is undefined, having no continuous parameters.

The other three decisions fix what three files are. **M-bias.txt**, dagitty's three-node ADMG, is refused as a DAG and expanded into **M-structure.txt**, five nodes with both latent causes explicit; once they are measured the trap disappears (the optimal set for (E, D) is $\{U_2\}$ either way), so it is a calibration fixture outside the frozen frame and the ledger. In **Didelez_2010** the node **S** is dagitty's **selected** indicator; it changes no orientation, and the 48 ledger rows with **S** as outcome are declared as a named stratum. **sachs** is byte-identical to bnlearn's file: the 17-arc network that Sachs et al. (2005) inferred, which differs from the 20-edge Tetrad ground truth by one reversal (**Plcg-PIP3**) and three added edges. Its CPDAG is fully undirected, and on that one edge five knowledge conditions answered in the map's direction and two in the file's. The commission count on **sachs** depends on the choice, and the paper names the object and never calls it the consensus network.

Without looking (answer 42, in the appendix)

What is **sachs** in the repository, how does it differ from the Tetrad ground truth, and why does the difference change a measured number?

Do not move on

In Example 9.1, point to the edge that makes the row dangerous and say why retracting $W \rightarrow X$ cannot reach it. If you pointed to the claim $W \rightarrow X$, reread the example before the exercises.

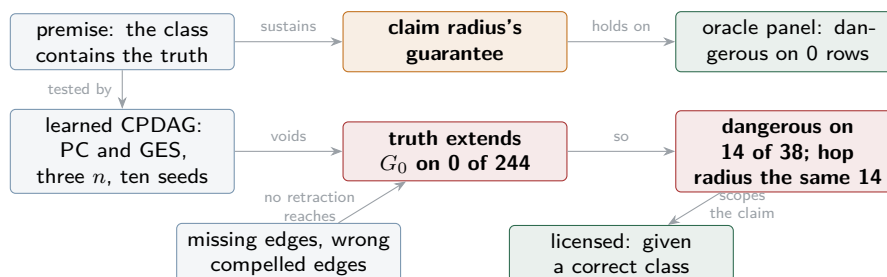
Exercise 9.1. *P3 was registered on rows whose claims all name true edges in the true direction and whose outcome is a true descendant of the treatment. A colleague argues that on such rows the claim radius cannot be dangerous, since every claim is true. Refute it with four nodes, using a missing edge rather than the wrong collider of Example 9.1: give the truth, the learned CPDAG, the claim, the committed set, and the bucket. Which measured mechanism does your construction reproduce?*

Solution. Truth: $W \rightarrow X, X \rightarrow Y, X \rightarrow C, C \rightarrow Y$, so C is a mediator. The learned skeleton misses $X - C$; with X and C non-adjacent, $X \rightarrow Y \leftarrow C$ is compelled as a v-structure and $W - X$ stays open. The analyst asserts $W \rightarrow X$, which is true, and Y is a true descendant of X , so the row passes P3's test. On G_0 the optimal set is $\text{pa}(Y) \setminus \{X\} = \{C\}$, valid because C is not a descendant of X there. Retracting $W \rightarrow X$ leaves $\{C\}$ valid, so the certificate reports no break and covers $w = 0$. At the truth C lies on the causal path $X \rightarrow C \rightarrow Y$, is forbidden, and $\{C\}$ is invalid: dangerous. It reproduces the missing-edge mechanism of GTEMP – GW on magic-irri; Table 7 counts 3 rows of the supplier that pass P3's test and are dangerous.

Exercise 9.2. *Rewrite the first two sentences of the paragraph quoted in the trap of this chapter («Assumptions inherited by every number», Section 7 of your draft) so that they carry what P-EST measured, in at most five sentences, in the paper's register: no percentages in parentheses on every clause, no forbidden phrase, and the premise stated as the scope of the certificate.*

Solution. One acceptable version: *The radius certifies against error in the analyst's claims given a correct equivalence class; the skeleton and the v-structures are treated as given. We measured what happens when that condition is dropped. On four networks with fitted Gaussian parameters we learned the class with PC from 5,000 samples and rebuilt every stage on it: no learned class contained the true graph, and the committed set was invalid at the truth on 28 of the 38 certifiable queries of the primary knowledge source. Both the claim radius and the hop radius over-certified on 14 of them, and truthful claims did not prevent it, because a missing edge or a wrong compelled orientation lies outside every retraction of claims. What changed: the premise became the scope of the certificate, «nothing here speaks to it» became a measurement, and «usually sparser» went, since the learned classes both lose and add edges and orient more than the truth.*

Chapter map



Part V

Pricing knowledge: the Rosenbaum analogue

Chapter 10

Rosenbaum's Γ , and Table 1 on real structure (M1)

Chapters 6 to 9 audited a certificate. This chapter is about what the paper is for, and about the first measurement that answers that question on real structures.

The paper in two sentences

Rosenbaum did not test what the data cannot test: he asked how strong hidden confounding would have to be to overturn the conclusion. We ask the same question of background knowledge, and the unit of violation is the number of the analyst's orientation claims that would have to be withdrawn.

Before reading

- The first measurement-paper draft of 12 September had correct numbers. What was wrong with it?
- What does Γ bound, in what design, and what does $\Gamma = 1$ mean?

10.1 12 September: the subject was wrong

On 12 September, with Table 4 of the ledger just out, I wrote a full draft of the measurement paper, *Claim Radius: Certifying Covariate Adjustment Sets Against Misspecified Background Knowledge*. Its numbers came from the result files, and its centre was the supplier ledger: knowledge suppliers from 7B to 72B, scrambled names, a scale axis as a table of its own. That was the wrong paper. The evaluation protocol had been built first, and its most elaborate arm had become the subject. What we set out to write is an uncertainty measure for background knowledge, the analogue of Rosenbaum's Γ for orientation claims. Elicited knowledge is one of four knowledge processes in the evaluation (with controlled reversals, chance orientations and truthful controls, the last asserting nothing false), never the title, the abstract's lead or Table 1. Your related-work paragraph (`sections/03_related_work.tex`) already has the sentence: the radius asks the structural analogue of how strong a confounder must be. The change is that this sentence becomes the spine of the paper.

What Rosenbaum does. In a matched study each pair joins a treated and a control unit with the same observed covariates; with perfect matching either member is equally likely to be the treated one. An unmeasured covariate breaks this. Rosenbaum does not estimate it: he lets the two members differ in their odds of treatment by at most a factor Γ , redoes the inference in the worst case the model allows, and reports the Γ at which the conclusion is lost.

Example 10.1 (Illustrative: the numbers are constructed for this page, not measured in any study). Take 40 discordant pairs with a binary outcome; in each, exactly one member had the event, and in 30 of them it was the treated one. With no treatment effect and $\Gamma = 1$, the count T of pairs where the treated member had the event is $\text{Bin}(40, 1/2)$ and the one-sided p-value is $P(T \geq 30) \approx 0.001$. With $\Gamma > 1$ the chance that the event member is the treated one may be as large as $p^+ = \Gamma/(1 + \Gamma)$, and the worst-case p-value is the upper tail of $\text{Bin}(40, p^+)$. For $\Gamma = 1, 1.25, 1.5, 1.6, 1.75$ it is 0.001, 0.009, 0.035, 0.053, 0.089 (exact binomial tails, computed for

this page). Significance at 5 % is lost near $\Gamma^* \approx 1.58$, and the report reads: the conclusion survives a hidden confounder that multiplies the within-pair odds of treatment by up to 1.58, and falls above that.

The example is the sign test, the simplest case of this model.

Definition 10.1 (Sensitivity model and sensitivity value). Let π_1, π_2 be the treatment probabilities of two members of a pair with the same observed covariates. The model with parameter $\Gamma \geq 1$ admits every assignment with

$$\frac{1}{\Gamma} \leq \frac{\pi_1(1 - \pi_2)}{\pi_2(1 - \pi_1)} \leq \Gamma.$$

Given that exactly one member is treated, the probability that it is the first lies in $[1/(1 + \Gamma), \Gamma/(1 + \Gamma)]$, and $\Gamma = 1$ is randomisation within the pair. The *sensitivity value* Γ^* is the smallest Γ at which the worst-case p-value over the model exceeds the level α .

Without looking (answer 43, in the appendix)

State what Γ bounds, why $\Gamma = 1$ is the randomised experiment within pairs, and why the worst-case p-value can only grow with Γ .

Before reading

- Which object of ours plays the worst-case p-value at each Γ , and which plays Γ^* ?
- Which of our two radii has no counterpart in Rosenbaum?

10.2 The translation, and where it breaks

Inside a Markov equivalence class every graph generates the same observational distributions, and Meek's rules certify that claims are consistent, not true: no test on the data separates an orientation from its reverse. That is the position of Rosenbaum's unmeasured covariate, and it takes the same answer, price the error instead of measuring it. The red rows are where the analogy breaks.

Rosenbaum: matched pairs		Our paper: orientation claims
an unmeasured covariate u within pairs equal on the observed x	what the data cannot test	the orientations in K inside the class: Meek checks consistency, not truth
$\Gamma \geq 1$: within-pair odds ratio of treatment	the parameter	$r = 0, 1, 2, \dots$: the number of the analyst's claims allowed to be wrong
the worst-case p-value over the assignments Γ admits	the report at each value	I_r : hull of the 95 % bands of the effects over the graphs of $B_r(K)$
Γ^* : smallest Γ at which significance is lost	the breakdown value	r_0 : smallest r at which the band contains zero
continuous: $\Gamma^* \approx 1.58$ is an ordinary report	the scale	an integer; on real structures r_0 is almost always 1 or beyond 3
a parameter of a model for treatment assignment	the reading	a count with no probability reading: it does not estimate how many claims are wrong
no counterpart	a second radius	r_{val} : smallest r at which the committed set Z stops being valid

In symbols, with $G_0 = \text{Meek}(\hat{C}, K)$ and the committed set $Z = O^*(G_0)$ read off the analyst's graph, the right-hand column is this.

The translation, formally

$$B_r(K) = \{\text{Meek}(\hat{C}, K \setminus S) : S \subseteq K, |S| \leq r\}, \quad I_r = \text{hull} \bigcup_{G \in B_r(K)} \text{band}_{95}(G),$$

$$r_0 = \min\{r : 0 \in I_r\}, \quad r_{\text{val}} = \min\{r : Z \text{ is not valid in some } G \in B_r(K)\},$$

and each endpoint of I_r comes with the set S that produces it, named in the domain's variables. Both events are upward closed in r because $B_r(K) \subseteq B_{r+1}(K)$.

A real analysis shows the translation end to end. The query $G311 \rightarrow G43$ on *magic-niab*, whose coefficients are fitted to real measurements, has two claims in the treatment's chain component under the larger supplier's knowledge, and both are true.

One report, results/interval/fig1_candidates.json, relaxed list (outside the registered rule), row 1

True effect -0.255 ; estimate -0.251 $[-0.259, -0.244]$ at $n = 20\,000$. The band barely moves at $r = 1$ (width ratio 1.06) and reaches zero at $r = 2$, when $G1853 \rightarrow G311$ and $G311 \rightarrow G43$ are both retracted: $r_0 = 2$ on 20 of 20 seeds, $r_{\text{val}} = 2$. The supplier made six claims in the network, three false, all outside the component.

Read as Rosenbaum would: the conclusion that $G311$ lowers $G43$ rests on two named claims and survives the withdrawal of either. Three breaks in the analogy have to be said once, in the paper. The scale is an integer and, on real structures, almost without a middle (next section). There is no probability model behind r : this row has $r_0 = 2$ with both local claims true and three false claims elsewhere, so r counts claims whose withdrawal changes the conclusion, not claims that are wrong. And the report is honest only if the class contains the truth, which fails once the CPDAG is estimated (Chapter 9).

Without looking (answer 44, in the appendix)

Complete the translation: Γ becomes what, the worst-case p-value becomes what, Γ^* becomes what, and which object of ours has no counterpart in Rosenbaum?

Before reading

- What do the rows of Table 1 have in common, and what do they differ in?
- With one wrong claim, how much does the stated report cover, and at what width does the one-claim interval restore coverage?
- What does an L_{95} above 1 mean?

10.3 M1: Table 1 on real structure

Before M1 the headline table existed only on synthetic generators, uncalibrated. Its pre-registration (`results/interval/PREREGISTRATION.md`, P1 to P9) was written on 12 September before `experiments/interval_panel.py` produced a row. A run takes about 90 seconds; five runs with 3 to 7 processes gave a byte-identical `rows.csv`.

Design. The frozen frame, 528 queries. Panel A: the 23 other networks of the ledger, real structure, semi-synthetic coefficients (magnitude uniform on $[0.5, 1.5]$, random sign, unit noise), 448 queries. Panel B: *ecoli70*, *arth150*, *magic-niab*, *magic-irri* with fitted coefficients, 80 queries. Knowledge: controlled reversals of $b = 0, 1, 2$ true claims inside the treatment's chain component (a reversal outside it moves no interval, a reduction tested against brute force on 7,770

pairs), and the 7B and 72B suppliers' elicited claims stratified by false local claims. Samples at $n = 1000$ and 20000, twenty seeds. **One estimator** in every row, the OLS coefficient of X given a set; rows differ only in the graphs the hull is taken over: the committed report (PLAIN), a matched control retracting one claim far from X and Y (FAR1), I_1 , I_2 , and the whole class (BLANKET, discarding the knowledge). The metric L_{95} stretches each procedure's intervals by the smallest common factor giving 95 % coverage in the cell and divides the mean stretched width by BLANKET's (Chapter 12 opens a cell). **Above 1 means wider than throwing the knowledge away.**

Table 10.1: M1, informative queries, $n = 1000$: L_{95} [network-bootstrap 95 % interval] / raw coverage. The three columns are different populations (see the rows line). Source: `results/interval/TABLE1_L95.md`.

procedure	$b = 0$		$b = 1$		$b = 2$	
	L_{95}	cov.	L_{95}	cov.	L_{95}	cov.
<i>Panel A, semi-synthetic coefficients, 23 networks</i>						
PLAIN (committed)	0.464 [0.35, 0.57]	0.967	5.123 [3.24, inf)	0.597	4.082 [2.71, inf)	0.554
FAR1 (control)	0.655	0.967	10.771	0.640	7.539	0.614
I_1	0.957 [0.87, 0.99]	0.982	0.851 [0.76, 0.94]	0.980	4.234	0.781
I_2	0.984	0.984	0.980	0.984	0.847 [0.74, 0.94]	0.984
BLANKET (class)	1.000	0.984	1.000	0.985	1.000	0.988
rows (networks)	391 (23)		359 (23)		131 (9)	
<i>Panel B, fitted coefficients, 4 networks</i>						
PLAIN	0.840 [0.76, 0.88]	0.969	3.316 [0.81, 4.59]	0.885	2.588	0.891
I_1	0.979	0.979	0.974 [0.94, 1.00]	0.980	0.887	0.947
I_2	0.997	0.980	0.989	0.983	0.945	0.979
rows (networks)	69 (4)		44 (4)		35 (4)	

On Panel A, with no wrong claim the committed report is the narrowest (0.464) and the one-claim interval pays a premium (0.957): insurance costs when nothing is wrong (P1). With one wrong claim the committed report covers 0.597 and would need 5.1 class widths to reach 95 %, while I_1 covers 0.980 at 0.851 of the class width. With two, I_1 is not enough (0.781) and I_2 pays. Figure 10.1 draws the crossing.

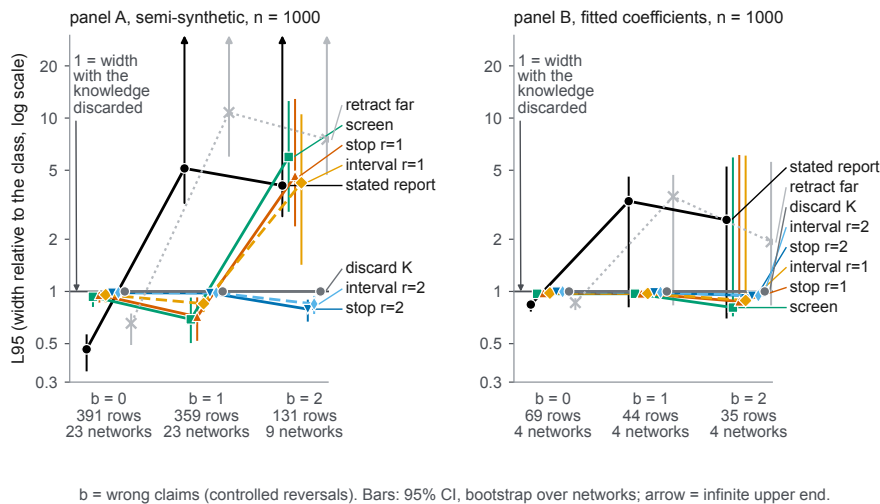


Figure 10.1: Panel A: with one wrong claim the committed report costs more than five class widths and retracting a far claim does not help; with two, only the budget-two procedures stay below 1. Panel B's four networks keep the order at smaller magnitudes. The columns are different populations, so the lines between them do not follow the same analyses.

The crossing and the saving, results/interval/TABLE1_L95.md and PREREGISTRATION.md A.2

Against discarding the knowledge, I_1 at $b = 1$ saves 14.9 % of the width at $n = 1000$ and 24.2 % at $n = 20000$ on Panel A (0.851 and 0.758), and 2.6 % and 4.8 % on Panel B (0.974 and 0.952). With errors nobody planted (elicited, one false local claim, suppliers pooled, $n = 1000$) the order repeats: PLAIN 0.600 at L_{95} 6.92 against 0.867 for I_1 on Panel A (208 rows), 0.696 at 2.41 against 0.901 on Panel B (23 rows). P1, P2, P4, P5, P7, P8, P9 supported; P3 partial; P6 falsified.

The saving is modest, and the paper should say so: the interval covers where the committed report does not, at less than the price of discarding the knowledge.

Without looking (answer 45, in the appendix)

Panel A, $n = 1000$, one wrong claim: give the coverage and L_{95} of the committed report and of I_1 , and say what 5.12 means.

Before reading

- Where an effect is established with the full knowledge, what values does r_0 take on real structures?
- How often does the hull of point estimates state a null radius above the population one, and how often does the band?

10.4 What M1 changed in the report

The translation promises a scale: how many wrong claims until the conclusion falls. On real structures the scale has almost no middle.

r_0 by knowledge condition, counted in results/interval/rows.csv ($r0_pop$)

Frame queries with an effect established under the full knowledge ($r_0 \geq 1$); “beyond 3” pools >3, **never** and censored. Panel A, true claims ($b = 0$): 221 queries, $r_0 = 1$ on 71, 2 or 3 on 16, beyond 3 on 134, so 1 or beyond 3 on 92.8 %. The same share is 88.8 % at $b = 1$, 74.2 % at $b = 2$ (66 queries), 93.6 % (7B), 94.1 % (72B) and 97.7 % under the recovering set (6 of 257 at 2 or 3). Panel B: 25 of 25 at $b = 0$, 33 of 38 under 72B. Restricted to informative queries the $b = 0$ share on Panel A is 148 of 164 (90.2 %): a non-informative query with a nonzero effect is “never” by construction.

Figure 10.2 shows it as a distribution. In the analyst's words, *the conclusion rests on no claim, or on one claim that can be named*. That changed decision D2 in the outline of 13 September: lead with the knowledge interval and Table 1, print r_0 beside it as “rests on no claim / on this claim” with its distribution, and retire the title candidate that named the null radius.

A reading trap: $r_0 = 0$ (227 of 391 informative Panel A queries under true claims, in `summary.json`) means there is no established effect to break, not a break at the first retraction.

The band, not the hull of point estimates, is the estimator (P7). At $n = 1000$, over 23,120 replicates of the controlled conditions, the plug-in hull states a null radius above the population one on 39.7 % of replicates and the band on 1.7 % (39.0 % and 1.8 % at $n = 20000$). A possible effect near zero lands on one side of it by sampling error, and the plug-in errs towards claiming robustness.

A confounding bound at its own oracle (M2, P9). The Cinelli and Hazlett bound, given the true partial R^2 of the block missing from Z , is defined on 666 of 808 committed-set queries

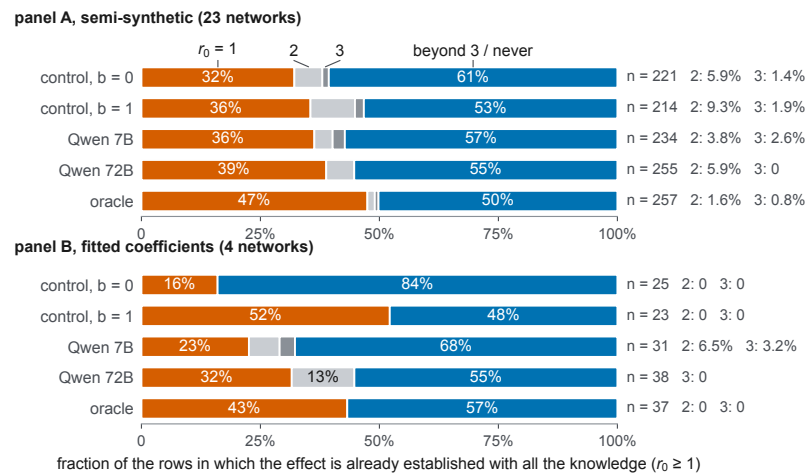


Figure 10.2: Under every knowledge source and on both panels, where an effect is established the conclusion rests on one claim or on none within depth three: the mass at 2 and 3 is a sliver, and even complete oracle knowledge does not spread it.

(on 101 the wrong claim put a true descendant into Z ; on 41 no valid set exists). There it is narrower than I_1 (Panel A, $b = 1$, $n = 20\,000$: L_{95} 0.445 against 0.632), being handed what the analyst lacks. On 102 queries whose set holds a true descendant and is invalid it covers 13.4 % at $n = 20\,000$: a wrong orientation is outside its model class.

Rosenbaum 2002; Cinelli and Hazlett 2020; Masten and Poirier 2020

Rosenbaum, *Observational Studies*, 2nd ed.: the chapter on hidden bias, where our report's form comes from. Cinelli and Hazlett, JRSS B 82(1):39–67: the bound M2 implemented. Masten and Poirier's breakdown frontiers, *Quantitative Economics* 11(1):41–111: a breakdown value with inference, outside graphs, not yet in your bibliography.

Table 2 came out thinner than planned. Seven of the nine declared queries have no claim in the treatment's component, so r_0 is “never” (`results/interval/TABLE2_REPORTS.md`); only `mediator` has a finite one (2). The informative reports are the 17 real-coefficient queries, mostly $r_0 = 1$ with the claim named (Chapter 12).

The analogy stated once, with the place it breaks

Following Rosenbaum, we report how many of the analyst's orientation claims would have to be withdrawn before the conclusion changes, beside the interval of effects those withdrawals leave possible. Unlike a sensitivity value, this count is an integer with no probability reading; on the real structures measured here it is almost always one or more than three, so the report names the claim on which the conclusion rests.

Without looking (answer 46, in the appendix)

Give the measured count that shows r_0 near-binary, with its population, and say what it changed in the way the paper reports.

Do not move on

Name the step where “significance is lost” becomes “the band contains zero”, and say why the band and not the plug-in hull, with the P7 numbers and their sample size.

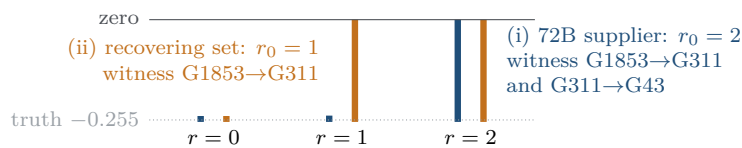
Exercises

Exercise 10.1. *Illustrative, population level. Local claims $K = \{a, b, c\}$, committed effect 0.80; effects each retraction leaves possible: $\emptyset$: $\{0.80\}$; $\{a\}$: $\{0.80\}$; $\{b\}$: $\{0.80, 0.35\}$; $\{c\}$: $\{0.80\}$; $\{a, b\}$: $\{0.80, 0.35\}$; $\{a, c\}$: $\{0.80, 0.00\}$; $\{b, c\}$: $\{0.80, 0.35, 0.00\}$; $\{a, b, c\}$ (the class): $\{0.80, 0.35, 0.00, -0.10\}$. Z is invalid after retracting b and valid after retracting a or c alone. (a) Compute I_0 to I_3 . (b) Give r_0 and r_{val} , with the witnesses. (c) What share of the class width does I_1 cover?*

Solution. (a) $I_0 = [0.80, 0.80]$; $I_1 = [0.35, 0.80]$ (from $\{b\}$); $I_2 = [0.00, 0.80]$ (from $\{a, c\}$ or $\{b, c\}$); $I_3 = [-0.10, 0.80]$, the class. (b) $0 \notin I_1$ and $0 \in I_2$, so $r_0 = 2$, witnessed by $\{a, c\}$ (and $\{b, c\}$); no single claim reaches zero, but a and c together do, an interaction. $r_{\text{val}} = 1$, witnessed by b . Here the set breaks before the sign does, as on **mediator** ($r_{\text{val}} = 1$, $r_0 = 2$). (c) $0.45/0.90 = 50\%$. The listed sets respect the pre-registered reduction “retracting more never removes an extension” (each superset retraction contains its subsets’ effects), which is what makes I_r non-decreasing.

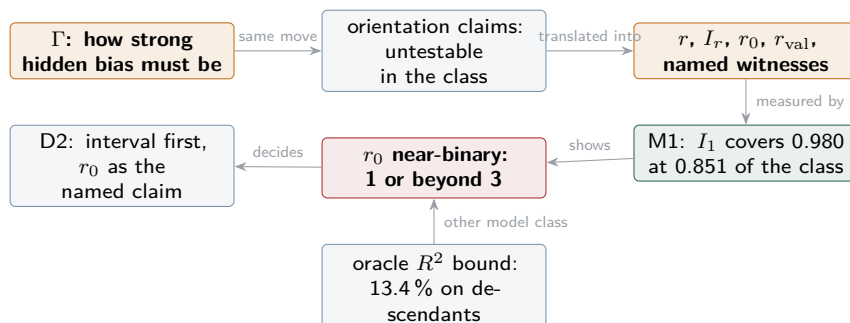
Exercise 10.2. *Draw the report for **magic-niab**, $G311 \rightarrow G43$ ($n = 20\,000$, seed 0), effect against $r = 0, 1, 2$, under two claim sets: (i) the 72B supplier’s two local claims, bands $[-0.260, -0.245]$, $[-0.260, -0.244]$, $[-0.260, 0.000]$; (ii) the recovering set, whose only local claim is $G1853 \rightarrow G311$, bands $[-0.260, -0.245]$ at $r = 0$ and $[-0.260, 0.000]$ at $r = 1$ and $r = 2$ (**results/interval/profiles.jsonl**). Mark r_0 and its witness, and explain how one query has $r_0 = 2$ in one report and 1 in the other.*

Solution.



Both start from a point interval at $r = 0$, so both analyst graphs orient $G311 \rightarrow G43$. In (ii) that orientation is not asserted, so it is propagated from $G1853 \rightarrow G311$, and retracting that one claim admits the graph where $G43$ is a parent of $G311$ and the effect is zero. In (i) it is also asserted directly, so it survives either single retraction and falls only when both go. r_0 belongs to the claims around the query, not to the query, so a Figure 1 caption must name the claim set.

Chapter map



Chapter 11

M1b: the stop rule and the free screen tie

M1b in two sentences

A stop rule uses the radius to decide when to widen the report; a free screen (leave a claim out, re-close, see whether the set changes) decides almost the same thing without computing any radius. M1b measured whether the rule beats such a screen, and at matched depth it does not.

Table 1 of Chapter 10 compares intervals that always widen with a narrow committed report. It never shows the measure being *used*. The table design of 7 September had asked for exactly that row, and I had not reread it when the outline was rebuilt on 12 September. M1b is that row, run on the M1 harness in the night of 13 September, with five predictions registered in Appendix B of `results/interval/PREREGISTRATION.md` before any policy quantity was computed. M1's own cells did not move: projected onto M1's columns the new `rows.csv` has the same hash.

Before reading

- Which comparison did the 7 September design fix in advance as the only one a radius could win against a free statistic?
- What sentence was written, before measuring, for the case of a tie?

11.1 What the 7 September design asked for

Three conclusions of that design are the ones M1b tested. **Rows are complete procedures**, and the natural use of the radius in that format is a stop rule: report the committed interval when the set survives every retraction within the budget, widen only when it does not. **A free screen is the same policy at one claim**: retract one claim, re-close under Meek, flag if the committed set changes. An unchanged set under every single retraction is a set that stays valid under every single retraction, so at depth one the screen cannot certify what the rule refuses. The only place the radius could beat a free statistic is at two wrong claims, where two retractions interact; the design fixed that comparison in advance (the starred comparison) and wrote the caption for the null result before any number existed: the work would then be “a measurement of when knowledge error is silent rather than a method”. **The price of the guarantee is a cell of its own**: the share of correct analyses the rule refuses when the knowledge is right.

Two things called “stop rule” in my September notes

My notes of 12 September (in Portuguese) call the ledger's halt-and-inspect clause, the one that found the bucket bugs and the back-door oracle of Chapter 8, a *regra de parada*. Here that clause is the **inspection trigger**; the **stop rule** is the reporting policy STOP_r .

Without looking (answer 47, in the appendix)

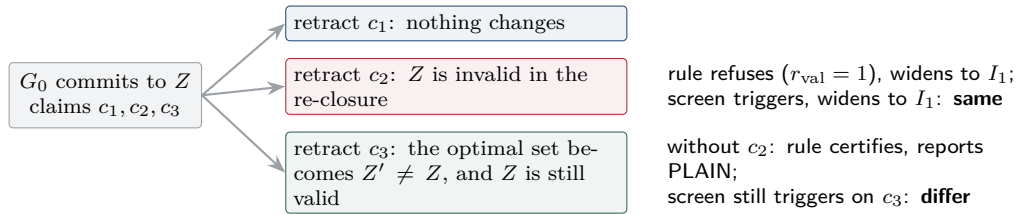
Which comparison did the 7 September design fix in advance, why only that one, and what sentence did it write for a tie?

Before reading

- In what class of cases do the stop rule and the one-claim screen decide differently?
- Why is “refusal implies trigger” an argument plus a measurement, and not a theorem?
- Why does comparing the two-claim rule with the one-claim screen mix two things?

11.2 Four policies, one argument, five predictions

Example 11.1 (Three retractions, two decisions (schematic)). G_0 commits to Z and holds three claims; what each single retraction does, and what each policy decides:



Where they differ, the screen widens without need and the rule is right to certify.

Appendix B fixed the policies before the run.

Definition 11.1 (The M1b policies). STOP_r , $r \in \{1, 2\}$: certify if G_0 commits to a set and $r_{\text{val}} > r$, then report PLAIN; otherwise report the band I_r . SCREEN: retract each claim, re-close, recompute the committed result; if any differs from G_0 's, report I_1 , else PLAIN. SCREEN2: the same over every retraction of one or two claims, widening to I_2 . The *committed result* is the set, or the verdict class when there is no set. Policy comparisons are scored on informative queries *with* a committed set; without one, the two rules read literally differ by construction.

One direction always holds: whenever the rule refuses, the screen fires.

Argument: refusal implies trigger, at the same depth

Premise P: every set the committed-set procedure returns for a graph is valid in that graph. If STOP_r refuses on a query with set Z , there is S with $|S| \leq r$ and Z invalid in $G_S = \text{Meek}(\hat{C}, K \setminus S)$. Either G_S commits to some Z_S , valid by P and therefore different from Z , or G_S commits to no set and the verdict changes. In both cases the depth- r screen triggers. The converse fails: the screen also triggers when the set changes and Z stays valid.

This is not a theorem proved here, and the paper should not present it as one. P is a property of the implementation (the closed-form optimal set, and one criterion both building the set and judging it, which the back-door oracle of Chapter 8 showed can fail). What carries it is the measurement: on all 2,212 policy rows every refusal at budget r is matched by a screen triggering at depth at most r , zero exceptions.

The predictions, written after M1 was read: Q1, rule and screen differ on at most 1% of committed queries at $b = 1$; Q2, at $b = 2$ on Panel A the two-claim rule covers more than the one-claim screen, with the interaction ceiling at most 0.31; Q3, at $b = 0$ the rule keeps the narrow report; Q4, it refuses fewer than half of correct analyses; Q5, it covers at least 0.95 at $b = 1$.

Without looking (answer 48, in the appendix)

State the premise on which “if the stop rule refuses, the screen triggers” depends, and say why the converse fails.

Before reading

- With correct knowledge, on what share of informative committed analyses does the stop rule refuse?
- What moves that number from 0.91 to 0.64, or to zero?

11.3 The price of the guarantee: 0.910

Refusal at $b = 0$, results/interval/TABLE1_L95.md, M1b section

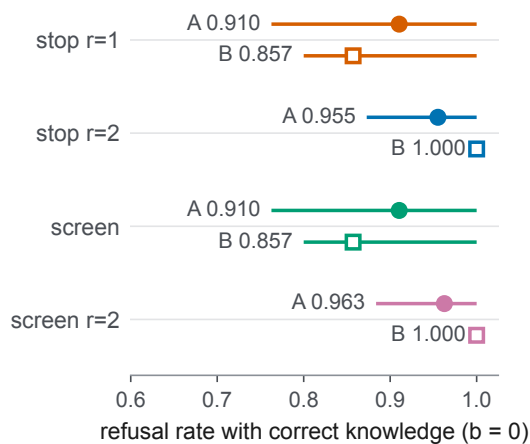
Share of informative queries with a committed set on which the policy widens, knowledge correct, network-bootstrap 95 % interval.

panel (queries, networks)	STOP ₁	STOP ₂	SCREEN	SCREEN2
A (134, 20)	0.910 [0.76, 1.00]	0.955	0.910	0.963
B (7, 3)	0.857 [0.80, 1.00]	1.000	0.857	1.000

Q4 failed, by a wide margin. With every claim true, the stop rule refuses nine informative committed analyses in ten. Where the knowledge carries identification, a single true claim, retracted, breaks the set: the same table reports $r_{\text{val}} = 1$ on 0.910 of these 134 queries, which is the refusal rate itself. The one-claim screen refuses on exactly the same queries. The left side of Figure 11.1 shows the four rates.

the price of the guarantee

rows with a committed set
A: 134 rows, 20 networks · B: 7 rows, 3 networks

**stop $r=2$ against the two screens**

brackets: paired difference per network
(stop minus screen), 95% CI

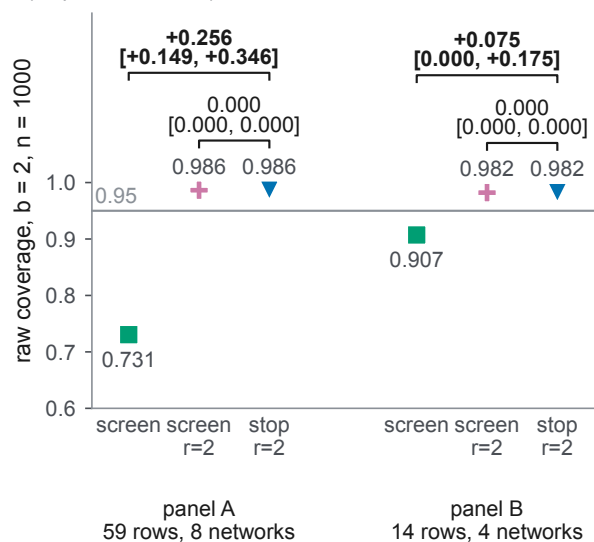


Figure 11.1: Left: with correct knowledge the stop rule and the one-claim screen refuse the same nine analyses in ten; the 0.91 does not measure how demanding the rule is, it measures that the conclusion rests on one claim. Right: at two wrong claims the two-claim rule covers more than the one-claim screen and exactly as much as the pair screen.

The denominator must travel with the number: on committed queries outside the informative

stratum the rule refuses none (57 on Panel A, 11 on Panel B), and over all committed queries it refuses 0.639 (A) and 0.333 (B). **Q3 holds on the letter and is empty:** $L_{95}(\text{STOP}_1)$ is below $L_{95}(I_1)$ in all four cells by 0.0003 to 0.0011, because read literally the rule widens on 97 to 99% of informative queries. **Q5 holds:** at $b = 1$ it covers 0.979 and 0.979 on Panel A ($n = 1000, 20\,000$), 0.980 and 0.966 on Panel B.

Without looking (answer 49, in the appendix)

Give the three refusal rates of STOP_1 at $b = 0$ on Panel A by denominator, and say which one Q4 was about.

Before reading

- How much more does the two-claim rule cover than the one-claim screen at $b = 2$, at which sample size, and on how many queries?
- Where does a third of that difference come from?
- Against the pair screen, what is the difference?

11.4 The starred comparison separates, then ties

The comparison fixed on 7 September, STOP_2 against SCREEN at $b = 2$ on Panel A (59 committed informative queries, 8 networks), separates as Q2 predicted: $+0.256 [+0.149, +0.346]$ in coverage at $n = 1000$ (0.986 against 0.731), $+0.269 [+0.163, +0.361]$ at $n = 20\,000$, and $+0.305 [+0.182, +0.410]$ in the population (1.000 against 0.695). The interaction ceiling is 0.237 (A) and 0.286 (B). Q2 is supported as written, and two facts qualify it.

Table 11.1: Where the $\text{STOP}_2 - \text{SCREEN}$ coverage gap comes from, Panel A, $b = 2$, $n = 1000$, committed informative queries. Source: `results/interval/TABLE1_L95.md`, M1b section.

stratum	queries	share of replicates	gap inside	contribution
interaction ($r_{\text{val}} = 2$, or a pair changes the result)	14	0.24	$+0.689$	$+0.164$
$r_{\text{val}} = 1$ (both policies refuse)	34	0.58	$+0.159$	$+0.092$
rest	11	0.19	$+0.005$	$+0.001$

About a third of the gap is the fallback, not the depth. On the 34 queries with $r_{\text{val}} = 1$ both policies refuse, but the rule reports I_2 and the screen I_1 , and with two wrong claims I_2 covers more: $+0.092$ of the $+0.256$ (Table 11.1). Only the 14 interaction queries measure depth. On Panel B (14 queries, 4 networks) the gap is $+0.075 [+0.000, +0.175]$ at $n = 1000$, all of it from 4 interaction queries, with an interval touching zero.

A free screen at depth two ties exactly. STOP_2 against SCREEN2, both falling back to I_2 : coverage $+0.000 [+0.000, +0.000]$ at $n = 1000$, $-0.002 [-0.003, +0.000]$ at $n = 20\,000$, zero on Panel B. They decide differently on 5 of the 754 committed informative queries of the five conditions ($141 + 162 + 73 + 161 + 217$), all of the example's class. The pre-fixed comparison separated only because its free baseline stopped at depth one.

Q1 failed by one query. At $b = 1$, rule and screen differ on 2 of 162 committed queries (1.23%), both of the expected class: `Kampen_2014`, ALN on PER, $Z = \{\text{AFF}, \text{SAN}\}$, $r_{\text{val}} = 3$; and `Schopf_2010`, A on S, Z empty, $r_{\text{val}} = 2$, where one retraction makes the closed form return a non-empty optimal set for which the empty set is still valid. Q2 and Q5 are supported, Q3 on the letter only, Q1 and Q4 falsified.

Without looking (answer 50, in the appendix)

Why does the +0.256 advantage of the two-claim rule over the one-claim screen not measure only the value of certifying at depth two? Give the sample size and the population line.

Before reading

- What does the pre-declared sentence turn the paper into?
- What survives M1b, measured?
- Which column is missing from the τ_b table of your draft?

11.5 What survives, and the column your ranking table lacks

Said without softening: **the sentence pre-declared on 7 September applies**. The starred comparison separated against the one-claim screen and tied against the pair screen; at no depth we measured does the stop rule compute something a leave- k -out screen does not. Nothing in the paper may suggest that the radius, the certificate or the stop rule beats such a screen.

The tie, in the licensed form

This is a measurement of when background-knowledge error is silent and what it costs. At every depth we measured, the validity half of the sensitivity report matches a leave- k -out re-closure screen, and the two differ only where a retraction changes the adjustment set while leaving it valid. We state this as a property of the measure; what the report adds is the depth that was needed and the named claims that witness it.

What survives is measured, each number with its population. One wrong claim costs the committed report 37 points of coverage (Panel A, $n = 1000$: 0.967 to 0.597; 42 in the population), and the knowledge interval restores coverage at 76 to 97 % of the class width ($L_{95}(I_1)$ at $b = 1$, 0.758 to 0.974 over panels and sample sizes). Where an adjustment depends on knowledge, it almost always rests on one claim (refusal 0.910; r_0 at 1 or beyond 3 on 92.8 % of Panel A's established effects), so the report's most useful output is the *name* of that claim; on the ledger (a different harness: primary supplier, 157 exact rows, oracle CPDAG) the named claim was false on 58 of the 62 queries whose set was invalid at the truth (**results/ledger/CHECKS.md**, Table 8 in Chapter 6). The outline of 13 September lists this framing as decision D9 and recommends accepting it rather than searching for a population where conclusions rest on more claims, which would be selected for the answer. I recommend the same.

The baseline your τ_b table is missing. The ranking table of your draft (**tab:tau** in **sections/06_simulation.tex**) sets $\tau_b(r_{\text{val}})$ against $\tau_b(\text{SHD})$ and $\tau_b(|K|)$, two weak baselines. A reviewer who knows the area proposes the screen: the depth at which leave- k -out re-closure first changes the set. By the argument above it never exceeds r_{val} and equals it except where the set changes while staying valid (5 of 754 queries in M1b). The column is unrun; your corruption states are stored, so it is cheap. I expect $\tau_b(\text{screen depth})$ close to $\tau_b(r_{\text{val}})$; that is a prediction, not a measurement.

Before the ranking table goes into the merged paper

Add $\tau_b(\text{screen depth})$ on the same stored states and bootstrap. If it ties, state the ranking as a property shared with the screen and say what the radius adds: the depth, the witness, the certificate that no failure lies nearer.

Do not move on

Rebuild the decomposition of the starred comparison: queries, contribution and mechanism per stratum. If you cannot say why the 34 queries with $r_{\text{val}} = 1$ contribute although both policies refuse there, reread the paragraph on the fallback.

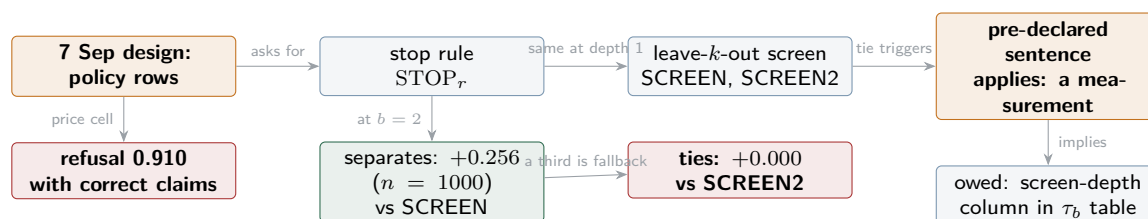
Exercises

Exercise 11.1. Before looking at the refusal table, you are given only the identity table for correct knowledge ($b = 0$), Panel A, 134 informative committed queries: $r_{\text{val}} = 1$ on 0.910 of them and $r_{\text{val}} = 2$ on 0.045; no query differs between STOP_1 and SCREEN . (a) Estimate the refusal rates of STOP_1 , STOP_2 and SCREEN . (b) Bound SCREEN2 's from below, and say what the gap between SCREEN2 and STOP_2 can consist of. (c) Describe the population on which Q_4 (refusal below 0.5) could have held.

Solution. (a) STOP_1 refuses iff $r_{\text{val}} \leq 1$: 0.910 (measured 0.910); STOP_2 iff $r_{\text{val}} \leq 2$: $0.910 + 0.045 = 0.955$ (measured 0.955); SCREEN equals STOP_1 , no query differing: 0.910. (b) Refusal implies trigger at the same depth, so SCREEN2 refuses at least wherever STOP_2 does: ≥ 0.955 . The excess can only be queries where a pair retraction changes the set while Z stays valid to depth two. Measured 0.963, so $(0.963 - 0.955) \times 134 \approx 1$ query. (c) One where committed sets mostly survive every single retraction, conclusions resting on two or more claims (larger chain components, denser knowledge near the query). Here that is about one query in twenty; choosing such a population now would select for the answer.

Exercise 11.2. A draft paragraph reads: “Our radius-based stop rule, which certifies the adjustment set up to two retractions, improves coverage over leave-one-out screening by 25.6 points (0.986 vs. 0.731), showing that the breakdown radius captures interactions between knowledge errors that a screen misses.” Diagnose every defect, with the number that exposes it, and say which sentence of Chapter 10 or this chapter should replace it.

Solution. (1) No population or sample size: $+0.256 [+0.149, +0.346]$ is Panel A, $n = 1000$, 59 committed informative queries on 8 networks; the population line is $+0.305$, and Panel B's $+0.075$ touches zero. (2) Depth is confounded with fallback: $+0.092$ comes from the 34 queries with $r_{\text{val}} = 1$ where both policies refuse and differ only in widening to I_2 or I_1 . (3) The conclusion is false at matched depth: against the pair screen the difference is $+0.000 [+0.000, +0.000]$, with 5 of 754 queries decided differently, none unsafe. (4) “Improves over” is the forbidden claim that the radius beats a leave- k -out screen. Replace the paragraph with this chapter's purple box, keeping the one-claim contrast only with its sample size and decomposition.

Chapter map

Chapter 12

16 September: what the cells mean, T3, and a bias in our own table

The chapter in two sentences

A cell of the price table answers one question: forced to cover honestly, how wide would this procedure have to be, compared with discarding the knowledge? And the table of named reports, read in full, shows that one retraction usually returns the analyst to the whole class, while the rule that chose its rows biased the one column that looked most like Rosenbaum.

Two documents were written this morning, both in English and both on the branch: `rho_breakdown/table1_dossier_2026-09-16/`, a dossier of every Table 1 the results support, and `rho_breakdown/panelB_T3_2026-09-16/`, which opens one cell of the price table, prints the named reports in full and audits how their rows were chosen.

Three numbering schemes

The dossier's **T3** is M1's Table 2 (`results/interval/TABLE2_REPORTS.md`), the paper's future Table 2. The "Panel B" of its recommended **T11** is the price table on M1's *Panel A* (semi-synthetic), not M1's Panel B. "Table 3" in the outline of 13 September is the audit under wrong knowledge. Below, T-numbers are the dossier's and panels are M1's.

Before reading

- In the one-wrong-claim cell, what are λ^* and the mean half-width of the committed report, and how do they give 5.123?
- How does a procedure earn an L_{95} above 1?

12.1 Opening one cell

Take the one-wrong-claim cell of Panel A at $n = 1000$: 359 queries, twenty seeds each, 7,180 replicates. For each replicate i a procedure reports an interval with midpoint m_i and half-width h_i , and $\lambda_i = |\tau - m_i|/h_i$ is the factor by which that interval must be stretched about its own midpoint to reach the true effect. Below 1, it already covers. Take λ^* , the 95th percentile of the λ_i over the cell: stretching every interval by λ^* makes 95 % of them cover. Average the stretched widths $2\lambda^*h_i$ and divide by the same quantity for the whole class.

The cell, recomputed from replicates by `make_figs.py`

Script `rho_breakdown/panelB_T3_2026-09-16/make_figs.py`, reading `results/interval/replicates.csv.gz`. Panel A, $b = 1$, $n = 1000$, 7,180 replicates.

procedure	λ^*	mean half-width	calibrated width	L_{95}
PLAIN, the committed report	20.416	0.3855	15.7397	5.123
I_1 , one revision	1.000	1.3064	2.6129	0.851
BLANKET, the whole class	1.000	1.5361	3.0721	1.000

15.7397/3.0721 = 5.123 and 2.6129/3.0721 = 0.851. At $n = 20\,000$ the same two cells read 14.368 and 0.758: more data narrows a wrongly centred interval without moving it.

The committed report is the narrowest interval in the cell (mean half-width 0.39 against 1.54 for the class) and the most expensive once it is required to be right. Figure 12.1 shows why: the stretch is about the interval's own midpoint, so a narrow interval centred on the effect the wrong graph implies has to be compensated on both sides.

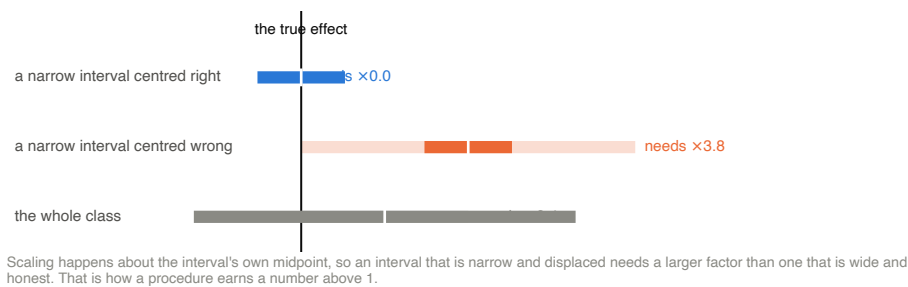


Figure 12.1: Schematic. A narrow interval centred away from the truth needs a larger factor than a wide honest one; that is the only way a procedure lands above 1, wider than discarding the knowledge.

The reproduction. The script reimplements the L_{95} pipeline from `replicates.csv.gz` and `rows.csv` and asserts that all 27 Panel A cells of `TABLE1_L95.md` at $n = 1000$ (nine procedures, three columns) come out to three decimals; I reran it today and it passes. The filter that makes it reproduce: a query enters a column iff it is informative *and* its realised b is non-empty (391, 359, 131 queries).

profiles.jsonl is the wrong join key for cells

My first attempt joined the replicates on `results/interval/profiles.jsonl` and got 0.180 for PLAIN at $b = 0$ instead of 0.464. That file holds 934 profiles, only for queries whose analyst graph commits to a set, so the join silently rebuilt the committed-set secondary table, whose cell really is 0.180 on 134 queries. A correct number for the wrong population; the comparison with the published cell is what caught it.

Without looking (answer 51, in the appendix)

In the Panel A, $b = 1$, $n = 1000$ cell, give λ^* and the mean half-width of the committed report and of the class, and compute L_{95} of the committed report.

Before reading

- Of the nine certifiable declared queries, how many break under some retraction of up to three claims, and how many of the seventeen fitted-coefficient queries break at the first?
- Why did the pre-registered Figure 1 rule return no candidate?
- Can a query have $r_{\text{val}} = 1$ and r_0 “never”?

12.2 T3 in full: a cliff, not a curve

T3 has 26 rows under the recovering set, estimates at $n = 20\,000$ and seed 0: the nine declared queries with a valid status and 17 frame queries on the four fitted-coefficient networks. Three declared queries are excluded with their reasons (**M-bias**: the published diagram is not a DAG; **Schipf_2010**, TT on T2DM: the outcome is not a possible descendant; **confounding**: the treatment has no undirected edge). Read twice, the table says four things.

The published analyses are robust and the sampled ones are not. Eight of the nine declared queries are unbroken by any retraction to depth three; the exception is **mediator**, a four-node textbook diagram. On the fitted networks, 16 of 17 queries have $r_{\text{val}} = 1$. The mechanism is the frame: a pair is admitted beside an undirected edge, where knowledge can matter, so the measured population is close to the complement of the queries the source papers were written to answer. That is an applicability statement for the paper’s evaluation section, not a result.

The sensitivity curve is a single step. On 16 of T3’s 17 informative rows (the 16 informative fitted queries and **mediator**) the interval after one retraction equals the whole-class interval to four decimals. The exception is again **mediator**, whose first retraction covers 57% of the class and whose second saturates it. All the identifying power the knowledge supplies sits in one claim, and withdrawing it returns the analyst to where they would be with no knowledge. This is why the pre-registered Figure 1 rule (both radii in $\{2, 3\}$ and a plateau before the break) found nothing on real coefficients: the shape it looked for is not in the data, and the declared and applied rows that did have radii 2 or 3 widened 27 to 34 times before breaking. The companion’s recommended figure (its candidate F-A) draws this cliff; the version on the full eligible pool is in the next section.

The null radius and the validity radius are different numbers. On **mediator** the set fails after one retraction and the interval reaches zero after two. On **ecoli70** $\text{eutG} \rightarrow \text{lacY}$ and both **magic-niab** G1217 queries, $r_{\text{val}} = 1$ and r_0 is “never”: the set breaks, the sign does not. On several fitted rows the band at seed 0 contains zero at budget zero (0 from the band, 1 from the population): no conclusion to break at that sample size. A sentence about r_0 must name its reading (population, one seed, range over twenty), and a figure’s axis label must say whether it shows conclusions or validity.

Why the one- and two-revision columns look identical. I_r grows with r and is capped by the class interval. Of the 303 committed queries under the recovering set, 281 (92.7%) print the same interval at both budgets: on 76 the class interval is a point, on 205 the first revision already returns the class. The second column carries information on 20 queries, on **insurance**, **Polzer_2012**, **Kampen_2014**, **sachs**, **Schipf_2010** and **mediator**, all but the last excluded from T3 by its real-coefficient rule. Replace the two-revision column with the class interval, and the identity becomes visible as the finding.

Without looking (answer 52, in the appendix)

State the one-claim cliff as measured on T3, name the exception, and say what it did to the pre-registered Figure 1 rule.

Before reading

- T3 prints 17 of the 294 eligible sampled queries. By which two rules, and which one is indefensible?
- What share of printed, dropped and eligible queries has a finite null radius?
- Which claim of T3 survives the selection audit?

12.3 A selection bias in our own table

T3 is a selection made by two rules. The funnel is the whole path under the recovering set; the counts are in `results/ledger/TABLE0_FUNNEL.md` or were recounted from `results/interval/rows.csv` and `profiles.jsonl` by `rho_breakdown/panelB_T3_2026-09-16/make_exhaustive.py`, which first checks the 26 printed rows cell by cell against `table2.json`.

count	stage	what leaves, and why
39 / 33 / 27	networks parsed / with a frame / swept	over 500 nodes, or every component over the questionnaire cap
80,132	candidate pairs, from observables only	
648	pairs sampled (at most 20 per network)	
537	rows: 528 frame rows in the swept networks + 9 declared	
303	commit to an adjustment set (9 declared, 294 sampled)	234: on the analyst's graph the outcome cannot descend from the treatment
37	rule 1: fitted coefficients only	257 on the 23 semi-synthetic networks; defensible , not in the caption
17	rule 2: finite r_0 first, then frame order, five per network	20; sorts on the outcome it illustrates: indefensible
26	T3 = 9 declared + 17 sampled	

Rule 1 is defensible: on the 23 semi-synthetic networks the effect size is one we drew, and a named report would present it as a real analysis. It belongs in the caption.

Rule 2 is not. The Table 2 block of `experiments/interval_tables.py` (line 1136) sorts the eligible queries of each network by (`r0_pop == CENSORED`, `frame_index`) and keeps the first five. A finite null radius is exactly what the table exists to illustrate.

The bias, `rho_breakdown/panelB_T3_2026-09-16/make_figs.py` (rerun 16 September)

Queries on the four fitted networks with a committed set: 37 eligible, 17 printed, 20 dropped (9 on `magic-niab`, 8 on `magic-irri`, 3 on `ecoli70`). Finite population null radius: 13 of 17 printed (76%), 3 of 20 dropped (15%), 16 of 37 eligible (43%).

I have to be exact about whose defect this is. The ordering was not chosen after seeing the answer: it is written in §6 of `results/interval/PREREGISTRATION.md` (“finite r_0 first, then frame order”), before any row existed, and the header of `TABLE2_REPORTS.md` states it. It was still the wrong rule, because it sorts on the outcome, and nobody measured its effect until this morning. The r_0 column of T3 is not representative of the population it was drawn from.

The fix, one line

In `experiments/interval_tables.py`, sort the candidates by `frame_index` alone (drop the `r0_pop == CENSORED` term), or better, print all 37 eligible queries so the inclusion rule has nothing in it a reader must trust. Put rule 1 and the eligible count in the caption. Draw the figure on the 37, not on the 17.

What the selection did not bias. The cliff survives. On all 26 informative queries of the 37 eligible, one retraction returns the whole-class interval (16 of 16 printed, 10 of 10 dropped). Over all 294 committed sampled queries on 27 networks, 226 are informative; on 205 the one-retraction interval equals the class interval to four decimals, and on 211 (93%) it covers more than 90% of the class width. Figure 12.2 draws F-A on the 37 eligible queries, printed ones marked, including the 11 rows where nothing moves, which T3 hides.

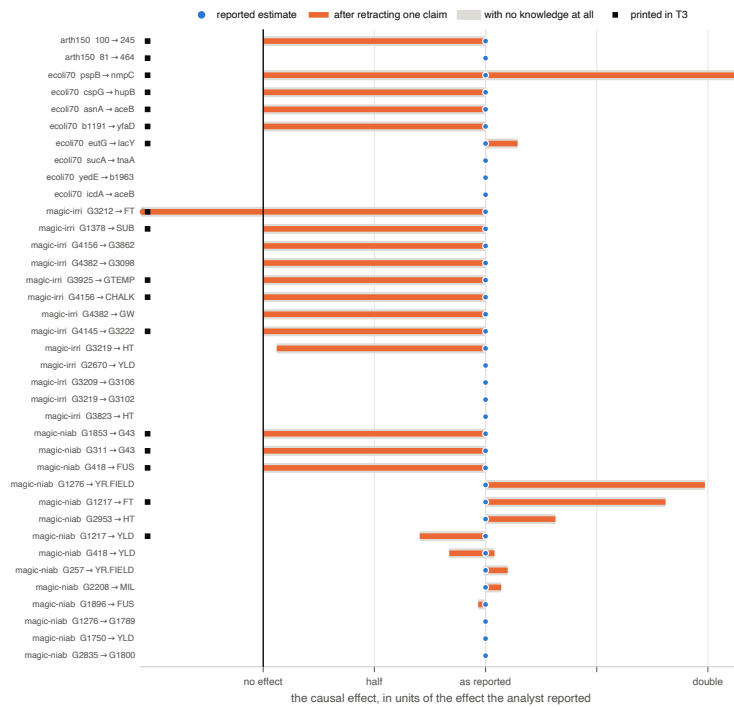


Figure 12.2: All 37 eligible fitted-coefficient queries, printed ones marked with a square: the cliff holds on the dropped queries too, and the rows with no bar (the class already pins the effect) are the honest counterweight the printed table leaves out.

Without looking (answer 53, in the appendix)

Give the three finite-null-radius shares of the T3 audit with their denominators, the line that produced the bias, and the fix.

Before reading

- Which candidate does the dossier recommend as Table 1, and what are its two panels?
- What blocks the abstract of 18 September, according to the dossier?

12.4 The Table 1 dossier

The dossier states eleven live claims before any table, each with evidence, significance, nearest prior art and falsifier (`rho_breakdown/table1_dossier_2026-09-16/sections/02_claims.tex`).

claim	what carries it
C1	nobody prices a structural assumption
C2	computable from the analyst's inputs
C3	claims are the safe unit, hops are not
C4	the price of a wrong claim
C5	the tie with the free screen
C6	refusal where knowledge matters
C7	locality in claims, not hops
C8	applicability, as a limit
C9	the class must be correct
C10	suppliers are an error process
C11	ranking by survival
	framing; novelty search never run in this wording
	monotonicity audit: 5,000 pairs, 0 violations (bound 0.06 %)
	3,014 certifiable rows: 0 over-certifications against 39
	I_1 at 0.851 of the class, premium at zero (Ch. 10)
	+0.000 at matched depth (Ch. 11)
	0.910 of 134 committed informative queries
	primary supplier: hop reading above claim depth on 21 of 157
	8 of 9 declared queries unbroken to depth 3
	estimated CPDAG: over-certification on 14 of 38 (Ch. 9)
	invalid set 0.298 at 72B, 0.288 at 7B
	your τ_b table; screen column owed

Thirteen candidates (T1 to T13, with a variant T1b) are typeset with the numbers that exist and scored against three requirements (readable after the abstract; verifies the claims we make; compares with the literature inside the float) and six constraints each earlier design paid for: one calibrated metric, one population per panel, a row that loses where it must, no discrimination column, no model family as subject, nothing internal in the float. **The recommendation is T11**, one float with two panels on two named populations. Its first panel is the certificate ladder over the 3,014 certifiable rows of eleven knowledge conditions on 27 networks (902 committed sets invalid at the truth): counting claims asserted over-certifies on 729, the separation rule on 15, the hop radius on 39, the claim radius on 0. Its second panel is the price table at $n = 1000$ with the leave-one-out screen as a row, and the refusal rate (0.910 of 134) under the float with its own denominator. T3 becomes Table 2, the policy table T8 becomes Table 3 or prose, your ranking table T4 stays in its section with the screen column added, and the decision table T5 is the best candidate for the first figure.

Two defects were found and fixed while the dossier was written, each the kind it warns against. The first draft of T8 and of T11's second panel filled the two-wrong-claim cells of the screen and stop-rule rows from M1b's 59 committed queries beside cells from the price table's 131: two populations in one row. And the ladder's 15 over-certifications had been attributed to the leave-one-out screen; they belong to the separation rule (the screen never over-certifies relative to the stop rule, 0 exceptions in 2,212 rows). A reconciliation, not a defect: my review notes quoted the depth-one screen gap as +0.305, the dossier as +0.256; both are in TABLE1_L95.md, the population line and the $n = 1000$ line, and every quotation now carries its sample size.

Blocking before the abstract on 18 September

C1's novelty search has never been run in the wording the claim uses. The searches on record were pointed at latent-causal representation learning and at Taeb, Guo and Henckel, not at the sensitivity-analysis lineage C1 says does not already price this, and the statement that Guo and Perković never perturb the knowledge rests on a paper I have not read in full. Either the search runs before the abstract, or C1 is narrowed to what was searched and the abstract says the narrower thing. **The author list must be settled by 18 September;** Chapter 14 quotes the rule.

Without looking (answer 54, in the appendix)

Which Table 1 does the dossier recommend, what are its two panels with their populations, and what blocks the abstract of 18 September?

Do not move on

Name the sort key that biased T3, say what it sorts on, and explain in one sentence why it biases the r_0 column but not the claim that one retraction returns the class interval. If you cannot, reread the paragraph "What the selection did not bias".

Exercises

Exercise 12.1. *Five rows of T3 (recovering set; bands at $n = 20\,000$, seed 0; class interval from the population; r_0 as population / band at seed 0 / range over 20 seeds).*

query	true	one revision	two revisions	class	r_0	r_{val}
mediator, $X \rightarrow Y$	-2.603	[-2.623, -1.109]	[-2.623, 0.000]	[-2.603, 0.000]	2 / 2 / 2	1
ecoli70, $\text{eutG} \rightarrow \text{lacY}$	0.309	[0.277, 0.377]	[0.277, 0.377]	[0.309, 0.354]	never / >3 / >3	1
magic-niab, $G418 \rightarrow FUS$	0.003	[-0.022, 0.011]	[-0.022, 0.011]	[0.000, 0.003]	1 / 0 / 0..1	1
ecoli70, $\text{pspB} \rightarrow \text{nmpC}$	-0.108	[-0.531, 0.026]	[-0.531, 0.026]	[-0.525, 0.000]	1 / 1 / 1..3	1
arth150, $81 \rightarrow 464$	0.256	[0.097, 0.265]	[0.097, 0.265]	[0.256, 0.256]	>3 / >3 / >3	≥ 2

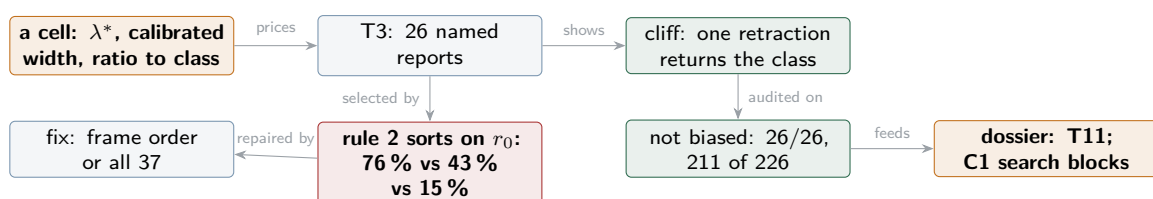
(a) On which row do the two revision columns differ, and what does that say about the cliff?
 (b) Which row has $r_{\text{val}} = 1$ and r_0 “never”, and why is that not a contradiction? (c) On which row does the null radius depend on the reading, and how should a sentence quote it? (d) Which row must be left out of “one retraction returns the class interval”, and why?

Solution. (a) mediator: one revision reaches -1.109, two reach zero, the class. It is the only row where the first retraction stops short of the class (57%), so the only one with a second step. (b) $\text{eutG} \rightarrow \text{lacY}$. r_{val} is about the set (one retraction makes Z invalid); r_0 is about the sign (every graph of the class gives an effect in [0.309, 0.354]). Validity breaks, the conclusion does not. (c) $G418 \rightarrow FUS$: population 1, seed 0 gives 0, range 0 to 1. The true effect is 0.003, so at $n = 20\,000$ the band usually contains zero before any retraction; quote “population null radius 1” or the range, never a bare “1”. $\text{pspB} \rightarrow \text{nmpC}$ (1 to 3 over seeds) needs the same care. (d) arth150, $81 \rightarrow 464$: the class interval is a point, so the query is not informative; its band is wide from sampling alone.

Exercise 12.2. *T3’s current caption reads, in part: “Sensitivity reports on twenty-six named analyses. ... r_0 is the null radius, the number of revisions after which the interval contains zero ... r_{val} is the breakdown radius ... Knowledge: the recovering set; estimates at $n = 20\,000$, first seed.” Rewrite it as it should be printed once the fix is applied (all 37 eligible sampled queries shown), so that a reviewer can check the inclusion rule, the counts and the two radii without opening any file. No internal vocabulary.*

Solution. “Sensitivity reports on named analyses. Above the line, the nine published analyses whose authors declared a treatment and an outcome (three more are excluded: one diagram is not acyclic, in one the outcome cannot descend from the treatment, in one no claim bears on the treatment); structure as published, parameters drawn. Below, all 37 sampled queries on the four networks with coefficients fitted to real data whose analyst graph commits to an adjustment set; networks with drawn parameters are not shown, since their effect sizes would be ours. Intervals: every effect identified after retracting at most one claim, and over the whole equivalence class. Null radius: retracted claims until the 95% band contains zero (population, and range over twenty samples). Validity radius: retracted claims until the adjustment set is no longer valid. Estimates at $n = 20\,000$.” What changed: both inclusion rules and the eligible count are stated, nothing is selected on the outcome, the class interval replaces the two-revision column, the null radius carries its reading, and “breakdown radius” is renamed so it cannot be confused with the null radius.

Chapter map



Part VI

Your draft and ours

Chapter 13

Your draft, read against the measurements

I read your draft of 16 September as a referee would and against everything in the previous chapters, and its theory was re-implemented from the definitions and searched for counterexamples. None was found: the mathematics looks right, and every finding below is about what the text says. The numbers printed in your draft were not audited in that pass. Each item gives where it sits, why it matters, and the fix.

The review in two sentences

Keep the object, the fixpoint definition and the reporting discipline as they are. Fix the positioning against the paper you build on, the conditions the text drops, and prose that claims more than its table.

Before reading

- Which of your structural results does the foundation paper state to be false for MPDAGs in general?
- Is there a formula for $d(G, H)$ on your space that needs no search?

13.1 What the draft gets right

The object, in the right geometry. Definition 1 by the fixpoint test is a real contribution: Appendix A.1 shows with minimal witnesses (the collider on a three-node chain, K_4 minus an edge with one claim) why the two obvious filters are unsound, and covers are computed from model inclusion rather than from a conjectured characterisation.

The clinician sentence in §4.4 is the best-written sentence of the draft, and it makes the object contestable by a domain expert. It belongs on page one.

The reporting discipline of §5.4 is already the one our protocol uses: statuses on a frozen instance list rather than filters, censoring rather than dropping, no numeric radius for an unreachable failure, the two corruption arms never merged. §6 states the observables-only invariant of Chapter 1 in one sentence. And Appendix B's E3 is designed to refute rather than to confirm («agreement proves nothing; only disagreement would be a theorem»), which is the right epistemics for a verified-not-proved property.

Your theory, re-implemented from Definitions 1 to 3

The space, the order, the covering relation and d were rebuilt using only MPDAG, `apply_orientations` and `enumerate_dag_extensions` from `src/bkrobust/demo/`, never the search or SAT code. Scale: 1,375 spaces, about 61,000 knowledge states, 9,648,249 ordered pairs, 46,698 covering pairs, 188,116 anti-exchange triples (39,402 of them on states whose undirected component is not chordal). Retraction-only radius equal to the brute-force radius on 8,161 of 8,161 instances; 599 of 599 nearest failing states reachable by retractions; the SAT clause set has 1,858 models for 1,858 space elements. **No counterexample to**

any checkable statement. The check shares your Meek and extension primitives, so a bug there would be invisible to it. From my review notes of 16 September; not in the branch.

Before reading

- Where in the draft is the anti-exchange property stated?
- Which paragraph of the draft computes radii for sets other than the optimal one?

13.2 The theory: what the text says

Figure 13.1 is the dependency chain as the appendix proves it. Three results are unconditional; four inherit the open case of the anti-exchange property; and the property itself is the one statement the reader is never shown.

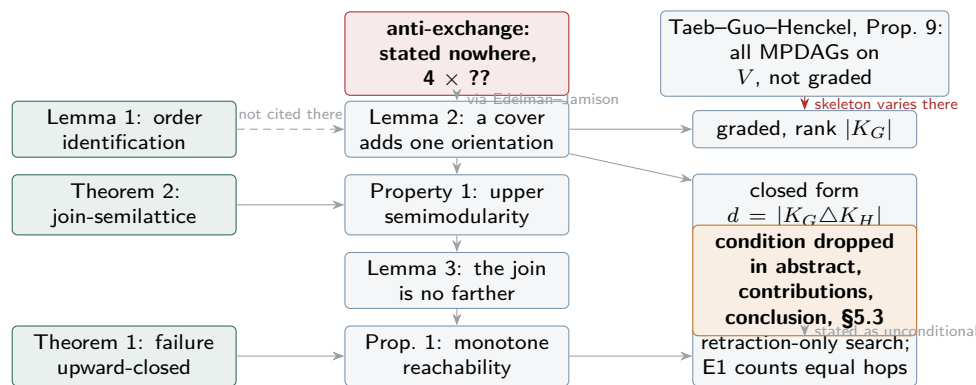


Figure 13.1: Green results hold unconditionally; every white box inherits the anti-exchange property, which the draft never states. The foundation paper’s non-gradedness is about a larger poset, and the draft does not yet say so.

1. The foundation paper states non-gradedness for a larger poset

Where. Appendix A calls $\rho(G) = |\text{dir}(G)| - |\text{dir}(\hat{C})|$ a rank function «so the order is graded», Property 1 is upper semimodularity, and Theorem 2 makes the space a join-semilattice. *Why.* Proposition 9 of Taeb, Guo and Henckel states that the model-oriented poset of MPDAGs on $|V| \geq 4$ vertices is not graded, and the paragraph after it says the poset is neither upper nor lower semimodular, has no semilattice structure, and has no explicit graphical characterisation of its covering relation; a referee who knows that paper reads Appendix A as a contradiction. *Fix.* Say why both hold, in one paragraph of §5.1 and one sentence of Appendix A.

Taeb, Guo and Henckel, arXiv:2511.10625v3, §4.5 and Appendix D.3

Their poset contains every MPDAG on V , from the empty graph to the complete undirected one, and covers can change adjacencies: in their Example 2 two neighbours of the same graph differ from it by one edge and by $n - 1$ edges, and the proof of Proposition 9 compares two maximal chains of lengths 3 and 5 in which a covering step removes an edge. On a common skeleton their Proposition 8 reduces the order to $[G] \subseteq [H]$, which is yours, and their pseudo-rank (Definition 7, directed edges plus twice the undirected ones) equals $2m - |\text{dir}(G)|$ on a skeleton with m edges: your rank, negated and shifted.

The paragraph needs five sentences, and it turns the largest liability of the draft into its clearest positioning statement.

A paragraph for §5.1

These results do not contradict Proposition 9 of Taeb, Guo and Henckel (2025), which shows that the model-oriented poset of all MPDAGs on four or more vertices is not graded, is not a semilattice and is not semimodular. Their poset lets a covering step change the skeleton. The space studied here fixes the skeleton and the v-structures of $\bar{C}$, so only orientations vary. On it the join exists unconditionally (Theorem 2), and gradedness and upper semimodularity hold under Conjecture 1, with their pseudo-rank reducing to $2m - |\text{dir}(G)|$.

2. The anti-exchange property is never stated

Where. Four `\Cref{prop:antiexchange}` calls (§7, the proof of Lemma 2, the heading of Appendix A.3, the E3 paragraph of Appendix B) point at a label that exists nowhere, so the compiled PDF prints «??» four times, including the heading «A.3 Status of ??». *Why.* The reproducibility statement promises that the single unproved property is isolated in A.3, and the reader finds a section about a statement they have never seen. *Fix.* One environment, which also repairs the four references.

The statement, as THEOREMS.md §4c gives it

Add to Appendix A a `conjecture` environment (your macros define it) carrying `\label{prop:antiexchange}`: *for a closed set S of orientations and distinct orientations $x, y \notin S$, if $y \in \text{cl}(S \cup \{x\})$ then $x \notin \text{cl}(S \cup \{y\})$* . Its reading in words: no two distinct orientations are perfectly correlated across the represented DAGs. Calling it a conjecture matches A.3, where Case B is open.

3. Your theorems make the distance a closed form

Where. Definition 2 defines d as a shortest path, and Appendix C's «Against a naive count» argues that counting the assertions that differ understates d . *Why.* Lemma 2 and Theorem 2 give $d(G, H) = |K_G \Delta K_H|$: each covering step changes K by exactly one orientation, so $d \geq |K_G \Delta K_H|$; and $J = G \vee H$ has $K_J = K_G \cap K_H$, so the two maximal chains up to J have $|K_G \setminus K_H|$ and $|K_H \setminus K_G|$ steps and $d \leq |K_G \Delta K_H|$. The re-implementation found the identity on all 3,047,289 ordered pairs on which it computed d (a subset of the 9,648,249 above), with no exception and no unreachable pair, so the naive-count paragraph argues against a count the theory makes equal. That paragraph reports a Spearman correlation of 0.73 to 0.77 between d and its naive count (90_appendix.tex); the other natural reading, a flip counted once, does not give 0.73 to 0.77 either. *Fix.* State the closed form as a corollary of Lemma 2 (your E2 paragraph already writes $|K_{G_0} \Delta K_G|$), and either drop the naive-count paragraph or name exactly which count it compares.

Without looking (answer 55, in the appendix)

State the closed form of d on the space of knowledge states, the two results it follows from, and which inequality each gives.

4. Proposition 1 says more than its proof, and the condition disappears where it is read

Where. Proposition 1 reads «The nearest failing state is reachable...», while its proof replaces the nearest failing state G by $J = G \vee G_0$. §5.1 promises «We state this caveat wherever a radius is reported», and the abstract («enough order structure for the radius to be computed without enumerating it»), the contributions («so the search can be restricted to retractions»), the conclusion («which confines the search to retractions»; «exact radii») and §5.3 («guaranteed to return identical radii») state it without the condition. *Why.* The proof delivers some failing state at distance at most r_{val} reachable by retractions only, which is what the algorithm needs; the two proved results named in the contributions do not confine the search, the conditional proposition does, and Appendix B itself says all three encodings are upper bounds. *Fix.* Write «some nearest failing state», and add «under Conjecture 1» in those four places. The strong reading survived on 599 of 599 nearest failing states, which is evidence, not proof.

5. The Edelman–Jamison appeal lacks its hypotheses

Where. The proof of Lemma 2. *Why.* Their theorem is about a closure operator defined on every subset of a finite ground set, while here cl is undefined on inconsistent orientation sets (937 of 4,137 two-orientation subsets, 22.7%, over 150 spaces), the full ground set was closed in none of those 150 spaces, and the identification of your order with reverse inclusion of K_G is Lemma 1, not cited at that point. *Fix.* Prove Lemma 2 directly from the anti-exchange statement, or say how cl is extended to a total operator and cite Lemma 1 there.

6. Two adjustment criteria, and where they part

Where. §2 cites the generalized adjustment criterion; the fast oracle in `src/bkrobust/mpdag_criterion/criterion.py` decides the back-door strengthening, as its docstring says; Appendix C's «Robust selection» computes r_{val} for all fourteen valid sets. *Why.* `THEOREMS.md` §14 proves the two criteria coincide on the optimal set of the graph being judged. Your radius judges one fixed set, $O^*(G_0)$, in other knowledge states, where it is not their optimal set, so the proof does not cover it; your own BFS check in §14 found identical radii on 5,304 instances, and the ledger audit found the back-door function rejecting 4 of 1,169 committed sets at the truth (Chapter 8). The main text is therefore very likely unaffected, by measurement rather than by proof. On arbitrary sets the two disagree on 23,034 of 372,430 cases (6.2%) in the re-implementation, none of them at the optimal set of the judged graph, and `THEOREMS.md` §14 records radii that differ on 29.14% of 16,926 arbitrary-set instances, so «Robust selection» is exactly the regime where the choice shows. The criterion for maximally oriented graphs is introduced in `perkovic2017cpdag`, already in your bibliography, while §1 and §2 cite `perkovic2018complete`, whose title restricts it to ancestral graphs. *Fix.* Name the criterion the code decides, say which one the fourteen sets were scored with, and move the MPDAG attribution to `perkovic2017cpdag`.

Without looking (answer 56, in the appendix)

Why is the main text of the draft very likely unaffected by the difference between the GAC and the back-door criterion, why is that not a proof, and which paragraph is clearly affected?

Without looking (answer 57, in the appendix)

What does Proposition 9 of Taeb, Guo and Henckel state, and why does it not contradict the rank function of your Appendix A?

Before reading

- Which cells of your Table 1 sit under the sentence «a strong and consistent predictor»?
- Where does the word «breakdown» come from in econometrics?

13.3 The results section and the manuscript

7. The prose contradicts the table on the same page

Where. §6.1, «Ranking quality» and «Over-claiming does not pay», printed on page 9 with Table 1. *Why.* The sentences say more than the table, as the box shows. *Fix.* Uncomment the «Limits of the present evidence» paragraph of `07_discussion.tex`, whose claim is the right one («the radius shares significance with the baselines in most strata ... not that it dominates them»), and restrict «more claims, less robustness» to the tiered arm.

The sentence against the cells printed under it

«A strong and consistent predictor»: $\tau_b(r_{\text{val}}) = -0.091 [-0.204, 0.030]$ in *flip*, *cov 1.0*, *bw 0.00* (a cell your caption marks unstable), and the confirmed null 0.032 $[-0.112, 0.168]$ in *flip*, *cov 0.5*, *bw 0.25* at 1,000 draws. «Standard baseline metrics fail»: in *flip*, *cov 1.0*, *bw 0.25*, SHD scores 0.582 and $|K|$ scores 0.557 against the radius's 0.355. «More claims mean less robustness»: $\tau_b(|K|)$ is positive in five of the six flip strata at 200 draws, and negative in the three tiered strata and in the 1,000-draw row (-0.200). «Strictly increasing the risk»: a claim that Meek already derives leaves G_0 , Z and your r_{val} unchanged.

Without looking (answer 58, in the appendix)

Give the two Table 1 cells that contradict «a strong and consistent predictor», and the stratum where both baselines score above the radius.

8. The bibliography is broken in the compiled PDF

Where. `keel2024fuzzy` and `robust2026imperfect` have no author field, so they print from their titles and are cited as «(kee, 2024)» and «(rob, 2025)»; `ejaz2025lges` prints the wrong title and first name. *Why.* These are visible on page 3 and page 12 of the PDF, and one of them puts a researcher's work under a name that is not theirs. *Fix.* From the arXiv records: arXiv:2405.08699 is by Wenrui Li, Wei Zhang, Qinghao Zhang, Xuegong Zhang and Xiaowo Wang; arXiv:2511.06790 by Zidong Wang, Xi Lin, Chuchao He and Xiaoguang Gao; arXiv:2506.22331 is «Less Greedy Equivalence Search», by Adiba Ejaz and Elias Bareinboim. Your own bib comments already flag all three with «VERIFY».

9. Citations a referee will ask for

Where. Related work (§3). *Why.* The central word is «breakdown», the distance is evaluated on adjustment, and the evaluation runs a tiered arm, so four literatures are one question away. *Fix.* Cite Masten and Poirier, «Inference on breakdown frontiers» (Quantitative Economics 11(1):41–111, 2020); the adjustment identification distance (Henckel, Würtzen and Weichwald, UAI 2024) and the structural intervention distance (Peters and Bühlmann, Neural Computation 27:771–799, 2015); Bang and Didelez on tiered knowledge, both entries already in your bibliography and never cited; and b-LOAD (`ahn2026bload`), also in the bibliography and never cited. Chapter 5 adds the minimal correction set.

10. The main text is eleven pages

Where. The conclusion ends on page 11 of the compiled draft. *Why.* The ICLR 2027 author guidelines say «At the time of submission, the main text should be 9 pages or fewer» and «Papers with main text beyond the page limit will be desk-rejected». *Fix.* Chapter 14, where the cut is part of the merge.

Smaller items. «What is required is model inclusion» in §4.3 is necessary and not sufficient (Exercise 13.1). «All 133 CPDAGs at four nodes» in Appendix B mixes two populations: `rho_breakdown/theory_checks_2026-09-16/check_four_node_classes.py` counts 185 labelled four-node classes, 126 of them with an undirected edge, and 133 is those 126 plus the 7 three-node classes with one, which is what A.1 says. An unreachable failure has three conventions (∞ in Definition 3, a status in §5.4, `UNREACHED = -1` in the code); pick one.

Do not move on

Write out the upper bound $d(G, H) \leq |K_G \Delta K_H|$ and name the exact place where Lemma 2 is used in it. If you can only write the lower bound, reread item 3.

Exercises

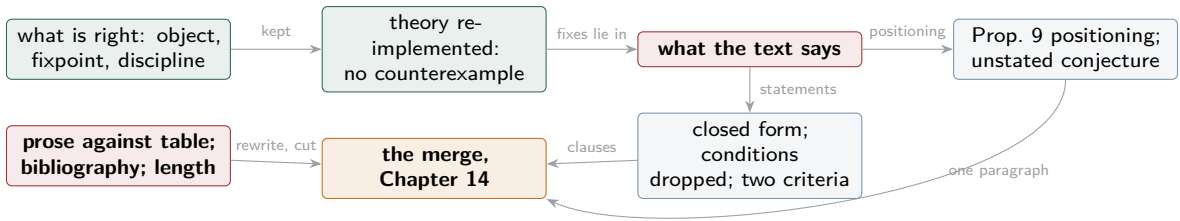
Exercise 13.1. Your §4.3 says «What is required is model inclusion, $[G] \subseteq [\hat{C}]$, which the fixpoint test enforces». Give the smallest counterexample showing that model inclusion alone does not make G a knowledge state, check it with the fixpoint test, and name the property it lacks.

Solution. Take $\hat{C} = A - B - C$ (three DAGs) and G with $A \rightarrow B$ and $B - C$. Then $[G] = \{A \rightarrow B \rightarrow C\}$ is a non-empty strict subset of $[\hat{C}]$, so model inclusion holds. But $\text{Meek}(\hat{C}, \{A \rightarrow B\})$ applies R1 (A and C are non-adjacent) and returns $A \rightarrow B \rightarrow C \neq G$, so the fixpoint test fails. G lacks maximal orientation: $B - C$ is undirected although every DAG of $[G]$ orients it $B \rightarrow C$. The test enforces inclusion *and* maximal orientation, and the second conjunct does the work.

Exercise 13.2. A referee writes: «The authors claim their radius ranks survival better than standard baselines, and that asserting more knowledge makes results more fragile. Table 1 supports neither.» Using only your Table 1, diagnose which part is right, which cells carry it, and the smallest true replacement for each sentence.

Solution. The first part is right; the second holds for the flip strata and fails for the tiered ones. The radius is not consistent (-0.091 in flip, cov 1.0, bw 0.00; the null 0.032 $[-0.112, 0.168]$ at 1,000 draws) and does not dominate (0.582 and 0.557 against 0.355 in flip, cov 1.0, bw 0.25). A true replacement is the commented-out paragraph of §7: the radius shares significance with the baselines in most strata and does not dominate them. For fragility: $\tau_b(|K|)$ is negative in the three tiered strata (-0.423 , -0.382 , -0.377) but positive in five of the six flip strata at 200 draws, so the true sentence is «under tiered, block-correlated errors, more asserted claims go with lower survival», without «strictly».

Chapter map



Chapter 14

One paper by Friday: the merge

The abstract and the author list lock on Friday 18 September, AoE; the paper is due on Friday 25 September, AoE. Your draft has the object, the geometry, the theory and the algorithm. Our side has an evaluation that already ran, on real structure, with the audit opened after the rows were frozen. This chapter maps one onto the other: what your open items correspond to, what the measurements add, where the two drafts disagree, three ways to merge them, and the order of work for the next nine days.

The merge in one sentence

Keep your object, geometry and theory as they stand; replace the evaluation section with the protocol that already ran; move the title and abstract to the pricing frame of Rosenbaum's Γ ; and cut to nine pages.

Before reading

- Your draft owes two experiments in `\todo` markers. Which of them has already run, and with which endpoint?
- Your commented-out caveat says $r_{\text{claim}} = 1$ on all full-coverage rows by construction. What removed that construction?

14.1 Your open items are the runs of 11 to 13 September

Your draft carries three `\todo` markers. The first asks for a single end-to-end speedup figure; nothing on our side answers it. The other two correspond to the two measurements of Chapters 6 to 8, the second one almost word for word.

§6.2. «The ranking evaluation of Section 6.1 has not yet been run on these 831 real instances; survival and the paired cross-arm design exist only on synthetic structure.» What ran on the committed real rows is the corrupted-knowledge calibration of Chapter 6, the nearest thing to what the marker asks. It flips claims of the recovering set at four rates and two coverages on the same real rows, and asks of every instrument whether its certificate held, scored in the four buckets of Chapter 1 rather than by τ_b against the survival area. The τ_b design itself has not been run on the real rows.

§6.3. «The elicitation arm: knowledge collected by asking for a causal order over each chain component rather than for pairwise edge directions, from which r_{claim} , r_{val} and φ_1 are measured on the same frozen instance frame, with a matched random control and an audit opened only after the rows are frozen.» That is the ledger, exactly as specified.

The two markers, closed: `results/stage0/` and `results/ledger/`

Calibration (`results/stage0/calibration_summary.json`): 16,496 draws, 8,911 scored, 24 networks, flip rates 0, 0.1, 0.25 and 0.5 at coverages 1.0 and 0.5, five repetitions per nonzero rate, seeds derived by SHA-256. `results/stage0/STAGE0.md` still quotes 8,553 scored rows from the run before the seeding fix; the JSON is the rerun. **Elicitation and ledger**: the questionnaire covers 120 chain components, 369 open pairs and 540 rendered

items, frozen and hashed before any supplier saw it (`docs/PROTOCOL_RUN.md`); `results/ledger/rows.csv` holds 8,308 rows over 14 knowledge conditions, with the chance control `D_RAND` matched on the supplier's edges and r_{claim} , r_{hop} , φ_1 on 528 frame rows per supplier condition (`results/ledger/TABLE4.md`).

Your commented-out caveat in §6.3 is also right, and the protocol removed its cause. `docs/EVALUATION_PLAN.md` confirms that the admissibility gate makes $r_{\text{claim}} = 1$ on 540 of 540 full-coverage rows. The frame of Chapter 6 is drawn from the CPDAG alone, 80,132 candidate pairs over 33 networks, and contains every one of the 543 committed pairs, so r_{claim} is now measured rather than fixed by construction.

Without looking (answer 59, in the appendix)

Map each of your two experimental `\todo` markers to the artefact that closes it, and say how the first one differs from what you asked for.

Before reading

- Your survival table and our bucket audit measure the same object. May they share a table?
- Which baseline would a referee add to your τ_b table first?

14.2 What the measurements add, and where the drafts conflict

Eight things exist on our side and not in your draft. Each has its chapter and its file.

1. **The knowledge interval and the null radius** (Chapter 10). With one wrong claim the stated report covers 0.597 and the interval at one claim covers 0.980 at an L_{95} of 0.851 (Panel A, $n = 1000$, `results/interval/TABLE1_L95.md`).
2. **The Γ framing**: price what cannot be tested, with the interval as the headline (Chapter 10).
3. **The unit, measured**: over 3,014 certifiable rows (true CPDAG) the hop radius over-certifies on 39, the free rule on 15, the claim certificate on none (`results/ledger/TABLE4.md`).
4. **The stop rule against the free screen** (Chapter 11): refusal 0.910 on 134 informative committed rows with correct knowledge (Panel A); different decisions from the one-claim screen on 2 of 162 rows at one wrong claim (both panels); and a coverage difference of +0.000 [+0.000, +0.000] against the pair screen at two wrong claims (Panel A, $n = 1000$) (`results/interval/TABLE1_L95.md`).
5. **The estimated class** (Chapter 9): the truth extends the analyst's graph on 0 of 244 certifiable rows, and both certificates are dangerous on 14 of 38 (`results/pest/TABLE7.md`).
6. **The audit oracle**: the back-door oracle rejected 4 of 1,169 committed sets that the GAC accepts, so the audit now uses `src/bkrobust/benchmarks/audit.py`, guarded by `tests/benchmarks/test_audit_oracle.py` (`results/ledger/PREREGISTRATION.md`, A.0b).
7. **The suppliers as error processes** (Chapter 7).
8. **The monotonicity audit**: 5,000 sampled pairs over 469 rows, 0 violations, a one-sided 95 % upper bound of 0.06 % (`results/ledger/monotonicity_audit.json`): the implemented failure predicate is upward-closed, which is your Theorem 1 as run.

Four conflicts follow, and a merged paper has to settle each of them in its text.

Framing. Your paper is a diagnostic radius; ours is a sensitivity analysis in the sense of Rosenbaum's Γ , with the interval leading. Your abstract already says the radius «prices the structural risk», which is the pricing move; what it prices is validity alone.

The unit. Your headline counts covering hops and treats claims as a refinement. The ledger

says the hop count over-certifies and the claim count does not, and on your own running example the two give 3 and 2 (Chapter 1).

A missing baseline. Your τ_b table has no leave- k -out screen, and M1b showed that at matched depth the stop rule built on the radius takes the screen's decisions, except where a retraction changes the set while leaving it valid.

Two designs on one object. Survival ranking on synthetic structure and the bucket audit on real structure answer different questions on related populations. They go in the paper as two panels with their populations named, never pooled, which is your own §5.4 rule.

Without looking (answer 60, in the appendix)

What must be added to your τ_b table, and what does M1b say it will show at matched depth and against the depth-one screen?

Before reading

- Which sections of your draft survive a merge almost untouched?
- Can the title still change after 18 September?

14.3 Three ways to merge, and the one I recommend

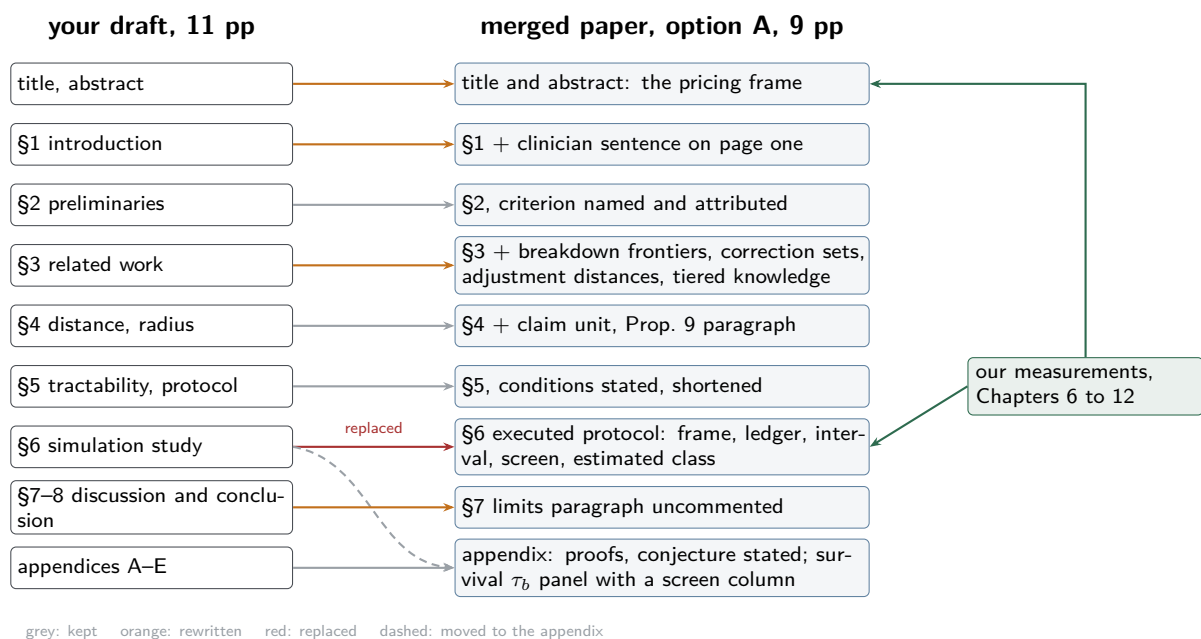


Figure 14.1: Under option A almost everything you wrote survives; the evaluation section is the one part that is replaced, and your synthetic survival table moves to the appendix as its own panel.

There are three ways to merge, and Figure 14.1 shows where each part of your draft goes under the first.

A. One paper, your skeleton, our measurements. Your §§2 to 5 stay almost as they are, with the fixes of Chapter 13; §6 is replaced by the executed protocol (frame, suppliers, ledger, interval and null radius, the screen, the estimated class); title and abstract move to the pricing frame; the main text is cut from eleven pages to nine. Cost: one new section, an abstract, a two-page cut.

B. One paper, my outline, your theory as its object section. The Γ framing leads, your geometry and tractability become §2 with proofs in the appendix. Cost: more rewriting of your prose, and the tractability results lose prominence.

C. Two papers. Yours at ICLR, ours elsewhere. Cost: both use one corpus and one definition, each is weaker alone, and the elicitation arm is what puts yours on real structure.

I recommend A. It keeps what you wrote, closes both experimental markers with measured results, and is the only option that fits two days before the lock.

Nine decisions for the text were listed in my outline of 12–13 September, and three of them change the abstract. **D1**, the title: name the analysis, not the radius. **D2**, the headline value: lead with the knowledge interval and report r_0 beside it as the number of claims the conclusion rests on, named. **D9**, the framing after M1b: accept the measurement framing declared on 7 September for exactly this tie. These are recommendations for the two of us to decide, not decisions taken.

Without looking (answer 61, in the appendix)

Under option A, which parts of your draft are kept, which are rewritten, and which one is replaced by what?

Before reading

- Name one sentence the paper must not contain, even though a result makes it tempting.
- After 18 September, what about the author list can still change?

14.4 Before Friday, and before the 25th

ICLR 2027 Author Guidelines, read on 16 September

«Please note that no authors can be added or removed after the abstract submission deadline (author order can be changed after the abstract submission deadline up to the full paper deadline).» «Abstracts submitted by the abstract submission deadline must be genuine; placeholder or duplicate abstracts will be removed.» «Please coordinate among the authors and make sure that everyone has an OpenReview profile before the abstract submission deadline.» «all submissions must have at least one author who is registered to review at least 3 papers». Titles «can be edited up to the paper submission deadline». «At the time of submission, the main text should be 9 pages or fewer.» «Required: AI use statement.»

So the author list, and with it everyone's OpenReview profile, must be settled by 18 September; only its order can move afterwards. In order, before Friday 18 September, AoE:

1. Settle the author list and check every author's OpenReview profile.
2. Decide the framing (D9) and the headline value (D2), and choose the merge option.
3. Narrow the novelty sentence to what was searched. The abstract's «nothing currently tells the analyst how much of the conclusion rests on them» is too broad: Taeb, Guo and Henkel's §6 already reports how large a misspecification ball around the analyst's graph each adjustment set tolerates (one example, in ADMGs, by adding edges to the specified graph). Chapter 12 lists the novelty search still owed before the abstract.
4. Write the abstract from the merged claims, one sentence each for the interval, the radius in claims, the tie with the screen and the class assumption, and check it against the list below.
5. Submit title, abstract, authors and subject areas, and register at least one author as a reviewer.

Before Friday 25 September, AoE: replace §6 by the protocol; make claims the reported unit, with 39, 15 and 0 as the reason; apply the ten items of Chapter 13; replace the admission in §7 that

nothing speaks to discovery error by the estimated-class result; move the clinician sentence to page one; cut to nine pages; write the AI use statement, which your `95_statements.tex` does not have yet.

Sentences the paper can print

- Counted in the analyst's claims, the certificate did not over-certify on any of 3,014 certifiable rows of eleven knowledge conditions on the true CPDAG: whenever the committed set was invalid at the truth, it licensed fewer withdrawn claims than the analyst had made false ones. This follows from the definition when the equivalence class and the validity criterion are correct, and is not an estimate. Counted in covering hops, the same audit found over-certification on 39 of those rows.
- With one wrong claim, the stated report covered 0.597, and the knowledge interval at one claim restored 0.980 at 85 % of the calibrated width of discarding the knowledge (23 networks, semi-synthetic parameters, $n = 1000$).
- At matched depth the report differs from a leave- k -out screen only where a retraction changes the adjustment set while leaving it valid; what it adds is the depth the analyst needed and the name of the claim the conclusion rests on.
- When the class is learned from data, the generating graph extended the analyst's graph on none of 244 certifiable rows, so the guarantee does not apply.
- The smallest retraction that breaks the set is a minimal correction set, a notion from the satisfiability literature, instantiated here on Meek-closed background knowledge.

Each of those sentences has a tempting neighbour that the evidence does not license, and the checklist below is the list of those neighbours.

Checklist: sentences the paper must not contain

- that the radius, the certificate or the stop rule *beats* a leave- k -out screen;
- that anything is certified robust under an *estimated* CPDAG;
- a continuity bound («a small knowledge error gives a small bias»);
- a probability reading of r , or that r estimates how wrong an expert is;
- that zero dangerous rows for the claim certificate is a statistic;
- that language models are the subject of the paper;
- the retracted numbers and phrases: «0/791», « O^* is a local functional», «up to $128\times$ »;
- that the MPDAG poset of Taeb, Guo and Henckel is graded.

Without looking (answer 62, in the appendix)

Give four sentences from the checklist that the merged paper must not contain, and for two of them the measurement that rules them out.

Without looking (answer 63, in the appendix)

What does the ICLR 2027 guideline allow to change about the authors after the abstract deadline, and until when?

Do not move on

Without looking, write the five steps before Friday in order, and name the one whose result can no longer be changed after 18 September. If you put the abstract before the framing decision, reread the section on the three options.

Exercises

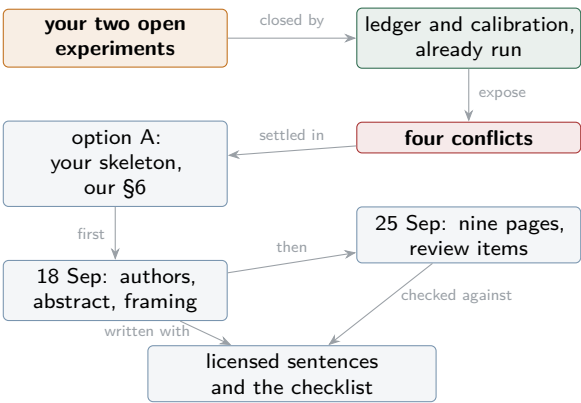
Exercise 14.1. Rewrite these two sentences of your abstract for option A, respecting the checklist and with every number carrying its population: «On synthetic and on published causal graphs we ask whether the radius ranks instances by whether the committed set survives corrupted background knowledge, and how the quantity is distributed. The radius is a diagnostic and not an objective: it prices the structural risk carried by an adjustment set, in the units the analyst reasons in.»

Solution. One possible rewrite. «On 27 networks with background knowledge from elicited and chance-level sources, audited only after every report was frozen, the report counted in the analyst’s claims never over-certified, while a count of covering hops did so on 39 of 3,014 rows. On 23 networks with semi-synthetic parameters, one wrong claim lowered the coverage of the stated report to 0.60, and the knowledge interval at one claim restored 0.98 at 85 % of the calibrated width of discarding the knowledge; at matched depth the report takes the decisions of a leave- k -out screen, and it names the claim the conclusion rests on. The guarantee assumes a correct equivalence class: with classes learned from data, the generating graph extended the analyst’s graph on none of 244 rows.» What changed: «ranks» goes, since ranking is the synthetic panel’s question; «in the units the analyst reasons in» becomes a measured contrast between claims and hops; the tie is stated rather than hidden; and the class assumption moves into the abstract.

Exercise 14.2. Classify each sentence as (L) licensed as written, (C) licensed only with its condition stated, or (F) forbidden. (a) «The claim certificate never over-certifies.» (b) «The breakdown radius outperforms a leave-one-out screen.» (c) «With an estimated CPDAG, the committed set remains certified robust.» (d) «One wrong claim lowered the stated report’s coverage from 0.967 to 0.597 (Panel A, $n = 1000$).» (e) «A radius of 2 means the expert is probably wrong about two orientations.» (f) «The smallest breaking retraction is a minimal correction set.» (g) «The space of knowledge states is graded.» (h) «Small errors in background knowledge produce small bias.»

Solution. (a) C: true given a correct class and a validity oracle that matches the instrument, and structural rather than statistical, so both conditions go in the sentence. (b) F: they tie at matched depth, and the radius-based rule separates only from the depth-one screen. (c) F: the truth extends the analyst’s graph on 0 of 244 certifiable rows. (d) L: measured, with its population. (e) F: a probability reading of r . (f) L, provided the citation is there: the name is conceded, not claimed (docs/PRIOR_ART.md). (g) C: on a fixed skeleton and under the anti-exchange conjecture; the general MPDAG poset is not graded (Proposition 9). (h) F: the radius is a breakdown value, not a continuity bound.

Chapter map



Part VII

The repository

Chapter 15

The repository, file by file

Everything in the previous fourteen chapters is on one branch of `bkrobust`, `ze/experiments`. This chapter is the map: how the branch is built, where each chapter's files are, what to run first, and the caveats I know about. The same map, in markdown, is `ZE_EXPERIMENTS.md` at the root of the branch.

15.1 How the branch is built

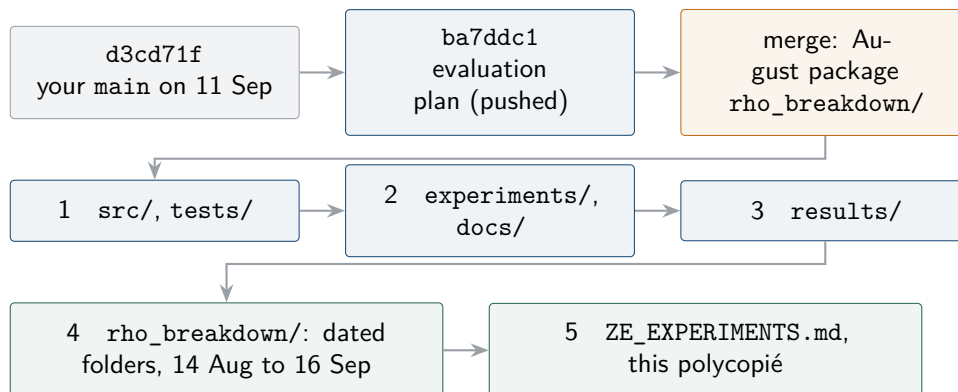
Before reading

- Why is the branch built on top of your history rather than as a fresh repository?
- Which of the five commits would you read first if you only cared about Table 4?

The idea in one sentence

The branch shares your history, so `git merge` works and `git diff` against your last common commit shows exactly what I added. Nothing on it rewrites a file of yours without a commit that says why.

The branch starts from `iclr2027-eval-plan` (commit `ba7ddc1`), the evaluation plan that is already on GitHub. On top of it come six steps, drawn in the diagram below: the merge of the August branch `rho-breakdown-evidence`, which I never pushed, then five commits.



Commits 1 to 3 are the protocol run of Chapters 5 to 12, as it sat uncommitted in my working copy of your repository: 158 files, split into library changes, scripts with docs, and results. Commit 4 brings in the experiment folders that lived outside the repository. Commit 5 is the index and this document.

The new tests, `tests/{benchmarks,criterion,estimation,knowledge}`

`pytest -q` on the nine new test files of the protocol run: **39 passed** in 4.4s, 16 September, with the library as committed on the branch. They cover the audit oracle (back-door against GAC), the discovery arm, measurement selection, the sortability probe, the closed-form O^* ,

the knowledge interval and the ancestral block.

That measurement covers the new tests only. Your older tests were not part of it, so run the whole suite once after merging, before trusting that the library changes left your results untouched.

Getting it

`git fetch origin ze/experiments`, then `git diff d3cd71f origin/ze/experiments -stat` to see what changed in your package. Your main has moved since d3cd71f, so merge, do not rebase; the library changes touch `benchmarks/measure.py`, `hybrid.py` and `search/exact*.py`, which are the places a conflict would appear.

Without looking (answer 64, in the appendix)

Which commit on the branch holds the pre-registrations of the protocol run, and why are they in the same commit as the tables?

15.2 Where each chapter lives

Before reading

- Which folder holds the script that reproduces the 27 cells of panel A?
- Where is the test that stops an audit from using the back-door criterion?

The table below is the index I would want if I were you. Paths are relative to the repository root.

chapter	folder	start with
2	<code>rho_breakdown/pilot/, e1prime/</code>	<code>REPORT.md</code> , <code>NUMBERS.md</code> , <code>e1prime/RESULTS.md</code>
3	<code>rho_breakdown/locality/</code> , <code>x3_skeleton/, rerank_2026-08-14/</code>	<code>locality/VERDICT.md</code> , <code>x2/cfirewall/</code> <code>STATUS.md</code> , <code>x3_skeleton/RESULTS.md</code>
4	<code>rho_breakdown/</code> <code>radius_redef_2026-09-07/</code> , <code>rnull_real_2026-09-09/</code>	<code>EXPERIMENTS.md</code> , <code>IMPROVEMENT.md</code>
5	<code>rho_breakdown/</code> <code>eval_protocol_2026-09-11/</code> , <code>docs/</code> , <code>results/stage0/</code>	<code>02_PROTOCOL.md</code> , <code>06_CODE_FIXES.md</code> , <code>docs/PRIOR_ART.md</code>
6	<code>results/frame/</code> , <code>results/ledger/</code>	<code>docs/PROTOCOL_RUN.md</code> , <code>TABLE0_FUNNEL.md</code>
7	<code>results/elicit/</code> , <code>experiments/remote/</code>	<code>PREREGISTRATION_SCALE.md</code>
8	<code>results/ledger/</code> , <code>src/bkrobust/</code> <code>benchmarks/audit.py</code>	<code>TABLE4.md</code> , <code>CHECKS.md</code>
9	<code>results/pest/</code> , <code>results/substrate/</code>	<code>TABLE7.md</code>
10, 11	<code>results/interval/</code>	<code>TABLE1_L95.md</code> , <code>PREREGISTRATION.md</code>
12	<code>rho_breakdown/</code> <code>panelB_T3_2026-09-16/</code> , <code>table1_dossier_2026-09-16/</code>	<code>PANELB-T3.pdf</code> , <code>TABLE1-DOSSIER.pdf</code>
13	<code>rho_breakdown/</code> <code>theory_checks_2026-09-16/</code>	the enumeration script

Five notes in the dated folders are in Portuguese, because I wrote them for myself on the day. `ZE_EXPERIMENTS.md` summarises each in a paragraph; their content is also in Chapters 3, 4 and 5.

Absolute paths in the dated folders

The scripts under `rho_breakdown/` were run where they were written and still carry absolute paths of my machine (for example `/Users/josecosta/bkrobust/src` added to `sys.path`). They will not run from a fresh clone without editing that line. `ZE_EXPERIMENTS.md` gives the mapping from each such path to its folder on the branch. The protocol-run scripts in `experiments/` do not have this problem: they resolve paths from the repository root.

Without looking (answer 65, in the appendix)

You want to check that the panel-A cells of Table 1 are what the pipeline produces. Which script, and what does it do when a cell disagrees?

Without looking (answer 66, in the appendix)

Which file must an audit of adjustment-set validity call, and which function in the package must it not call?

15.3 Caveats I know about

Before reading

- Which stage-0 file quotes a number its own data files no longer give?
- Which printed table has a selection rule that favours its own headline?

Four things about the files deserve a warning before anyone quotes from them.

Stage-0 calibration, `results/stage0/calibration_summary.json`

The corrupted-knowledge calibration on the committed rows was rerun on 13 September after a seeding fix: **8,911 rows scored of 16,496 draws**. `results/stage0/STAGE0.md` and `PREREGISTRATION.md` still quote the first run, **8,553 rows**, which seeded with Python's salted `hash()` and cannot be reproduced across processes. The calibration files on the branch are the rerun. It changes one reading: on the corrupted rows the hop radius is dangerous on 3 rows (was 0), the free rule on 9 (was 8), the claim radius on 0 (unchanged).

Salted `hash()` seeded six things: the chance arm of the ledger sweep, the Lever 0 differential, the calibration, the first frame-status run, the questionnaire builder and the source-overlap probe. `docs/PROTOCOL_RUN.md` heads its list «Four experiment scripts» and names five; Chapter 6 has the full account. All now seed with SHA-256 of a stable name. The frozen questionnaire does not regenerate byte for byte from its builder, so the committed file, with its hash, stays the authoritative input and the builder refuses to overwrite it.

The second caveat is the selection rule of T3 in `experiments/interval_tables.py`, analysed in Chapter 12: it sorts a finite null radius first before taking five rows per network. The fix is one sort key, or printing all 37 eligible rows. The third is the validity oracle above, which is still called in two places I have not changed. `experiments/declared_queries.py` scores validity at the truth with the back-door function; the committed set judged at the truth is not the truth's optimal set, so the two criteria can disagree there (they did on 4 of 1,169 ledger rows). Rerun it with the GAC audit before quoting its validity column. And the hop radius, through `hybrid.py`,

tests validity with `mpdag_criterion.is_valid_mpdag`, while the claim radius uses the generalised criterion; the stricter test can only make r_{hop} smaller, so the hop unit's over-certification counts in Chapter 8 are, if anything, conservative. The fourth is what is deliberately absent: `figures/*` and `results/axisa3/networks/acquisition_manifest.json` were regenerated on my machine by a reproduction run and are not on the branch, and neither is the pgmpy source tarball.

Without looking (answer 67, in the appendix)

State the one-line fix for the T3 selection rule and the alternative that avoids choosing at all.

Without looking (answer 68, in the appendix)

Why does the committed questionnaire stay authoritative even though its builder is now reproducible?

Do not move on

Pick one number from Table 4 in Chapter 8. Write down the file you would open to check it and the script that wrote that file. If you cannot name both, reread the second table of this chapter.

15.4 What is not here

The paper drafts are not on the branch: your draft of 16 September is yours, and my outline of 13 September is a planning document I can send separately. Two third-party PDFs that one August experiment read (Guo and Perković, arXiv 2010.08611; Taeb, Guo and Henckel, arXiv 2511.10625) are left out; fetch them from arXiv. The response cache of the local suppliers is regenerable from the questionnaire and stays untracked, as your `.gitignore` already says.

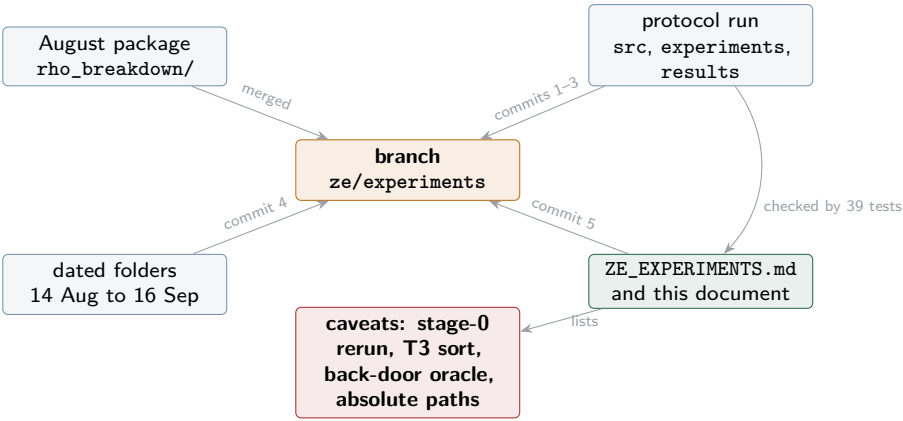
Exercise 15.1. *A reviewer asks where the claim «the certificate counted in claims is dangerous on no deployment row» comes from. Using the second table of this chapter, name the chapter, the folder, and the two files you would send, and say which of the two was written first.*

Solution. Chapter 8; folder `results/ledger/`; the files `TABLE4.md` (the table with the dangerous-row counts) and `PREREGISTRATION.md` (the predictions and the bucket definitions). The pre-registration was written first, before the rows were read; the table was built after the sweep was frozen.

Exercise 15.2. *Before merging, you want a rough idea of the conflict surface. Commit 1 touches eight existing files of the package and adds five. Without running git, estimate which of your recent branches is most likely to conflict, and what you would check first.*

Solution. The conflicts can only appear in the eight modified files: `dagitty.py`, `measure.py` and `parsed.py` under `benchmarks`, `conventions.py` under `core`, `hybrid.py`, the `__init__.py` of `knowledge`, and `exact.py` and `exact_fast.py` under `search`. The five added files cannot conflict. A branch of yours that changed the knowledge-selection or the search dispatch (for example work on `measure.py` or `hybrid.py` after 11 September) is the most likely conflict. Check first with `git merge-tree` or a trial merge on a throwaway branch, then look at `measure.py`, where both the thinning rule and the rejection labels live.

Chapter map



Answers

1. Chol→Statin. The analyst asserts BMI→Chol, the edge Chol–Statin is undirected in the CPDAG, and BMI and Statin are not adjacent, so Meek’s rule R1 orients Chol→Statin to avoid a new unshielded collider at Chol. G_0 therefore adds five orientations to the CPDAG, of which four are asserted.
2. $r_{\text{claim}} = 2$ with witness {Smoke → BMI, BMI → Chol}; $r_{\text{hop}} = 3$. Both reach the same failing state, where Smoke–BMI–Chol–Statin is undirected. The hop walk must also un-orient Chol→Statin, which R1 propagated from BMI→Chol; withdrawing the asserted claim removes that propagation with it, so the claim count does not pay for it.
3. Your r_{val} (Definition 3) counts covering hops in the space of knowledge states; this document writes it r_{hop} (instrument B6). A bare r_{val} elsewhere in this document is the validity radius counted in the analyst’s claims, the quantity the ledger computes as r_{claim} (instrument B7).
4. With w the number of the analyst’s claims false at the truth, a radius r covers the row when $w \leq r - 1$ (an unreached radius covers everything). Covered and Z invalid at the truth: dangerous, the failure. Covered and valid: held. Not covered and invalid: safe. Not covered and valid: slack, the price of caution.
5. Every reported quantity is a function of $(\hat{C}, K, X, Y, \hat{\Sigma}, n)$; the generating DAG enters only to draw the samples and, after the rows are frozen, to audit. `results/ledger/rows.csv` records it in `true_dag_on_path`, `chat_source_sha256` and `generating_dag_sha256`.
6. Its `sign` and `rel50` criteria compare each perturbed estimate with τ , the true total effect (`pilot/code/analyse.py`, `breakdown_radius`), which an analyst never has. Only `ident`, which asks whether some Meek-consistent member loses amenability, is read off the MPDAG alone; it says nothing about how far the estimate moves.
7. At $n = \infty$ the standard error is zero, so a problem is censored exactly when no consistent, amenable member of the full ball moves the estimate at all; that share is 0.515 (gate G6). Of the 1,074 censored problems at $n = \infty$, 94.6 % are such exact zeros and 5.4 % lose amenability under every perturbation; none is censored because Meek caught everything. Censored means the ball is inert, not that the analysis is robust.
8. The instrument fires when $d(\rho) > z \text{SE}_n$. SE_n shrinks like $n^{-1/2}$, but an inert ball has $d(\rho) = 0$ exactly, and zero exceeds no threshold, not even the zero threshold at $n = \infty$. The share of problems with $d(1) = 0$ is 64.6 % at every n in the grid.
9. 0.951, 0.720 and 0.568. They are computed among reversals Meek accepts: the check catches 23.6 % of single and 38.6 % of double reversals, which leave the sample. The scope is linear Gaussian iSCM, Erdős–Rényi graphs with p from 5 to 10, and the oracle CPDAG.
10. X1: a spurious required edge replacing a true claim at distance ≥ 1 moved O^* into invalidity 142 times in 3,639 (**original**). X2: 70 of 10,615 knowledge sets whose every reversed claim lies at distance ≥ 1 changed O^* , all into invalidity. What survives is narrower: a single reversed claim that does not touch X or Y never changed O^* in the complete enumeration at $p \leq 5$, and 95.8 % of those trials had an amenable base, where class invariance already forbids any change.
11. Distance ≥ 1 holds exactly when the claim’s edge touches neither X nor Y , and no graded distance does any work: the informative trials all sit at distance 1 and there are none at distance 2. Of 137,040 far trials with both orientations amenable, 131,280 have an amenable base, where O^* cannot move at any distance; 5,760 carry information.
12. For an MPDAG G_0 amenable for (X, Y) and a spurious claim $a \rightarrow b$ on a non-adjacent pair,

if the closure H is acyclic then either H is not amenable or $O^*(H)$ is valid in every DAG of $[G_0]$. The census found 0 violations in 70,375,325 trials, complete at $p \leq 5$; and the control arm, where the check fails and the edge is stamped anyway, has 21.8% invalid sets, so the detector can fire.

13. INCONCLUSIVE. The ratio of silent-bias rates, distance 0 against distance ≥ 1 , is 7.10 and passes the bar of 5, but the rate at distance ≥ 1 is 0.0231 and fails the bar of 0.02. The cut at two hops (0.1376 against 0.000423, 326 times) was chosen after seeing the data, so it is a hypothesis to re-register and re-run on fresh draws.

14. The fixed set $\{\text{BMI}, \text{Exercise}\}$ stops being valid in every DAG of the graph after two claim retractions ($r_{\text{val}} = 2$), while the interval of estimates is the single point -0.4191 up to radius two and opens to $[-0.8711, -0.4191]$ only at radius three, a width of 0.4520, or 108% of the effect.

15. For every budget b , the union of retractions of at most b claims equals the union of reversals of at most b claims. A world compatible with the re-closed graph after retracting S contradicts K on exactly one subset $\sigma \subseteq S$, so the retraction of S is the disjoint union of the reversals of its subsets. It merges the “flip claims” and “retract claims” families: reversal models the error, retraction builds the hedge.

16. L2: the hull over the load-bearing claims already equals the hull over all claims, so no screen narrows the interval. The idea pays in cost (the query-local screen needs 3.13 instead of 4.31 enumerations per problem, for a width of 0.621 against 0.630 of the blanket) and in elicitation (claims away from the query moved coverage by 0.2 points, so do not ask about them).

17. From $0.672 \times 1 + 0.328 \bar{x} = 0.710$, $\bar{x} \approx 0.116$. Where the selective hull does anything it costs about an eighth of the blanket; the average saving of 29% is carried by a third of the problems, and a headline must say so.

18. The unit: the harness radius (r_{hop}) is a breadth-first distance in covering hops, r_0 counts retracted claims. And the stopping predicate, which is the one that changes the meaning: r_{hop} stops when some graph makes the committed set Z invalid, a property of an intermediate step; r_0 stops when the hull of possible effects contains zero, a property of the conclusion. A set can become invalid while every compatible effect stays far from zero, and then the validity radius raises an alarm about nothing the analyst asserted to the world.

19. They come from different admission rules. My draws admitted every Meek-consistent query, including queries with no causal path, and counted unidentified queries as “never”. Your gate rejects those and, at full coverage, admits a pair only if one retraction breaks O^* , which makes an unreachable radius impossible; your gated random arm has 0 unreachable in 2,334. The contrast is between pipelines, not structures.

20. It may generate the sample, which is what nature does and all the analyst receives, and it may audit the certificate after the deployment rows are frozen and hashed. What makes them harmless is the order: both come after the freeze, so no reported observable quantity can have been chosen by looking at the truth.

21. Nothing: all 518 are non-amenable, so no definition of the optimal set can certify them; the deletion is right and the label `o_g0_not_identified` is wrong. The defect is the thinning in `select_knowledge`, which sorts the recovering set alphabetically and keeps an even stride; on 29 of 40 testable cases another subset of the same size would have kept the query amenable.

22. Because `fast_gate` admits a pair only if retracting one claim of the recovering set breaks the optimal set, and at coverage 1.0 the asserted set is the recovering set: admission is the definition of $r_{\text{claim}} = 1$. On `paths` one claim compels the whole chain, so the hop radius is 14 while the analyst can afford zero wrong claims; “13 safe moves” would read a Meek-cascade length as an error budget.

23. Predict the separation s (hops from X to the nearest member of O^* inside the undirected

component of X), and 1 where no member lies in the component. Exact on 636 of 831 rows, within one on 813. It over-certifies on 16 and under-certifies on 179: a conservative lower bound to beat on sharpness at equal validity, not a competitor on agreement.

24. The size of a minimum minimal correction set: the smallest subset of asserted constraints whose removal makes the property fail, failure being upward-closed because adding constraints cannot invalidate a fixed valid set. What is left is the instantiation (knowledge closed under Meek, adjustment validity in an MPDAG, a certificate stated against the analyst's own claims) and the measurement of what it buys over the free statistics once the truth is revealed.

25. Five networks above 500 nodes, excluded for cost: 100 rows. **pathfinder**, whose only chain component holding a frame treatment has 85 nodes, above the questionnaire's cap of twenty, so no supplier was asked: 20 rows. Both cuts use only the size of a network and of a chain component, which are facts of $\hat{C}$.

26. Two namings times two presentation orders: $120 \times 4 = 480$ open items; plus two control items (real and scrambled, first order) on each of the 30 networks with compelled edges: 60. An order cannot contain a cycle, whereas pairwise directions from an imperfect source produce cycles at scale; the inconsistency that remains is a new v-structure against $\hat{C}$, which the Meek repair handles.

27. The model never stated that pair; its order implied it, so the claim carried no confidence, and **repair()** drops unscored claims last. The nine claims stated at confidence 1 went first, then the other unscored claim, **Age** $\rightarrow$ **Disease**, and the closure succeeded with one claim left. It is false: **LungParench** is the parent of **C02**.

28. The search enumerates retraction sets in a fixed order and names the first one that breaks the set; when several single claims break it, only one is named. On the four **Kampen_2014** rows with treatment **AFF**, the named claim **AFF** $\rightarrow$ **ALN** is true, and the set is invalid because of the supplier's four false claims on that network. The name is a minimal correction set, not necessarily the culprit.

29. The hop radius, from 0 to 3 dangerous rows, all at corruption rate 0.5 (the free rule went from 8 to 9). The claim radius stayed at 0. Its zero is structural: given a correct class and a sound criterion, retracting the w false claims leaves a graph the truth extends, so $r_{\text{claim}} \leq w$ whenever the search reaches depth w , whatever the draw.

30. The two act in opposite directions on the certificate and have different denominators. A decline leaves an edge open, widens the analyst's graph and may make the query non-amenable; a false claim closes the edge the wrong way and may invalidate the committed set. Commission is a fraction of kept claims, omission a fraction of asked pairs, so their sum is a fraction of nothing.

31. A committed set is invalid when at least one wrong claim sits where the query needs it. The larger model declines far less, so it asserts more edges in each component and more queries become certifiable; every extra claim near a query is another chance that one of them is wrong. A lower rate per claim times more claims per query leaves the chance of at least one wrong claim where it was. Accuracy per claim and validity per query are different quantities, and the certificate sits between them.

32. At 72B: control minus open-edge commission $+0.004 [-0.153, +0.159]$, paired over 26 networks, against $-0.153 [-0.324, +0.009]$ at 7B. The 72B's error on compelled edges is 0.232, higher than the 7B's 0.191; the bridge is unbiased because its control error and its open-edge commission now agree, not because the control error fell.

33. Closure per claim depends on how much a source says. The 7B declines a lot (137 of 369 pairs) and keeps few claims per component, and each drags orientations along through Meek's rules, as the oracle's minimal set does. The 72B keeps claims on almost every pair (281 of 369); when nearly every edge of a component is asserted directly, the closure adds little and the ratio is

close to 1. “Unrealistic” only means something against a reference source, and the real sources disagree with each other.

34. Because $r = \infty$ covers every w , so only validity at the truth separates held from dangerous. For the claim radius such a row contradicts the proposition of Section 8.3 as long as the class is correct: retracting the w false claims leaves a graph the truth extends, where an invalid Z must fail, so some retraction would have broken Z . The row is a code defect or a violated premise, which is what the inspection trigger of Section 8.4 exists to separate.

35. On false claims yes, but only in a post hoc paired contrast: chance is +0.143 [+0.006, +0.287] higher on 24 networks (0.479 against 0.329). On sets invalid at the truth it is not shown: +0.082 [−0.020, +0.176] (0.378 against 0.288). L4 asked for separated intervals and is not supported as registered, so the elicited condition is reported as not shown to be informative on this substrate.

36. Retract exactly the w false claims: the rest are true, so if the truth is in the class of $\hat{C}$ it extends the re-closed graph, where an invalid Z fails, giving $r_{\text{claim}} \leq w$ (and any censored floor below depth w is at most w). The premise is a correct class, with certificate and audit on one criterion. The zero is a consequence, not a frequency: a bound like $3/229 = 0.0131$ describes a rate that more rows could tighten, while this zero fails only when the premise does.

37. The claim radius uses `is_gac_valid_mpdag`, the generalised criterion; the hop radius uses `mpdag_criterion.is_valid_mpdag`, with the back-door strengthening. Back-door validity implies GAC validity (THEOREMS.md §13), so back-door failure comes no later and that hop radius is never larger than a GAC one. A larger radius covers more, so a GAC hop radius would be dangerous on at least the same rows.

38. Because its skeleton precision there is 0.47 to 0.67: BIC on few samples returns a sparse, nearly fully oriented graph, so the low open fraction is a wrong graph with orientations, not information. Fewer open edges means fewer questions to the supplier and more certifiable queries, all built on orientations the truth does not have.

39. Its reference skeleton misses no edge and adds one, so the invalid sets of the supplier and of chance come from their claims, which the certificate sees: all 6 and 14 are detected. The truth still does not extend the analyst’s graph, because 14 of 62 compelled edges are wrong, and the audit conditions, which assert only true claims, are dangerous on 3 and 4 rows: an error on a compelled edge that no retraction reaches. The limit shows up exactly where the premise almost holds.

40. With certificate and audit on the same criterion, and supposing the true DAG belongs to the class of $\hat{C}$: if the committed set is invalid at the truth and w claims are false, retracting those w claims leaves a graph the truth extends, where the set fails, so $r_{\text{claim}} \leq w$ and the claim radius does not over-certify. The premise is that the class contains the truth. On the panel it held on 0 of 244 certifiable rows.

41. The medians reverse, 0.021 against 0.028. Without `magic-irri` the means reverse too, 0.047 against 0.050, because the pooled order comes from that network’s 14 chance rows at floor 1, averaging 0.344. And of the eight network-by-condition cells with rows at both floors, four go each way. On an invalid set the bias is structural anyway, and a sampling band says nothing about it.

42. bnlearn’s 17-arc network inferred in Sachs et al. (2005), with tables fitted to discretised data. The Tetrad ground truth has 20 edges: it shares 16, reverses `Plcg`–`PIP3` and adds three. The CPDAG of the 17-arc graph is fully undirected, so the reversed edge is a claim; five conditions answered `PIP3` → `Plcg` and two `Plcg` → `PIP3`, so which group is scored as wrong depends on which object is called the truth.

43. Γ bounds the ratio of the odds of treatment of two members of a pair that agree on the observed covariates: a hidden confounder can tilt that chance by at most this factor. At $\Gamma = 1$ both members have the same chance, which is a coin flip within the pair. The worst-case p-value is a maximum over the admitted assignments, and the set admitted at Γ' contains the set admitted

at any $\Gamma < \Gamma'$; a maximum over a larger set cannot decrease. So there is a single point where significance is lost, Γ^* .

44. Γ becomes r , the number of the analyst's claims allowed to be wrong. The worst-case p-value becomes the knowledge interval I_r , the hull of the 95 % bands over the graphs of the retraction ball $B_r(K)$. Γ^* becomes the null radius r_0 , the smallest r at which the band contains zero. The validity radius r_{val} has no counterpart: a matched design does not read an adjustment set off a graph that can break. The named witnesses of each endpoint are also ours.

45. PLAIN covers 0.597 with $L_{95} = 5.123$; I_1 covers 0.980 with $L_{95} = 0.851$ [0.76, 0.94], over 359 informative queries on 23 networks. The 5.12 means that to reach 95 % coverage the committed report's intervals would have to be stretched to a mean width of 5.1 times the calibrated width of the whole class: trusting one wrong claim costs more than discarding all the knowledge.

46. Panel A, true claims ($b = 0$), frame queries with an established effect: 205 of 221 (92.8 %) have $r_0 = 1$ (71) or beyond 3 (134); restricted to informative queries 148 of 164 (90.2 %); Panel B 25 of 25. So the paper leads with the knowledge interval and Table 1 and reports r_0 as “the conclusion rests on no claim, or on this named claim”, printing the distribution; the title that named the radius was withdrawn.

47. The two-claim stop rule against the leave-one-out screen, at two wrong claims. Only that one, because at one claim the screen and the rule are the same policy (a set unchanged by every single retraction is valid under every single retraction), so any difference has to come from two retractions interacting. The sentence: the work would be “a measurement of when knowledge error is silent rather than a method”.

48. Premise: every set the committed-set procedure returns is valid in the graph it was returned for. With it, if Z is invalid in some G_S with $|S| \leq r$, the committed result of G_S is either a valid set (so different from Z) or a no-set verdict, and the screen triggers. The converse fails because the screen also triggers when a retraction changes the optimal set while Z stays valid; there the rule certifies correctly and the screen widens without need.

49. 0.910 over the 134 informative queries with a committed set (20 networks), the population of Q4, which predicted below 0.5; 0.639 over all committed queries; zero over the 57 committed queries outside the informative stratum, where the class already identifies the effect.

50. Because the policies differ in two things: the depth of the certificate and the budget of the fallback interval. On the 34 queries with $r_{\text{val}} = 1$ both refuse, but the rule reports I_2 and the screen I_1 ; with two wrong claims I_2 covers more, and that stratum contributes +0.092. Only the 14 interaction queries (+0.164) measure depth, and against a screen that also tries pairs, with the same fallback, the difference is +0.000. The +0.256 [+0.149, +0.346] is at $n = 1000$ on 59 queries and 8 networks; the population line of the same contrast is +0.305 [+0.182, +0.410].

51. Committed report: $\lambda^* = 20.416$, mean half-width 0.3855, calibrated width $2 \times 20.416 \times 0.3855 = 15.74$. Class: $\lambda^* = 1.000$, mean half-width 1.5361, calibrated width 3.072. $L_{95} = 15.74/3.072 = 5.123$, over 7,180 replicates of 359 queries.

52. On 16 of the 17 informative T3 rows the interval after one retraction equals the whole-class interval to four decimals; the exception is `mediator`, where one retraction covers 57 % of the class and two saturate it. The pre-registered rule wanted both radii in $\{2, 3\}$ and a plateau before the break on a real-coefficient network under truthful knowledge; the curve is a single step, so no row qualified, and the rows with radii 2 or 3 widened 27 to 34 times before breaking.

53. Finite population r_0 on 13 of 17 printed queries (76 %), 3 of 20 dropped (15 %), 16 of 37 eligible (43 %). The sort key (`r0_pop == CENSORED, frame_index`) in `experiments/interval_tables.py` puts finite null radii first before taking five per network; the rule is in §6 of the pre-registration. Fix: sort by `frame_index` alone, or print all 37, and state the real-coefficient rule and the eligible count in the caption.

54. T11. Panel one is the certificate ladder over 3,014 certifiable rows (eleven knowledge conditions, 27 networks, 902 sets invalid at the truth): the claim count over-certifies on 729, the separation rule on 15, the hop radius on 39, the claim radius on 0. Panel two is the price table on M1's Panel A at $n = 1000$ (391, 359, 131 informative queries) with the leave-one-out screen as a row, and the refusal rate 0.910 of 134 printed under the float with its own denominator. Blocking: the novelty search for C1 has never been run in the wording of the claim, and the author list must be settled by 18 September.

55. $d(G, H) = |K_G \Delta K_H|$. Lemma 2 (a cover changes K by exactly one orientation) gives $d \geq |K_G \Delta K_H|$, since each step moves the symmetric difference by at most one. Theorem 2 gives the join J with $K_J = K_G \cap K_H$, and with Lemma 2 again the maximal chains from G and from H up to J have $|K_G \setminus K_H|$ and $|K_H \setminus K_G|$ steps, so $d \leq |K_G \Delta K_H|$. It inherits the anti-exchange condition through Lemma 2.

56. THEOREMS.md §14 proves the two criteria coincide on the optimal set of the graph being judged. The main text judges the fixed set $Z = O^*(G_0)$ in other states, where it is an arbitrary set, so the proof does not reach it; the evidence is measured instead (identical radii on 5,304 instances in §14's BFS check), and the ledger shows the criteria can part there (4 of 1,169 committed sets rejected by back-door at the truth). Appendix C's «Robust selection» computes radii for every valid adjustment set, which is the regime where the two criteria part (6.2% of 372,430 cases), so it must say which criterion it used.

57. The model-oriented poset of all MPDAGs on $|V| \geq 4$ vertices is not graded (and, by the paragraph after it, not semimodular and without a characterisation of its covers). Their poset lets covers change the skeleton; the space of knowledge states fixes the skeleton and the v-structures of $\hat{C}$, so only orientations vary. There their pseudo-rank equals $2m - |\text{dir}(G)|$ and the draft's rank is its negation shifted, so gradedness can hold (under the anti-exchange condition) without contradicting Proposition 9.

58. $\tau_b(r_{\text{val}}) = -0.091$ in flip, cov 1.0, bw 0.00, and the confirmed null 0.032 $[-0.112, 0.168]$ in flip, cov 0.5, bw 0.25 at 1,000 draws. In flip, cov 1.0, bw 0.25, SHD scores 0.582 and $|K|$ 0.557 against 0.355 for the radius.

59. The §6.3 elicitation arm is the ledger: a causal-order questionnaire over 120 components (369 open pairs, 540 items) frozen and hashed first, 8,308 rows over 14 conditions in `results/ledger/`, a matched chance control and an audit opened after freezing. The §6.2 real-structure evaluation is the corrupted-knowledge calibration on the committed rows (8,911 scored of 16,496 draws over 24 networks, in `calibration_summary.json` under `results/stage0/`), which scores instruments in the four buckets rather than by τ_b against the survival area; the τ_b design itself has not run on real rows.

60. A leave- k -out screen: withdraw one (or two) claims, re-close, and flag if the committed result changes. At matched depth the stop rule and the screen take the same decisions except where a retraction changes the set while leaving it valid: they differ on 2 of 162 rows at one wrong claim (both panels) and on 5 of 754 queries at depth two, and the coverage difference against the pair screen at two wrong claims is +0.000 $[+0.000, +0.000]$ (Panel A, $n = 1000$). Against the one-claim screen at two wrong claims, the two-claim stop rule covers +0.256 $[+0.149, +0.346]$ more at $n = 1000$ (+0.305 in the population row). The licensed reading is that the radius tells the analyst which depth was needed, not that it beats the screen.

61. Kept: §2 preliminaries, §4 distance and radius, §5 tractability, the appendices, each with the fixes of the review (criterion named, Proposition 9 paragraph and claim unit in §4, conditions stated in §5, the conjecture stated). Rewritten: title and abstract (pricing frame), §1 (clinician sentence on page one), §3 (missing citations), §7 (limits paragraph uncommented). Replaced: §6, by the executed protocol (frame, ledger, interval and null radius, screen, estimated class); the synthetic τ_b table moves to the appendix as its own panel with a screen column.

62. Any four of: the radius or stop rule beats a leave- k -out screen; robustness certified under an estimated CPDAG; a continuity bound; a probability reading of r ; zero dangerous claim rows presented as a statistic; language models as the subject; the retracted «0/791», « O^* is a local functional», «up to 128×»; the general MPDAG poset is graded. Measurements: the screen ties at matched depth (+0.000 [+0.000, +0.000], M1b); under a learned class the truth extends the analyst's graph on 0 of 244 rows; Proposition 9 of Taeb, Guo and Henckel proves non-gradedness for $|V| \geq 4$.

63. No author can be added or removed after the abstract submission deadline; only the order of authors can change, and only up to the full paper deadline. So the list, with every author's OpenReview profile, must be settled by Friday 18 September, AoE.

64. Commit 3, `results/`. Each results folder carries the `PREREGISTRATION*.md` written before its rows were read, next to the tables built from those rows, so that a reader can check the predictions against the outcomes in one place (for example `results/ledger/PREREGISTRATION.md` with `TABLE4.md`).

65. `rho_breakdown/panelB_T3_2026-09-16/make_figs.py`. It reimplements the pipeline from `results/interval/replicates.csv.gz` and `rows.csv` and asserts, on every run, that it reproduces the 27 panel-A cells of `results/interval/TABLE1_L95.md`; a disagreement stops the script.

66. It must call the generalised adjustment criterion in `src/bkrobust/benchmarks/audit.py`. It must not use `is_valid_adjustment_set_dag`, which implements Pearl's back-door criterion and rejects some sets that are valid under the generalised criterion. `tests/benchmarks/test_audit_oracle.py` guards this.

67. In the script that builds T3 (`interval_tables.py`, under `experiments`), remove the term that sorts a finite null radius first, so the five rows per network are not chosen by the outcome. The alternative: print all 37 eligible rows on the four networks with fitted coefficients instead of choosing.

68. Because the committed questionnaire was the input every supplier answered and it was hashed before any model saw it. The old builder seeded with salted `hash()`, so the file does not regenerate byte for byte; rebuilding it would produce a different questionnaire from the one the answers refer to. The builder therefore refuses to overwrite it.